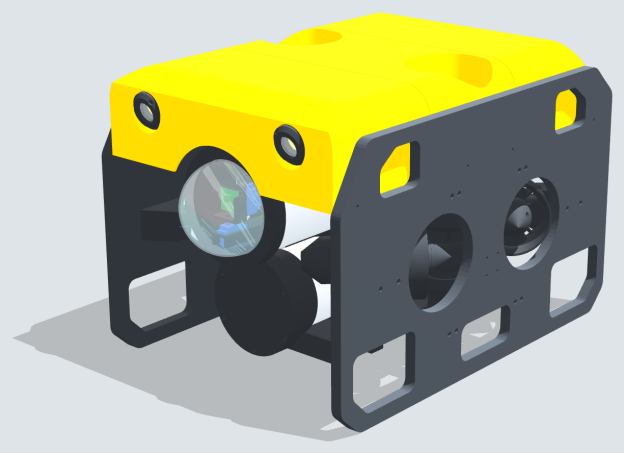
TinkerSub KB-1 Build Guide

A printable, open-source ROV you can build and change. Start-to-finish build: 4" electronics hull, 3" battery pod, HDPE side plates, yellow top cowl and six vectored 1200 GPH thrusters



Colours in every picture match the build: yellow = top cowl (yellow PETG); black = every other printed part (black PETG) and the HDPE side plates; white / grey = PVC pipe and bought parts.

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#	Section
1	Overview and build order
2	What's in the package
3	Tools and ground rules
4	Shopping list
5	Print the parts
6	Cut the side plates from HDPE
7	Build the thrusters (x6)
8	Build the electronics sled
9	Electronics: parts, schematic, wiring and tests
10	Build and test the hull
11	Build the battery pod
12	Build the lights
13	Station frames: join hull and pod
14	Side plates and thrusters
15	Belly pan
16	Junction deck and wiring
17	Top cowl
18	Leak test, trim and first dive
19	Reference views and layout
A	Appendix A: side plate cutting templates, 1:1 (pages 60-68)
B	Appendix B: heat-set insert chart and picture-by-picture assembly (pages 69-78)
C	Appendix C: updating parts from an earlier build (page 79)

1 Overview and build order

TinkerSub KB-1 is an open-source, mostly 3D-printed ROV. It is two sealed tubes held together by two printed station frames: a 4" PVC hull on top with the camera, computer and drivers, and a 3" PVC battery pod underneath. Two 1/2" HDPE side plates bolt to the stations' arms and carry the six oil-filled thrusters. A yellow printed cowl closes the top (lights and foam built in) and a belly pan under the pod holds the lead ballast.

Phase	What you do	Sections
1 Order	Buy the hardware, the HDPE sheet, pipe and both filaments. Long-lead items first: motors, dome, LEDs, drivers.	4
2 Print	Two Bambu Studio projects, 92 parts. Black PETG for everything except the 6 cowl sections (yellow). Start with the Core project: the thrusters take longest to finish.	5
3 Cut	Both side plates from one 24 x 24" HDPE sheet with the paper templates in Appendix A (or have a shop CNC them).	6, A
4 Sub-assemblies	Thrusters, electronics sled and its wiring, hull, battery pod and lights. Each is finished and checked on its own.	7-12
5 Frame	Stations on the hull and pod, side plates with the thrusters, belly pan, junction deck and wiring, cowl.	13-17
6 Test	Vacuum and dunk tests, thruster checks, trim in a pool, then dives at 10, 30 and 60 ft.	18

Key numbers

Item	Value
Size	351 W x 328 H x 508 L mm (13.8 x 12.9 x 20.0 in)
Weight (est.)	about 12.5 kg in air before foam and lead; about 0.8 kg buoyant in fresh water before trimming.
Trim (est.)	1.6 kg lead + 1.0 L foam: about 0.3 kg buoyant, centre of buoyancy 31 mm above the centre of gravity. 2.25 kg + 1.7 L: 41 mm (steadier). Fine-tune in a pool.
Thrusters	4 horizontals at 45 deg in the corners (forward/back, sideways, turn) + 2 verticals at the centre of gravity
Power	3S8P 18650 pack (about 11-12.6 V) in the pod; Cat6 tether carries data only
Depth	Target 60 ft. The printed hull and pod caps carry 100-150 kg there: go deeper in steps (section 18.2), or machine those caps from aluminium.

Directions: port = left and starboard = right when you stand behind the ROV looking forward. The front is the dome end.

2 What's in the package

File / folder	Contents
TinkerSub_KB-1_Assembly_Guide.pdf	This guide.
TinkerSub_KB-1_Frame.3mf	Bambu Studio project, 28 parts: station frames, junction deck, corner brackets, thruster pylons, belly pan (black) and the top cowl (yellow, on its own plates).
TinkerSub_KB-1_Core.3mf	Bambu Studio project, 64 parts, all black: thrusters, hull flanges and caps, sled, camera mount, battery pod, lights, junction box and lid.
TinkerSub_KB-1_Reprint_2026-10-20.3mf	The parts changed on 2026-10-20, for builders who printed before then (Appendix C).
BOM_printed_parts.csv	Every printed part: project, file, quantity, filament colour, print notes, filament weight.
Parts_to_buy.csv	The shopping list (section 4): every bought part with quantity, exact size, use, rough price and source.
Pipe_cut_list.csv	4" hull and 3" pod lengths.
STL/	Every printable part in print orientation (mm), in 8 folders in build order. The cowl folder is marked YELLOW.
CAD/	Side plate templates (PDF) and DXFs: which file for which way of cutting is in section 6. CAD/STEP: a STEP file for most printed parts and the whole ROV as a mesh (TinkerSub_KB-1_assembly_mesh.3mf). The full assembly STEP (named, coloured components, for Fusion 360 or any CAD app) is too big for GitHub: it is the TinkerSub_KB-1_assembly_step.zip download on the KB-1 release.
Images/	Renders, assembly pictures, part pictures, diagrams and the plate maps of both projects.
Source/	The Python generators for every part, the 3MFs, templates, STEP files, renders and this guide, with their input files. One command rebuilds it all: python3 Source/build.py all (Source/README.md).
Firmware/	Pico firmware (source and ready-to-flash UF2), the Pi Zero bridge, the topside program and their tests.
docs/	Battery_Pack_Build.md: cell matching, welding, BMS wiring, first charge and voltage limits for the 3S8P pack.
LICENSE, LICENSES/	Hardware CERN-OHL-S-2.0, software MIT, documentation CC BY-SA 4.0. CHANGELOG.md lists every change to the files.

3 Tools and ground rules

Tools

Tool	Used for
Bambu A1 (256 mm bed)	Every part fits. The station halves and cowl sections are the largest.
Soldering iron with heat-set insert tip	M3, M4 and M5 brass inserts (Appendix B has a chart of every one)
Taps: M12 x 1.5 (fine pitch), M4 x 0.7, M3 x 0.5	Tail-cone glands, oil ports, light set screws (the cap threads are printed)
Jigsaw + fine blades, drill + 3.5 / 4.5 / 5.5 / 10 mm bits, centre punch, clamps	HDPE side plates
Drill + 1/8" (3.17 mm) bit or reamer	Open the brass coupler's 3 mm end for the motor shaft
Drill press or lathe + files	Aluminium light discs (or have them cut)
PVC saw or cutter, deburring tool	4" and 3" pipe
Hex keys 2-4 mm, 7 / 8 mm nut drivers	General assembly. The 8 mm driver fits inside the prop hub.
Syringe (60 ml) with a short tube	Oil fill
Hand vacuum pump + gauge (generic brake-bleeder type) + 6 mm hose	Hull and pod vacuum tests through the M10 barb
Spot welder (or a shop that has one)	Nickel strip on the 18650 pack. Never solder directly to cells.
Multimeter	Wiring and pack checks

Ground rules

1. Measure your motor shaft before printing the props: the design assumes 3.17 mm (1/8").
2. PVC cement only bonds PVC to PVC. Printed parts are bolted, or epoxied where they meet the pipe. Nothing sticks to HDPE: it is always bolted.
3. Grease every O-ring lightly with silicone grease. Never use petroleum grease on nitrile.
4. Keep the 12 V wiring and the 3.3 V / 5 V logic apart: the Pico's pins are 3.3 V only (section 9).
5. Label left/right parts as they come off the printer: most frame parts come as a port and a starboard version.

4 Shopping list

Every bought part for one ROV: total quantity, exact size, what it is for, and a rough US price to budget with (check current prices before ordering). The screw, nut and insert counts come from the build's fastener table, which is checked against the 3D model, and the filament weights from the models. Tools are in section 3. The same list is Parts_to_buy.csv.

Group	Qty	Item (exact size)	Used for	Price	Source
Printed parts	2.9 kg	PETG filament, black	every printed part except the cowl (includes ~10 % for brims and reprints)	$45-60	Bambu / Amazon
	1.5 kg	PETG filament, yellow	the six top cowl sections	$20-25	Bambu / Amazon
Fasteners	6	M2.5 x 25 mm socket head cap screw, stainless	light head: ring + disc to body	$3-5 (pack)	Amazon / hardware store
	18	M3 x 8 mm button head screw, stainless	nose cone to prop hub	$3-5 (pack)	Amazon / hardware store
	4	M3 x 8 mm socket head cap screw, stainless	junction box to deck	$3-5 (pack)	Amazon / hardware store
	4	M3 x 8 mm countersunk screw, stainless	junction box lid	$3-5 (pack)	Amazon / hardware store
	10	M3 x 10 mm socket head cap screw, stainless	junction deck to station rib; radial flange screw through the pipe	$3-5 (pack)	Amazon / hardware store
	6	M3 x 12 mm socket head cap screw, stainless	dome ring	$3-5 (pack)	Amazon / hardware store
	22	M3 x 16 mm socket head cap screw, stainless	corner pylon to bracket; hull rear cap	$3-5 (pack)	Amazon / hardware store
	28	M3 x 20 mm socket head cap screw, stainless	station top halves, centre rib; cowl centre seam; cowl section seam; cowl foot to side plate	$3-5 (pack)	Amazon / hardware store
	8	M3 x 30 mm socket head cap screw, stainless	vertical pylon through the plate	$3-5 (pack)	Amazon / hardware store
	4	M4 x 20 mm socket head cap screw, stainless	station middle to bottom (pod pair)	$3-5 (pack)	Amazon / hardware store
	16	M4 x 30 mm socket head cap screw, stainless	pylon pinch bolt; station top to middle (hull pair)	$3-5 (pack)	Amazon / hardware store
	4	M4 x 50 mm socket head cap screw, stainless	belly pan hanger	$3-5 (pack)	Amazon / hardware store
	6	M5 x 14 mm socket head cap screw, stainless	pod rear cap	$3-5 (pack)	Amazon / hardware store
	6	M5 x 16 mm socket head cap screw, stainless	pod front cap	$3-5 (pack)	Amazon / hardware store
	24	M5 x 20 mm socket head cap screw, stainless	side plate to station arm; side plate to corner bracket	$3-5 (pack)	Amazon / hardware store
	28	M3 nyloc nut, stainless	screw joints with a nut	$3 (pack)	Amazon / hardware store
	8	M4 hex nut, stainless	screw joints with a nut	$3 (pack)	Amazon / hardware store
	8	M3 washer, 1 mm thick (shim)	vertical pylons	$2 (pack)	Hardware store
	42	M3 brass heat-set insert (for 3D prints)	Appendix B chart	$8-10 (pack)	Amazon
	16	M4 brass heat-set insert (for 3D prints)	Appendix B chart	$8-10 (pack)	Amazon
	36	M5 brass heat-set insert (for 3D prints)	Appendix B chart	$8-10 (pack)	Amazon

Group	Qty	Item (exact size)	Used for	Price	Source
Thrusters	6	1200 GPH bilge pump motor cartridge	T1-T6	$20-25 each	Amazon / marine store
	6	Brass shaft coupler, 3 mm to 5 mm, 9 mm OD x 20 mm	motor shaft to prop stud	$12 (6)	Amazon
	6	M5 x 20 mm set screw (grub screw), stainless	prop stud in each coupler	$4 (pack)	Hardware store
	6	M5 hex nut, stainless	jam nut, drives the prop hub	$2 (pack)	Hardware store
	6	M5 nyloc nut, stainless	holds the prop hub on	$2 (pack)	Hardware store
	12	M4 x 4 mm set screw (grub screw), stainless	oil fill and bleed ports, 2 per tail cone	$3 (pack)	Hardware store
	6	O-ring 32 x 2 mm (ID x cross-section), nitrile	tail cone spigot	$6 (kit)	Amazon
	6	O-ring 5 x 1.5 mm, nitrile	holds each compensator bladder on its nipple	in the 1.5 mm kit	Amazon
	6 (+4 spare)	Nitrile finger cot	compensator bladders	$6 (pack)	Pharmacy / Amazon
	500 ml	Mineral oil (white, food or USP grade)	oil fill	$6	Pharmacy
	6	Cable gland M12 x 1.5, nylon, clamping range 3-6.5 mm	motor lead out of each tail cone	$8 (10-pack)	Amazon
Hull	1	4" Sch40 PVC pipe, 406 mm (16")	electronics hull	$15	Hardware store
	1	Acrylic dome 100 mm, 115 mm flange (UQISOVI, Amazon B0DR2HP4YQ)	camera dome	$15-25	Amazon
	4	O-ring AS568-238 (94.84 x 3.53 mm / 3.734 x 0.139"), nitrile 70A	flanges 201 and 203, two radial grooves each	$5 (pack)	Amazon / McMaster
	1	O-ring AS568-237 (91.67 x 3.53 mm / 3.609 x 0.139"), nitrile 70A	rear cap face seal, on 201	$5 (pack)	Amazon / McMaster
	1	O-ring AS568-240 (101.19 x 3.53 mm / 3.984 x 0.139"), nitrile 70A	dome face seal, on 203	$5 (pack)	Amazon / McMaster
	2	Cable gland M16 x 1.5, nylon, clamping range 5-10 mm	MOTORS and BATTERY in the rear cap	$8 (5-pack)	Amazon
	2	Cable gland M12 x 1.5, nylon, clamping range 3-6.5 mm	LIGHTS and TETHER in the rear cap	$8 (10-pack)	Amazon
Hull + pod	2	M10 x 1.0 x 12 mm hex bolt (fine pitch), stainless	VENT plugs in 202 and 302	$4	Hardware store / Amazon
	4	M10 bonded sealing washer (Dowty)	under the vent bolts and the test barb (2 spare)	$6 (pack)	Amazon
	1	M10 x 1.0 male to 6 mm (1/4") hose barb, brass	vacuum tests	$6	Amazon
Sled	2	M3 threaded rod, 1 m, stainless	cut 4 sled rods of 340 mm	$4 each	Hardware store
	24	M3 hex nut, stainless	sled rods (16) and pan-tilt base (8)	$3 (pack)	Hardware store
	12	M3 x 8 mm male-female standoff, brass	IBT_2 boards	$10 (standoff kit)	Amazon
	12	M3 x 8 mm female-female standoff, brass	IBT_2 boards	in the standoff kit	Amazon
	24	M3 x 6 mm screw	IBT_2 boards	$3 (pack)	Hardware store
	4	M3 x 10 mm male-female standoff, brass	Pico breakout	in the standoff kit	Amazon
	4	M3 x 6 mm screw	Pico breakout onto its standoffs	$3 (pack)	Hardware store
	4	M2.5 x 8 mm male-female standoff, brass	Pi Zero	$6 (kit)	Amazon
	4	M2.5 x 6 mm screw	Pi Zero onto its standoffs	in the M2.5 kit	Amazon
Camera	1	Arducam for Pi Zero (Pi Camera v2 size board) + 150-200 mm Pi Zero ribbon	camera	$20-30	Arducam / Amazon

Group	Qty	Item (exact size)	Used for	Price	Source
	2	MG90S metal-gear micro servo (with its screws and horns)	pan and tilt	$10 (2)	Amazon
	9	M2 x 6 mm self-tapping screw	servo horn (4), camera (4), tilt lever (1)	$6 (M2 kit)	Amazon / hobby shop
	1	M2 x 8 mm self-tapping screw	cradle pivot	in the M2 kit	Amazon / hobby shop
	4	M2 x 2 mm spacer / standoff	under the camera board	in the M2 kit	Amazon / hobby shop
	1	M2 flat washer	tilt lever	in the M2 kit	Amazon / hobby shop
	100 mm	1 mm steel music wire	tilt pushrod	$3	Hobby shop
Battery pod	1	3" Sch40 PVC pipe, 300 mm	battery pod	$12	Hardware store
	4	O-ring 72 x 2.5 mm, nitrile	pod caps, two per cap	$5 (pack)	Amazon
	1	Cable gland M16 x 1.5, nylon, clamping range 5-10 mm	battery lead out of the rear cap	$8 (5-pack)	Amazon
	24	18650 Li-ion cell, matched (same model and batch)	3S8P pack	$3-5 each	Battery supplier
	1	3S BMS, 40 A, with balancing	pack	$12	Amazon
	1	M4 threaded rod, 1 m, stainless	cut 4 pack tie rods of ~225 mm	$4	Hardware store
	16	M4 hex nut, stainless	pack tie rods	$3 (pack)	Hardware store
	3 m	Pure nickel strip, 8 x 0.2 mm (not nickel-plated steel)	cell links	$10	Amazon
	1 + 1	Inline ATO fuse holder, 12 AWG + 35 A ATO fuse	pack fuse	$5	Auto parts store
	1 set	Fish-paper insulator rings (18650), Kapton tape, silica gel packs	pack	$15	Amazon
	1	2-part marine epoxy (JB Weld Marine or West System)	bonds the pod flanges to the pipe	$10	Hardware store
Lights	2 (+1 spare)	Cree XHP50.2 LED, 6 V version, 5000-6500 K, on a 20 mm copper DTP star	light heads	$8-12 each	Amazon / Satisled / KaiDomain
	2	Mean Well LDD-700HW constant-current LED driver	inside the hull	$8-10 each	Amazon / Jameco / Digi-Key
	2	Glass lens 28 x 3 mm, AR coated	light windows	$8 (pack)	Amazon
	1	6061 aluminium round bar, 1-3/4" x 6"	two 5 mm heat-sink discs	$12	Amazon
	1	Thermal adhesive (Arctic Alumina)	LED star onto the disc	$10	Amazon
	2	O-ring 30 x 1.5 mm, nitrile	front ring face groove	$10 (1.5 mm kit)	Amazon
	2	O-ring 24 x 1.5 mm, nitrile	window, under the front lip	in the 1.5 mm kit	Amazon
	2	M3 x 8 mm set screw (grub screw), stainless	locks each light head in the cowl	$2 (pack)	Hardware store
Frame	1	Black HDPE sheet, 1/2" (12.7 mm), 24 x 24" (at least 605 mm one way)	both side plates	$40-60	Amazon / plastics supplier
	1.6-2.3 kg	Lead shot in sealed bags (or lead bar)	belly pan ballast	$15-25	Sporting goods
Cowl	1.0-1.7 L	Depth-rated closed-cell foam (PVC core foam such as H80, or 2-lb closed-cell PU)	buoyancy blocks in the cowl	$20-35	Composites supplier / Amazon
Junction box	6 pairs	SP13 IP68 2-pin connector: panel socket (rear nut) + cable plug, 4-6.5 mm cable	thrusters T1-T6	$18-25	Amazon / AliExpress

Group	Qty	Item (exact size)	Used for	Price	Source
	2 pairs	SP13 IP68 3-pin connector: panel socket (rear nut) + cable plug, 4-6.5 mm cable	lights L1, L2	$6-9	Amazon / AliExpress
	1	MG Chemicals 832B epoxy potting compound	junction box and light cavities	$35-40	Amazon
Wiring	4 m	2-core 18 AWG marine cable, OD 5-6 mm	6 thruster leads + 2 light leads	$15	Amazon
	0.8 m	12-core control cable, 0.5 mm2 cores, OD 8-10 mm	junction box to the hull (motors)	$10	Amazon
	0.5 m	4-core cable, 0.5-0.75 mm2 cores, OD 3-6.5 mm	junction box to the hull (lights)	$4	Amazon
	0.6 m	2-core 12 AWG silicone or marine cable, OD 5-10 mm	battery pod to the hull	$6	Amazon
	as needed	Cat6 cable, slim (OD 6.5 mm or less), + strain relief	tether (data only)	$20-40	Amazon
Electronics	6	IBT_2 (BTS7960) H-bridge motor driver	U1-U6	$40-60 (6)	Amazon
	2	Mini MP1584EN buck board, fixed 5 V, 1.8 A	U9A, U9B	$8 (pack)	Amazon
	1	Raspberry Pi Pico with headers + 57 x 63 mm screw-terminal breakout board	U10	$12	Pi reseller / Amazon
	1	Raspberry Pi Zero 2W with header	U11	$15-18	Pi reseller
	1	Waveshare ETH/USB HUB HAT (RTL8152B) for Pi Zero	U12	$15-20	Waveshare / Amazon
	1	Cat6 patch lead 0.3 m, one end 90-degree	HAT to the tether	$6	Amazon
	1	RJ45 inline coupler	patch lead to the tether	$3	Amazon
	1 + 1	Inline ATO/ATC fuse holder, 12 AWG + 30 A fuse	F1	$7	Auto parts store
	1 + 1	Inline mini-blade fuse holder + 3 A fuse	F2	$4	Auto parts store
	1	Capacitor 1000 uF 25 V low-ESR electrolytic	C1	$1	DigiKey / Mouser
	1	Capacitor 470 uF 10 V electrolytic	C2	$0.50	DigiKey / Mouser
	1	1N5817 Schottky diode	D1	$0.25	DigiKey / Mouser
	3	Resistor 10 k, 1/4 W	R1, R2, R4	$0.10 each	Any
	1	Resistor 33 k, 1/4 W	R3	$0.10	Any
	2	Wago 221-415 lever connector (5-way)	12 V and GND bus	$6 (2)	Amazon
	1 m + 1 m	14 AWG silicone wire, red + black	bus to the top driver boards	$8	Amazon
	0.5 m + 0.5 m	16 AWG silicone wire, red + black	jumpers to the bottom driver boards	$6	Amazon
	1 m + 1 m	20 AWG silicone wire, red + black	aux branch	$5	Amazon
	1 set	Dupont female housings + 22 AWG and 26 AWG hook-up wire	logic wiring: RPWM / LPWM wires and the ARM / VCC / GND chain	$8	Amazon
Supplies	1	Silicone O-ring grease (not petroleum)	every O-ring	$6	Hardware store
	1	Silicone dielectric grease	SP13 plug threads and pins	$6	Hardware store
	1	Clear silicone sealant (RTV), small tube	gland threads and oil-port set screws	$6	Hardware store
	1 set	Heat-shrink assortment incl. 20-25 mm clear, double-sided foam tape, zip ties 2.5-3.6 mm	splices, buck boards, harness	$8	Any

Pipe cut list

Piece	Qty	mm	in
4" Sch40 hull	1	406	15.98
3" Sch40 battery pod	1	300	11.81

5 Print the parts

Everything is in two Bambu Studio projects, already laid out on plates: open the 3MF, slice, print plate by plate. Each plate holds one group of parts and is named after them, with their part numbers (for example "Camera pan-tilt: 216, 217, 218, 219"); both plate maps are at the end of this section. About 2.6 kg of black PETG and 1.3 kg of yellow. Bambu Studio shows each plate's time after slicing.

Print settings

Setting	Value
Material	PETG for everything. Black for all parts except the top cowl, which is yellow.
Profile	Bambu A1 0.4 nozzle, 0.20 mm Standard, Bambu PETG Basic. Per-part walls, infill, brims and supports are already set in the 3MF files.
Walls / infill	4 walls, 25 % gyroid by default; 5 walls on thread, O-ring and bracket parts; 40 % on the corner brackets. SOLID (100 %) on the parts that hold water out, so it can't seep through the skin and past the O-rings: hull flanges 201 and 203, rear cap 202, pod flanges 301 and caps 302 / 303, light front rings 402.
Supports	None, except tree supports under the prop blades, inside the two middle cowl sections and under the camera cradle's (218) port cheek. Those parts, and all cowl sections, sit at least 18-28 mm from their neighbours so supports and brims can't collide.
After printing	Seal the thruster housings, tail cones and light heads: a thin coat of epoxy on the outside closes any layer gaps.

Heat-set inserts: 18 of the printed parts take them (42 x M3, 16 x M4, 36 x M5 in all). Appendix B at the back opens with a chart of every printed part showing which hole takes which insert, from which side; the others take none. Press them all as the parts come off the printer, before you start building.

1. TinkerSub_KB-1_Core.3mf first, plates 1-4: thruster housings and tail cones (1-2), then the props with their nose cones (3 right-hand, 4 left-hand; tree supports under the blades are already on). The thrusters' oil fill and testing take the longest, so you can build them while the rest prints.
2. Then TinkerSub_KB-1_Core.3mf plates 5-10: hull rear flange, cap and sled bulkheads (5); dome flange and camera pan-tilt (6); dome ring, lights and junction box with lid (7); sled rails and deck (8); battery pod (9); battery pack plates and spacers (10).
3. TinkerSub_KB-1_Frame.3mf plates 1-6 in black: station tops (1), station middles (2), station bottoms (3), thruster pylons and corner brackets (4), junction deck (5), belly pan (6).
4. Change to yellow PETG for TinkerSub_KB-1_Frame.3mf plates 7-9: the six top cowl sections, rear, middle and front. Each plate's name says YELLOW and no plate mixes colours.
5. Check every part against the list below as it comes off. Pictures of every part are in the Appendix B chart.

TinkerSub_KB-1_Frame.3mf (28 parts)

Part #	File	Qty	Print notes	Used for	g
Stations + brackets					
601	601_rov_station_top_port.stl	2	Flat, no supports. 2x M5 + 1x M3 heat-set inserts.	Station top, port half: hull top + upper arm	80
602	602_rov_station_top_stbd.stl	2	Flat, no supports. 2x M5 + 1x M3 heat-set inserts.	Station top, starboard half	80
603	603_rov_station_mid.stl	2	Flat, no supports. M4 nuts in the pockets.	Station middle: hull bottom + pod top	81
604	604_rov_station_bot_port.stl	2	Flat, no supports. 2x M5 + 1x M4 inserts.	Station bottom, port half: pod bottom + lower arm	77
605	605_rov_station_bot_stbd.stl	2	Flat, no supports. 2x M5 + 1x M4 inserts.	Station bottom, starboard half	77
606	606_rov_junction_deck.stl	1	Flat. 4x M3 heat-set inserts.	Carries the junction box (501)	129
607	607_rov_bracket_corner_a.stl	2	Plate face down. 5 walls, 40 %. 4x M3 + 2x M5 inserts.	Front-port + rear-starboard thruster bracket	53
608	608_rov_bracket_corner_b.stl	2	Plate face down. 5 walls, 40 %. 4x M3 + 2x M5 inserts.	Front-starboard + rear-port thruster bracket	53
Thruster pylons					
106	106_pylon_clamp.stl	6	Rear end down, no supports. M4 + M3 inserts.	Clamps each thruster body	90
Belly pan					
701	701_rov_belly_pan.stl	1	Floor down, no supports.	Lead ballast tray under the pod	107
Top cowl (YELLOW)					
801	801_rov_cowl_rear_port.stl	1	Rear end down. The 3 mm shell prints solid.	Top cowl, rear, port	196
802	802_rov_cowl_rear_stbd.stl	1	Rear end down.	Top cowl, rear, starboard	196
803	803_rov_cowl_mid_port.stl	1	Seam face down. Tree supports (vertical thruster sleeve).	Top cowl, middle, port	231
804	804_rov_cowl_mid_stbd.stl	1	Seam face down. Tree supports.	Top cowl, middle, starboard	231
805	805_rov_cowl_front_port.stl	1	Nose down (light sleeve vertical).	Top cowl, front, port	218
806	806_rov_cowl_front_stbd.stl	1	Nose down.	Top cowl, front, starboard	218

TinkerSub_KB-1_Core.3mf (64 parts)

Part #	File	Qty	Print notes	Used for	g
Thrusters					
101	101_housing_no_foot.stl	6	Threaded rear end down, brim, 5 walls.	Thruster body + Kort duct	410
102	102_tail_cone_comp.stl	6	Tip down, brim. Tap the tip M12 x 1.5.	Tail cone with oil compensator	80
103	103_prop_hub_RH.stl	3	Back face down, tree supports under the blades.	Right-hand prop	81
104	104_prop_hub_LH.stl	3	Back face down, tree supports under the blades.	Left-hand prop	81
105	105_nose_cone.stl	6	Base down.	Prop spinner	15
107	107_coupler_gauge.stl	1	Tool only, any material.	Sets the coupler height	3
Hull + sled					
201	201_br_flange_rear.stl	1	Ring face down. SOLID: 100 % infill (holds water out).	Hull rear flange	86
202	202_rov_endcap_rear_5hole.stl	1	Flat. SOLID: 100 % infill (holds water out).	Hull rear cap: 4 glands + vent	66
203	203_br_flange_front_dome.stl	1	Ring face down. SOLID: 100 % infill (holds water out).	Dome flange	107
204	204_dome_retaining_ring.stl	1	Flat.	Dome clamp ring	30
207	207_sled_bulkhead_front.stl	1	Flat.	Sled front bulkhead	27
212	212_sled_rail_port.stl	1	Flat outer face down, no supports.	Sled side rail, port (12 V side)	34
213	213_sled_rail_stbd.stl	1	Flat outer face down, no supports.	Sled side rail, starboard (logic side)	34
214	214_sled_deck.stl	1	Flat. 4 walls, 40 %.	Sled deck: boards on both faces	43
215	215_sled_bulkhead_rear_v2.stl	1	Front face down, pockets up.	Sled rear bulkhead: 12 V bus + LED drivers	39
216	216_pantilt_base.stl	1	Plate face down, no supports. 5 walls.	Pan-tilt base v2: on the sled rods, carries the pan servo	25
217	217_pantilt_yoke.stl	1	Base down, no supports. 5 walls, 40 % infill.	Pan-tilt yoke v2: on the pan horn, tilt servo in a closed pocket	6
218	218_pantilt_cradle.stl	1	On its side, axle up. Tree supports ON (under the port cheek). 100 %.	Pan-tilt camera cradle	1
219	219_pantilt_lever.stl	1	Flat. 100 %.	Tilt lever on the cradle axle	0
Battery pod					
301	301_rov_pod_flange_3in.stl	2	Socket end down. SOLID: 100 % infill. 6x M5 inserts each.	3" pod flange, one at each end	144
302	302_rov_pod_endcap_3in_rear.stl	1	Plate down. SOLID: 100 % infill (holds water out).	3" pod rear cap: M16 gland + vent	92
303	303_rov_pod_endcap_3in_front.stl	1	Plate down. SOLID: 100 % infill (holds water out).	3" pod front cap, blank (no holes)	110
305	305_rov_cell_plate_8_v2.stl	6	Grooved face down.	18650 holder plate	110
306	306_rov_cell_spacer_3mm.stl	8	Flat.	Rod spacer between groups	1
Lights					
401	401_rov_light_rear_body.stl	2	Socket end down. 5 walls. (ASA also fine.)	Light head rear body	38
402	402_rov_light_front_ring.stl	2	Front face down. SOLID: 100 % infill (seals the window).	Light head front ring	14
403	403_rov_light_spacer.stl	2	Flat.	Window spacer	3
Junction boxes					

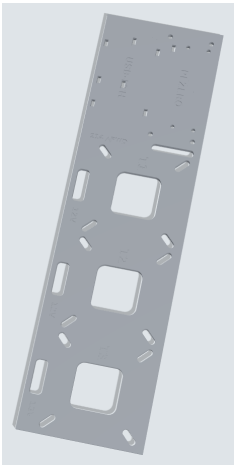
Part #	File	Qty	Print notes	Used for	g
501	501_rov_junction_box.stl	1	Base down, no supports. 5 walls.	Junction box: 6 thruster + 2 light SP13 sockets, 12-core + 4-core out	38
503	503_rov_junction_lid.stl	1	Top face down (as exported), no supports.	Junction box lid with the socket map	28

Part numbers and labels

Every printed part has its three-digit part number debossed (cut 0.6 mm in) on a flat face, in the part's own colour: no second filament, no painting. The first digit is the subsystem: 1 thrusters, 2 hull and sled, 3 battery pod, 4 lights, 5 junction box, 6 stations and brackets, 7 belly pan, 8 top cowl (yellow), 9 the two HDPE side plates (marked from the template). The STL and 3MF file names start with the same number, and the guide refers to parts by it.

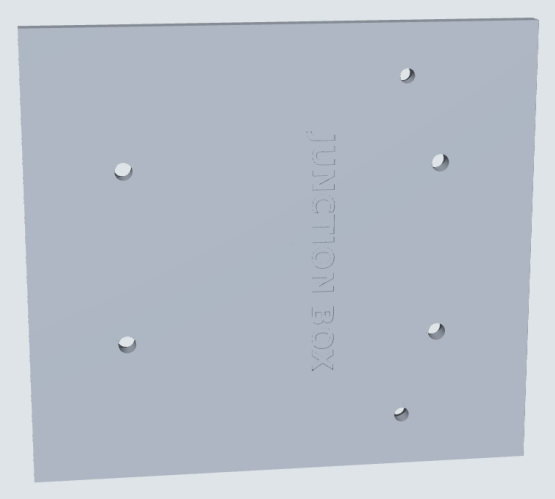
Parts that something mounts on also carry where it goes:

Part	Label(s)	Meaning
214 sled deck	PI ZERO, USB-ETH, T1-T3, 12V (top); PICO, BUCK, C1, T4-T6 (underneath); FWD arrow. BUCK and C1 are no longer used there; the two mini 5 V bucks now sit in the USB-ETH space	Where each board goes. Every power terminal faces port, every logic header starboard.
212 / 213 sled rails	PORT, 12 V SIDE / STBD, LOGIC SIDE; FWD arrows	Which rail goes where, and which side each harness runs.
215 sled rear bulkhead	+12V, GND, LDD 1, LDD 2, 12V and LOGIC windows, TOP	Wago bus and LED driver pockets on its rear face.
216 pan-tilt base	TOP, PAN SERVO	Which way up, and where the pan servo goes.
202 hull rear cap	MOTORS, LIGHTS, BATTERY, TETHER, VENT (outside face)	Which gland goes in which hole.
302 / 303 pod caps	REAR, BATTERY, VENT / FRONT (outside faces)	Pod ends and the rear cap's two holes.
305 battery holder plate	Number on the grooved face	Grooved face goes outward; tab notch at the top.
601-605 station parts	PORT / STBD	Which side each half goes.
606 junction deck	JUNCTION BOX (top face)	Where the junction box screws down.
501 junction box / 503 lid	T1-T6, L1, L2 under each socket; same map on the lid	Which plug goes in which socket.
607 / 608 corner brackets	A FP/RS, B FS/RP	Bracket A: front-port and rear-starboard; B: front-starboard and rear-port.
103 / 104 props	RH / LH (back face)	Prop hand.
801-806 cowl sections	REAR / MID / FRONT + PORT / STBD (inside)	Where each section goes.
901 / 902 HDPE plates	Template marks each hole: 601-605 ARM, 607 A / 608 B, T5 / T6 PYLON, COWL FOOT	Write them on the plate with a paint marker while the template is on.

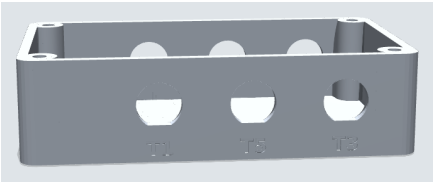


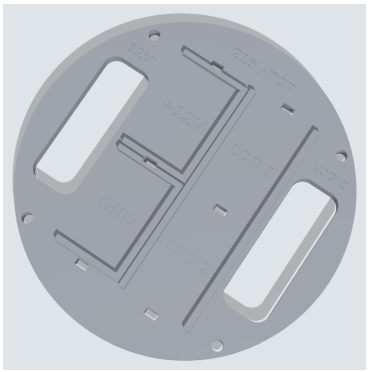
214 sled deck, top face: board positions engraved.

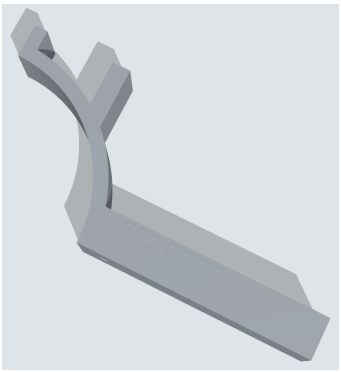




Left: 202 hull rear cap, outside face, one label per gland. Right: 606 junction deck, top face.







501 junction box, port face: T1, T5, T3; 215 sled rear bulkhead, rear face; 601 station top, PORT.

TinkerSub_KB-1_Frame.3mf plate map (the three yellow plates are the cowl)

Plate 1: Station tops - 601 ×2, 602 ×2

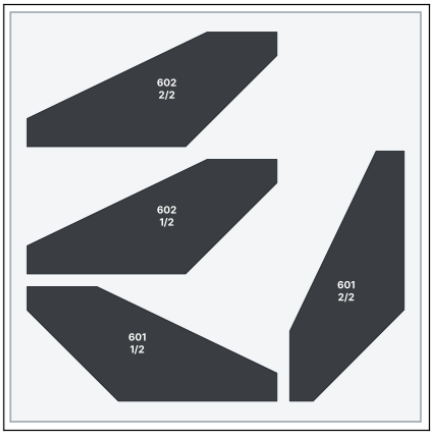


Plate 2: Station middles - 603 ×2



Plate 3: Station bottoms - 604 ×2, 605 ×2

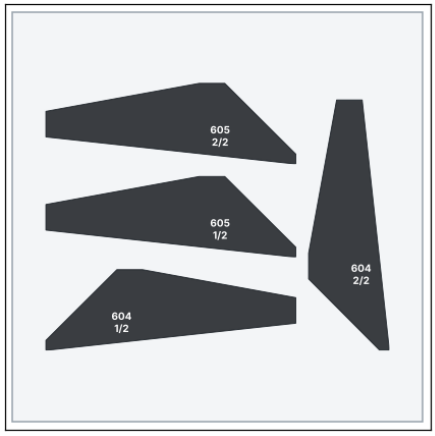


Plate 4: Thruster pylons + corner brackets - 106 ×6, 607 ×2, 608 ×2

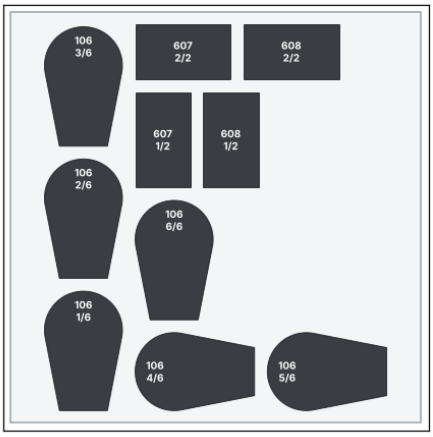


Plate 5: Junction deck - 606



Plate 6: Belly pan - 701



Plate 7: Top cowl rear - YELLOW - 801, 802

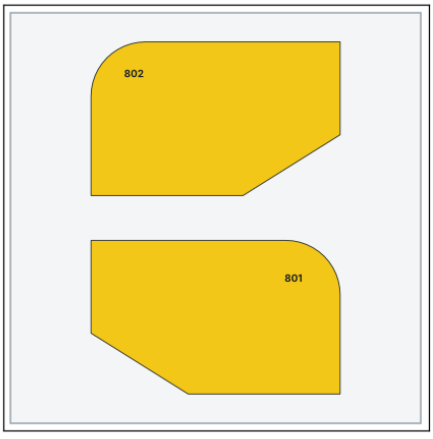


Plate 8: Top cowl middle - YELLOW - 803, 804

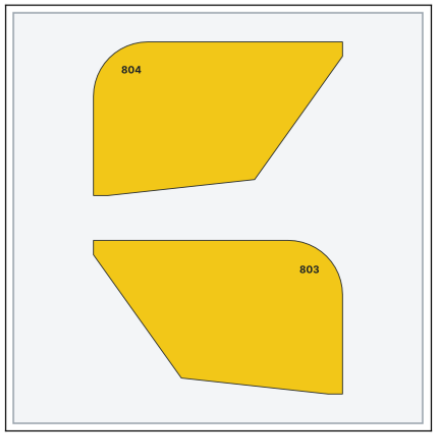
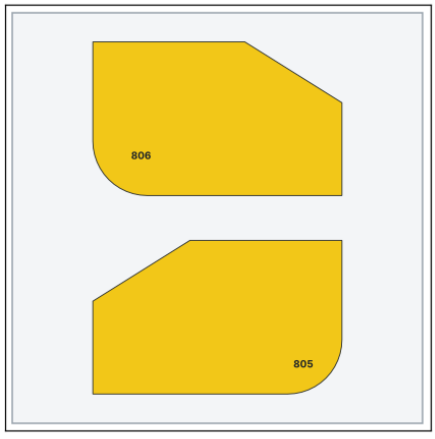


Plate 9: Top cowl front - YELLOW - 805, 806



TinkerSub_KB-1_Core.3mf plate map

Plate 1: Thruster housings + tail cones (1 of 2) - 101 ×4, 107

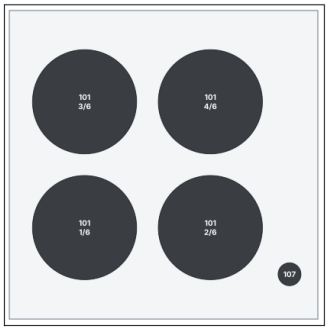


Plate 2: Thruster housings + tail cones (2 of 2) - 101 ×2, 102 ×6

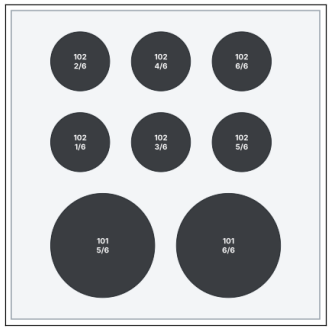


Plate 3: Props right-hand + nose cones - 103 ×3, 105 ×3

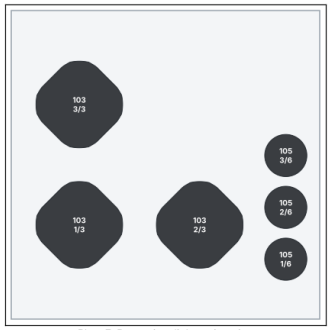


Plate 4: Props left-hand + nose cones - 104 ×3, 105 ×3

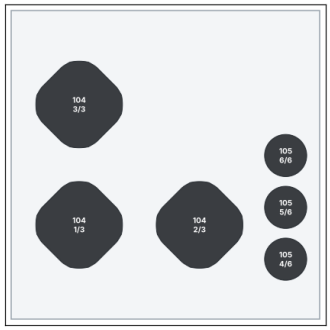


Plate 5: Hull rear flange, cap + sled bulkheads - 201, 202, 207, 215

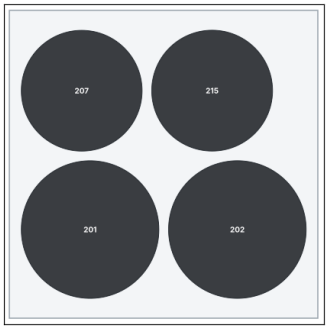


Plate 6: Dome flange + camera pan-tilt - 203, 216, 217, 218, 219

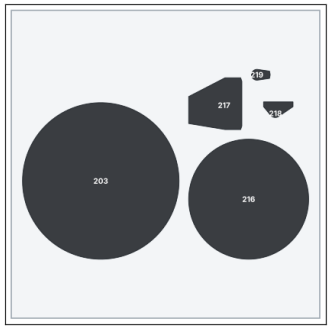


Plate 7: Dome ring, lights + junction box - 204, 401 ×2, 402 ×2, 403 ×2, 501, 503

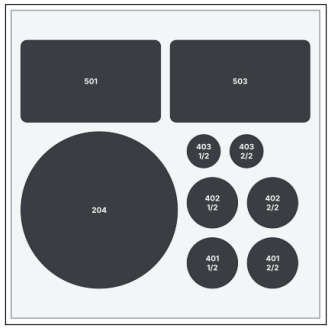


Plate 8: Sled rails + deck - 212, 213, 214

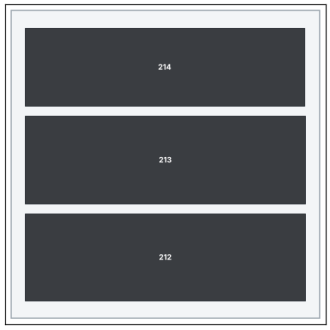


Plate 9: Battery pod - 301 ×2, 302, 303

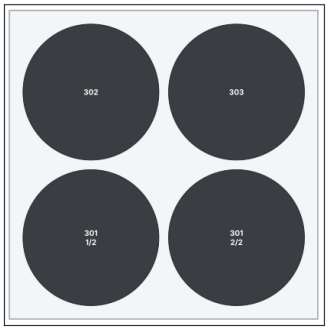
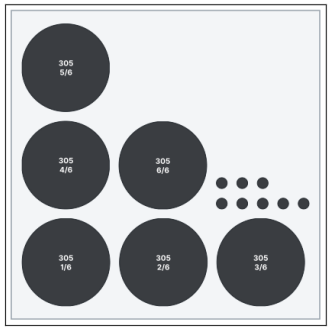


Plate 10: Battery pack plates + spacers - 305 ×6, 306 ×8



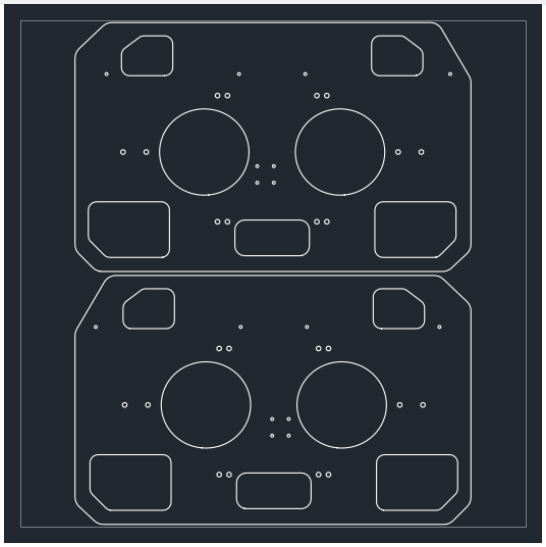
6 Cut the side plates from HDPE

Parts for this section

901	Side plate, port (1/2" HDPE, cut)	x1	**902**	Side plate, starboard (1/2" HDPE, cut)	x1

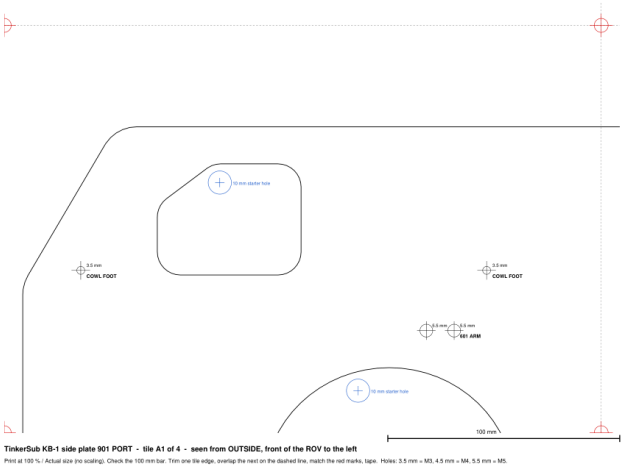
Both plates come out of one 24 x 24 x 1/2" sheet of black HDPE (marine board). Each plate is 478 x 300 mm (18.8 x 11.8 in) and its bottom edge is the skid. The two plates are mirror images. Pick one way to get the 1:1 templates, every hole marked with its drill size:

Template	File (in CAD/)
Letter landscape tiles, taped together (Appendix A of this guide)	rov_side_plate_templates.pdf
Letter portrait tiles	rov_side_plate_templates_letter_portrait.pdf
Each plate whole on one 24 x 18 in sheet: print shop, no taping (easiest)	rov_side_plate_templates_fullsize_24x18.pdf
No template: a shop with a CNC router cuts them; then skip to section 7	rov_side_plate_port.dxf, rov_side_plate_stbd.dxf (mm)

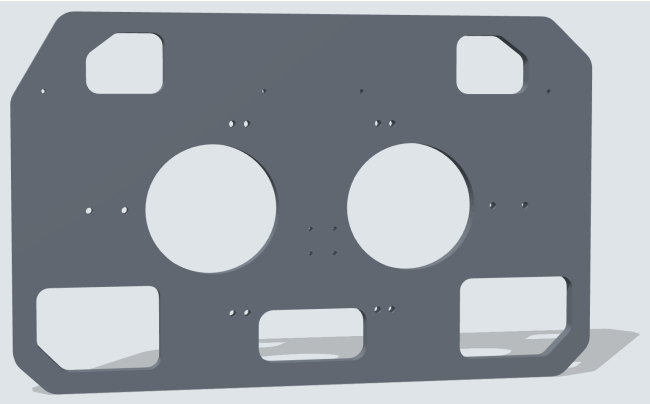


Both plates on the 24 x 24" sheet, 4.8 mm apart. The sheet has only about 2 mm to spare at each edge.

1. Measure the sheet: it needs at least 605 mm (23.8") in one direction. Undersize sheets are common.
2. Print the templates at 100 % / Actual size: turn off 'fit to page', 'shrink oversized pages' and any borderless scaling. Measure the 100 mm bar on every page before trimming anything; if it isn't exactly 100 mm, fix the setting and reprint.
3. Trim and tape the 4 tiles of each plate together on the dashed lines, matching the red marks.
4. Spray-glue (3M Super 77) or glue-stick both templates onto the sheet as on the layout, 5 mm apart, outside faces up.
5. Centre-punch every crosshair. Drill on a scrap-wood backer at slow speed: 3.5 mm (M3: vertical pylons, cowl screws), 4.5 mm (M4 where marked), 5.5 mm (M5: station arms, corner brackets).
6. Drill the blue 10 mm starter hole in each window and cut the windows with a jigsaw, fine-tooth blade, slow speed, just outside the line. Fast cutting melts HDPE and it welds shut behind the blade.
7. Cut the two outlines, then file and sand all edges and break the corners. Write 901 PORT / 902 STBD and the hole tags printed on the template (601-605 ARM, 607 A / 608 B, T5 / T6 PYLON, COWL FOOT) on each plate's outside face with a paint marker, then peel off the templates.



One template tile: outline, holes with drill sizes, starter holes, overlap and registration marks.



Finished port plate, outside face.

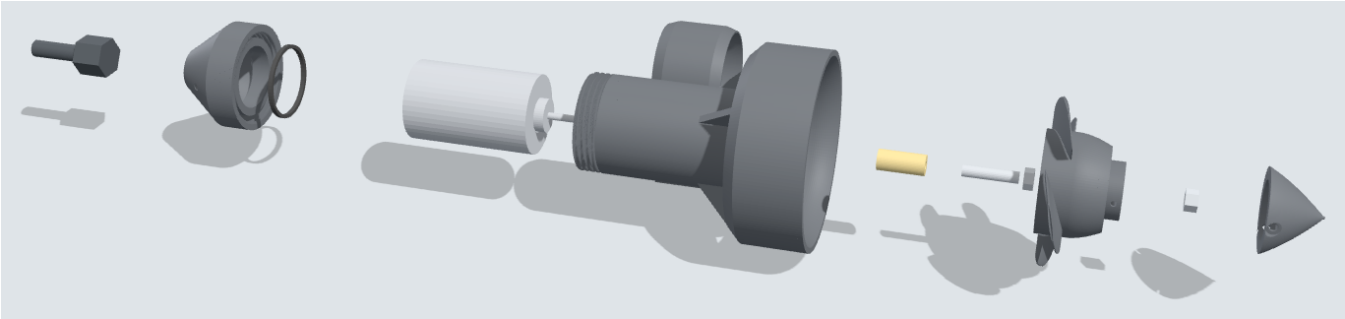
	Before you go on
[ ]	Two plates, mirror images, all holes drilled, windows cut, edges smooth, outside faces marked.

7 Build the thrusters (x6)

Parts for this section

101	Thruster body + Kort duct	x6	**105**	Prop spinner	x6
102	Tail cone with oil compensator	x6	**106**	Clamps each thruster body	x6
103	Right-hand prop	x3	**107**	Sets the coupler height	x1
104	Left-hand prop	x3			

Each thruster is a printed housing with the Kort duct built in, an oil-filled motor bay, a threaded tail cone with a built-in compensator, and a two-piece prop driven by a brass coupler. Build all six the same way; only the prop hand changes (3 right-hand, 3 left-hand: one of each per pair).



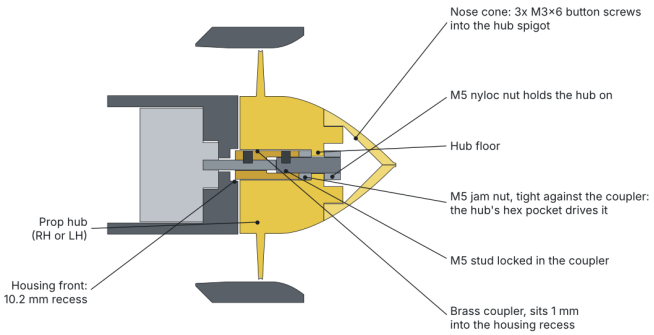
Exploded thruster, rear to front: motor lead gland, tail cone with its spigot O-ring, motor, housing (pylon clamp on it), brass coupler, M5 stud with jam nut, prop hub, M5 nyloc, nose cone.

7.1 Prepare the parts

1. Tap the tail cone (102) tip M12 x 1.5 and the two side ports M4. Thread the M12 gland in with a drop of sealant.
2. Ream the coupler's 3 mm end to 1/8" (3.17 mm) so it slides onto the motor shaft with a light push fit.

7.2 Motor, coupler and prop

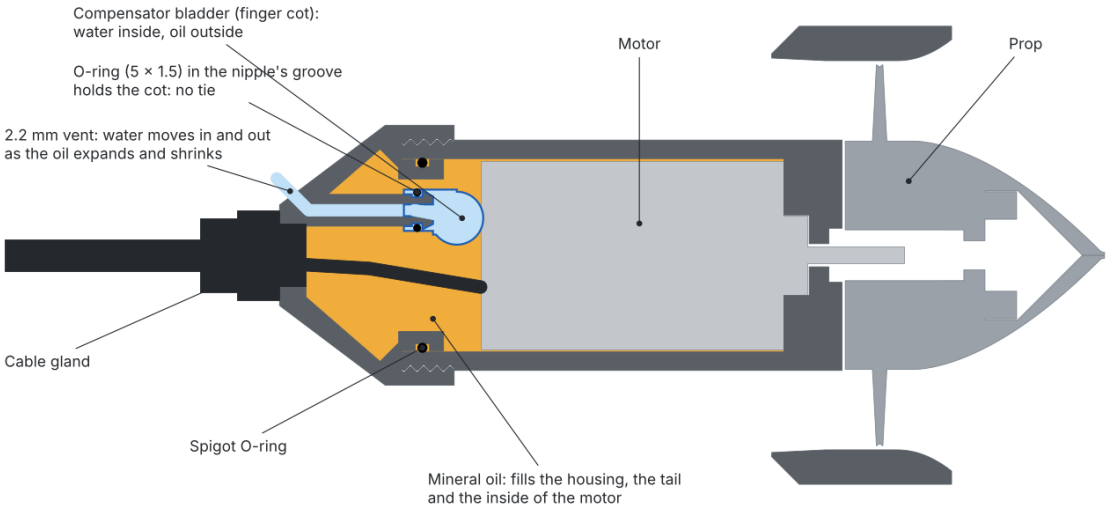
3. Lightly grease the motor's nose O-ring. Feed the motor lead out through the rear of the housing (101) and push the motor in from the back until the nose seats in the front bore.
4. Slide the bare coupler onto the shaft, 3 mm end down (no stud in it yet: the stud would hit the gauge's ceiling and set it too low). Stand the coupler gauge (107) flat on the housing's front face over it, push the coupler up until it touches the gauge's ceiling, and tighten the coupler's shaft-end set screw through a gauge window. Lift the gauge off. The coupler now sits 1 mm below the housing face, so it can never rub. No gauge? Set the coupler's top 19.0 mm above the housing face with a caliper's depth rod.
5. Thread the M5 x 20 set screw halfway into the coupler's 5 mm end and lock it with the coupler's own grub screw. Spin the M5 jam nut down onto it, flat side toward the coupler, and tighten it against the coupler.
6. Drop the prop hub (103 RH or 104 LH, marked on its back face) over the stud so the jam nut sits in the hub's hex pocket. Spin on the M5 nyloc with the 8 mm nut driver through the hub until it just seats; the hub must turn freely with no rub on the duct.
7. Slide the nose cone (105) over the hub and fit the 3 M3 x 8 button screws.



Section through the prop drive.

7.3 Tail cone, oil fill and compensator

8. Fit the compensator bladder: push a nitrile finger cot straight down over the nipple inside the tail cone (102) until it covers the groove (the nipple's tip is just below the rim). Roll a 5 x 1.5 mm O-ring down over the barb with a fingertip until it drops into the groove: it clamps the cot, no tie needed. Fold the rest of the cot down into the cone. Water enters it through the 2.2 mm vent; oil stays outside it.
9. Pass the motor lead through the tail cone gland, grease the 32 x 2 O-ring on the spigot and thread the tail cone on hand tight.
10. Stand the thruster nose down. Fill through one M4 port with a syringe of mineral oil while air leaves the other; rock it and spin the prop by hand to free bubbles. Top off, then seal both ports with M4 set screws and sealant.
11. Tighten the gland on the lead. Wait an hour and check for weeping at the thread and the ports.

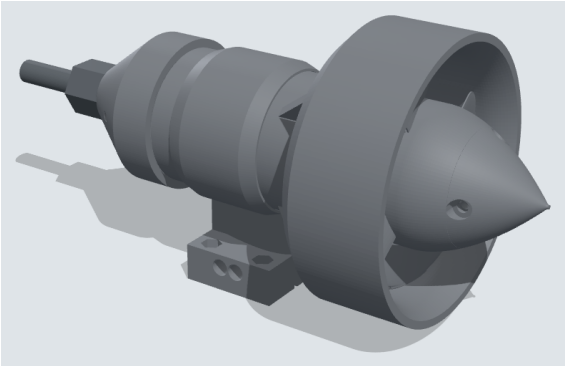


Section through the thruster axis, from the part geometry (motor, gland and bladder drawn schematically). Fill and bleed ports are on the sides, not in this cut.

Oil fill and compensator, section through the axis.

7.4 Pylon clamp

12. Slide the pylon clamp (106) over the housing body (101) and close it with the 2 M4 x 30 pinch bolts. Leave it loose: its final position is set on the ROV (section 14).



Finished thruster with its pylon clamp.

Label each thruster T1-T6 now and note its prop hand: T1/T2, T3/T4 and T5/T6 are each one right-hand + one left-hand. Reverse the left-hand units in software so both of a pair push the same way.

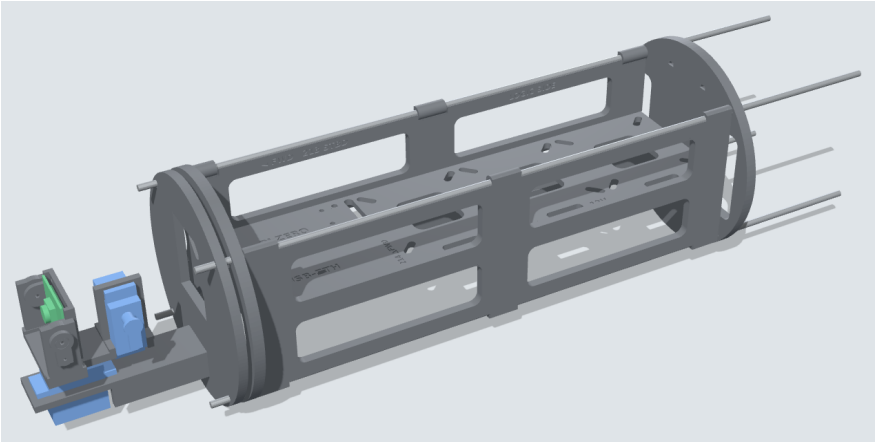
	Before you go on
[ ]	Six thrusters, oil full, no bubbles, no weeping after an hour, props spin freely, labelled T1-T6 with prop hand.

8 Build the electronics sled and camera pan-tilt

Parts for this section

207	Sled front bulkhead	x1	**216**	Pan-tilt base v2: on the sled rods, carries the pan servo	x1
212	Sled side rail, port (12 V side)	x1	**217**	Pan-tilt yoke v2: on the pan horn, tilt servo in a closed pocket	x1
213	Sled side rail, starboard (logic side)	x1	**218**	Pan-tilt camera cradle	x1
214	Sled deck: boards on both faces	x1	**219**	Tilt lever on the cradle axle	x1
215	Sled rear bulkhead: 12 V bus + LED drivers	x1			

The sled is a front bulkhead (207) and a new rear bulkhead (215) 234 mm apart on four M3 rods, with two side rails (212, 213) and a deck (214) between them. The deck carries the six IBT_2 motor drivers, three on each face, plus the Pi Zero with its Ethernet HAT and two small 5 V buck boards on top, and the Pico on its screw-terminal breakout underneath. The rear bulkhead carries the 12 V Wago bus and both LED drivers. 12 V runs down the port side, logic down the starboard side. The camera pan-tilt (216-219) sits on the same rods in front of 207, inside the dome. Section 9 covers the electronics: parts, layout, schematic, wiring and the bench test.



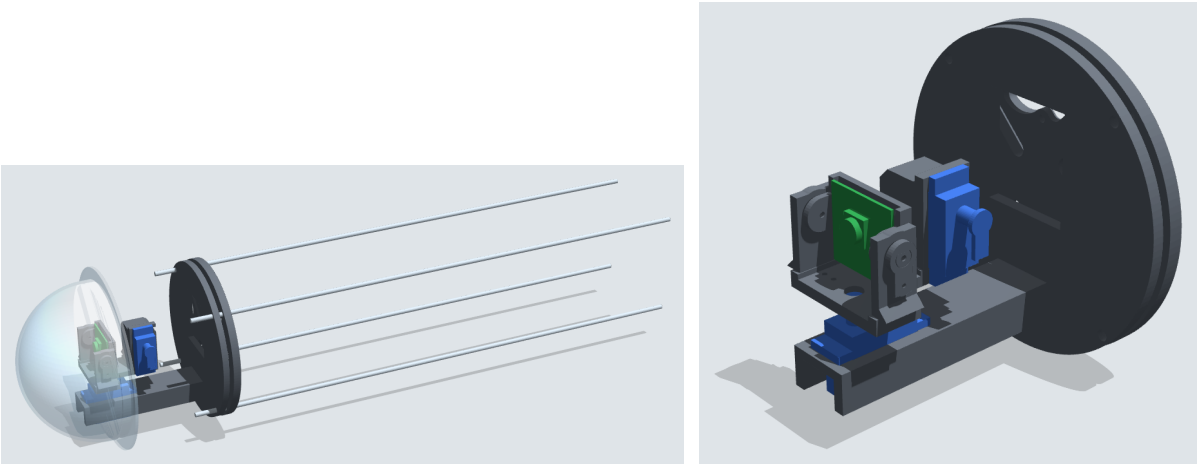
Sled with the pan-tilt on the front: rails, deck, rear bulkhead 215, front bulkhead 207 and the four rods.

8.1 Sled

1. Front bulkhead (207): its two rod holes sit on the horizontal axis, but the rods run at 45 deg. Turn it 45 deg and drill the two missing 3.2 mm holes on the 94 mm circle, so all four rods pass at 45 deg.
2. Strip the heatsinks off all six IBT_2 boards. The deck slots take any square hole pattern from 38 to 46 mm.
3. For each stacked pair, push four M3 x 8 male-female standoffs down through the deck (214) slots and thread an M3 x 8 female-female standoff on each from below. Screw the top board on with M3 x 6 screws, terminals to port. Flip the deck and fit the bottom board the same way. The deck is labelled T1-T3 on top, T4-T6 underneath.
4. Plug the camera ribbon into the Pi's CSI connector first: the HAT sits over it. Then fit the Pi Zero with its ETH/USB HUB HAT on top where it is labelled PI ZERO, on M2.5 x 8 standoffs (not 5 mm: the HAT's micro-USB link board hangs 6.5 mm below the Pi), header to starboard, so the HAT's RJ45 and USB ports face the hull centreline. The USB-ETH space beside it is now where the Ethernet plug turns aft. Bench-test both mini buck boards first: about 1 A load each (a USB load tester or a 5 ohm 5 W resistor) and check 4.9-5.2 V. Then slide clear heat-shrink over each and stick them to the top face beside the Pi, in the USB-ETH space, with foam tape: U9A in front, U9B behind it, IN ends to the front, about 3 mm in from the port rail. The Ethernet plug passes over them.
5. Underneath, mount the Pico breakout on four M3 x 10 standoffs, centred between the rails with its front edge 4 mm behind the deck's front edge, Pico USB to the front and pins 1-20 (GP0-GP15) on the starboard side. Its terminals face the rails: every wire reaches them through a rail window, so wire them before the deck goes into the rails, or leave the wires long and feed them through.
6. Slide the deck into the grooves of both rails: 212 (PORT, 12 V SIDE) and 213 (STBD, LOGIC SIDE), FWD arrows to the front. The rails' lower windows run right up to the front end, so the wires can reach the Pico breakout's front terminals.
7. Fit the two Wago 221-415 blocks and both LDD-700HW drivers into the pockets on the rear face of 215 (labelled +12V, GND, LDD 1, LDD 2; TOP up) and zip-tie them.
8. Cut the four M3 rods to 340 mm, file the ends square and take the sharp edge off. Thread them: nut, 215, nut, then through the rail bosses, then 207 and a nut each side. Set 234 mm between the bulkhead inner faces, with the rods centred: each rod sticks out 73.5 mm behind 215's rear face (and about 25 mm in front of 207, where the pan-tilt base goes in 8.2). Tighten the nuts on both bulkheads.
9. How the sled is held in the hull: the rod tips are locating pins. Each end of each rod drops about 6 mm into one of the four plain 4 mm holes in the end of a flange skirt (201 at the rear, 203 at the front, on the 94 mm circle at 45 deg), so the sled is trapped between the two flanges and can't turn. Leave those eight holes plain: no heat-set inserts.

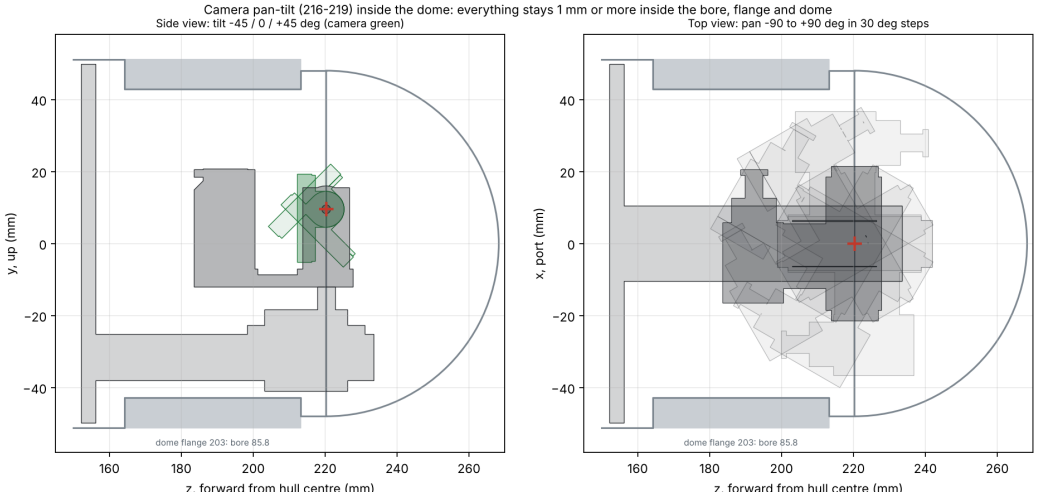
8.2 Camera pan-tilt

A 100 mm dome leaves little room: the pan servo hangs under the camera, and the tilt servo stands behind it in a closed pocket in the yoke and drives the cradle through a short wire pushrod. Both axes pass through the lens, which sits at the dome centre (the dome's sphere centre is 7.1 mm in front of its flange face), 9.5 mm above the hull axis. Checked in the model at every angle: pan to +/- 90 deg and tilt to +/- 50 deg stay at least 1 mm inside the hull bore, the dome flange bore (85.8 mm) and the dome (99.8 mm OD, 96 mm ID: inner radius 48 mm).



Left: pan-tilt in the dome (dome and flange see-through). Right: base 216 with the pan servo, yoke 217 with the tilt servo, cradle 218 with the camera.

9	Centre both servos (1500 us pulse) before fitting any horn. MG90S metal-gear servos are worth the extra dollar here.
10	Pan servo into base 216: body down through the shelf window, shaft up and at the front end, tabs on the shelf (labelled PAN SERVO). Two M2 servo screws through the tabs.
11	Screw the double-arm horn into the pocket under yoke 217 with four M2 x 6 self-tappers. Tilt servo: feed its lead down through the slot in the yoke base first, then slide the servo into the box-shaped pocket from the port side, bottom first and shaft to port, until both tabs sit against the pocket face. One M2 servo screw through the upper tab; the lower tab hangs free beside the base. The pocket walls take the pushrod load, the screw only stops the servo sliding out.
12	Fit the yoke on the pan servo's spline, square to the front, and fix it with the horn screw through the access hole.
13	Screw the Arducam to cradle 218 with four M2 x 6 on 2 mm standoffs, lens forward, ribbon down through the slot.
14	Hang the cradle in the yoke: its D-axle through the port arm, an M2 x 8 through the starboard arm into the cradle cheek. It must swing freely.
15	Push lever 219 onto the D-axle outside the port arm, pointing down, and hold it with an M2 x 6 and washer into the axle end.
16	Pushrod: 1 mm steel wire with a Z-bend at each end, 28 mm between bends, from the tilt horn's hole 8 mm below the shaft to the lever's hole 8 mm below the axle. With the servo centred, the camera should look straight ahead; adjust by re-bending.
17	Slide 216 onto the four rods in front of 207 with one M3 nut behind it and one in front (the rods need about 6 mm more thread). Run the ribbon and both servo leads back through the window in 216 and the windows in 207 to the Pi and Pico.
18	Set the firmware limits to pan +/- 90 deg and tilt +/- 45 deg.



Fit check against the hull: side view at three tilt angles, top view across the pan range. Grey bands: dome flange bore.

Before you go on

- [ ] Sled square on its rods; pan and tilt move through their full range by hand without touching anything; everything powers up on the bench from a 12 V supply.

9 Electronics

Everything electrical inside the hull: the parts and their reference numbers, how the boards sit on the deck, the schematic, the wiring diagram with every wire numbered, and the order to wire and test the sled on the bench. The cables from outside connect, and the sled goes into the hull, in section 16.

Two rules hold everywhere: **no Pico pin ever touches 12 V** (the IBT_2 logic VCC comes from the Pico's 3V3 OUT), and **12 V wiring runs down the port side, logic down the starboard side**. Check B+ / B- polarity with a meter before the battery ever goes on: the IBT_2 has no reverse-polarity protection, and a swapped lead kills all six boards.

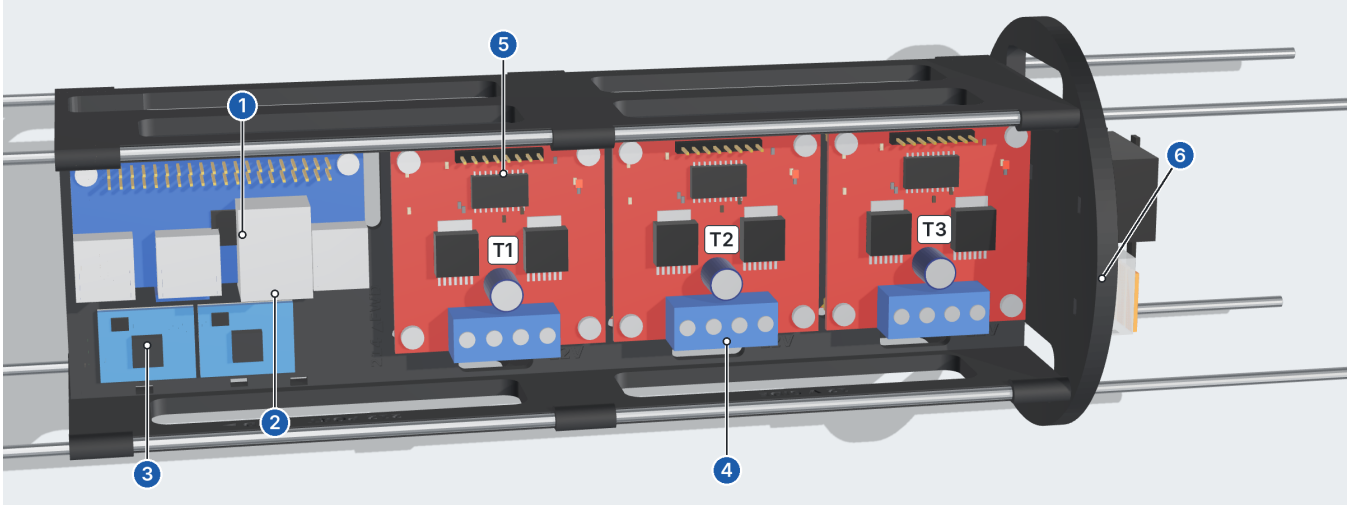
9.1 Parts and reference numbers

The reference numbers used on the schematic and in the wire list. Quantities and prices are in the shopping list (section 4); the firmware (Pico UF2, Pi bridge, topside program) is in the Firmware folder.

Ref	Part	Where
U1-U6	IBT_2 (BTS7960) H-bridge driver	One per thruster, heatsinks removed. Logic VCC from the Pico 3V3
U9A, U9B	Mini MP1584EN 5 V 1.8 A buck boards (5-30 V in, fixed 5 V out)	Top face, port side beside the Pi: U9A for the Pi, HAT, camera and Pico; U9B for the servos only
U9 mounting	Clear 20-25 mm heat-shrink + double-sided foam tape	One sleeve over each mini buck, taped to the deck
U10	Raspberry Pi Pico, headers fitted, on a 57 x 63 mm screw-terminal breakout board	Underneath at the front, centred between the rails, USB to the front
U11	Raspberry Pi Zero 2W, header fitted	Top face, header to starboard
U12	Waveshare ETH/USB HUB HAT (RTL8152B: 1 x RJ45 + 3 x USB-A) with its micro-USB link board	On the Pi Zero, 11 mm above it; tether to topside
Ethernet	Cat6 patch lead, 0.3 m, one end 90-degree (left or right angle), + RJ45 inline coupler	HAT RJ45 to the tether: the plug has to turn aft within 25 mm
U7-U8	Mean Well LDD-700HW LED driver	Rear face of 215 (shopping list: Lights)
J1	Arducam for Pi Zero + 150-200 mm Zero ribbon	On the pan-tilt cradle (shopping list: Camera)
SV1-SV2	MG90S micro servo	Pan and tilt (shopping list: Camera)
F1	Inline ATO/ATC fuse holder, 12 AWG, + 30 A fuse	Battery red, into the +12V Wago
F2	Inline mini-blade holder + 3 A fuse	Aux branch: LED drivers and both 5 V bucks
C1	1000 uF 25 V low-ESR electrolytic	Across the 5 V bucks' 12 V feed, in the port trough
C2	470 uF 10 V electrolytic	Across +5V / GND at the servo plugs
D1	1N5817 Schottky diode	U9A 5 V to Pico VSYS, band toward the Pico
R1, R2	10 k resistor, 1/4 W	ARM and DIM pull-downs
R3, R4	33 k and 10 k resistors, 1/4 W	Battery-voltage divider to GP26
Bus	Wago 221-415 lever connectors	+12V and GND, rear face of 215 (shopping list: Electronics sled)
Wire	14 AWG silicone wire, red + black	Bus to the three top boards
Wire	16 AWG silicone wire, red + black	Jumpers to the bottom boards
Wire	20 AWG silicone wire, red + black	Aux branch: F2 to LED drivers and both 5 V bucks
Wire	Dupont female housings + 22 AWG hook-up wire, crimp splices	RPWM / LPWM wires and the ARM / VCC / GND chain (Dupont at the IBT_2 headers, bare ends into the Pico terminals), UART, DIM, 5 V
Misc	Zip ties 2.5-3.6 mm, heat-shrink assortment	Boards, harness ties, every splice

9.2 How it fits on the deck

The deck (214) carries boards on both faces. Each top board sits directly over its bottom partner on the same four standoffs, so one 12 V feed serves a pair: T1 over T4, T2 over T5, T3 over T6. On both faces every screw terminal faces port and every logic header faces starboard. The deck is engraved with each board's name.



Top face, seen from above, front of the ROV to the left. T1-T3 are the front port, front starboard and rear port thrusters.

1 Pi Zero 2W + ETH/USB HUB HAT on M2.5 x 8 standoffs, header to starboard

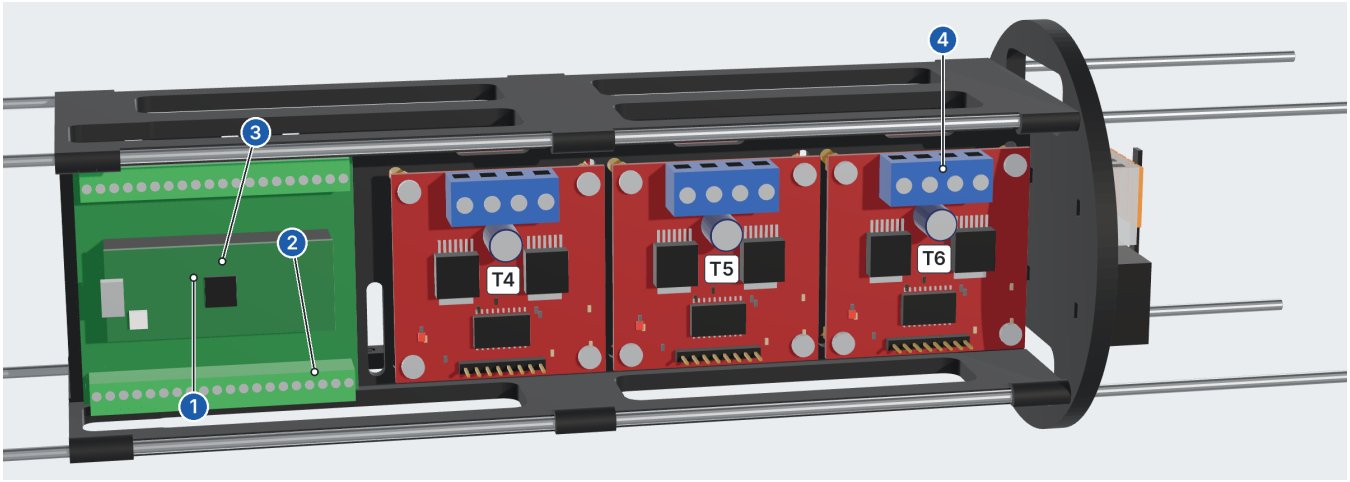
2 HAT's RJ45 and USB ports face the hull centreline: the Cat6 plugs in here with a 90-degree plug

3 Two mini 5 V buck boards, heat-shrunk, on foam tape: U9A (front) for the Pi and Pico, U9B for the servos

4 Screw terminals B+ B- M+ M- face port

5 8-pin logic header faces starboard

6 Rear bulkhead 215: 12 V Wago bus and both LED drivers on its rear face



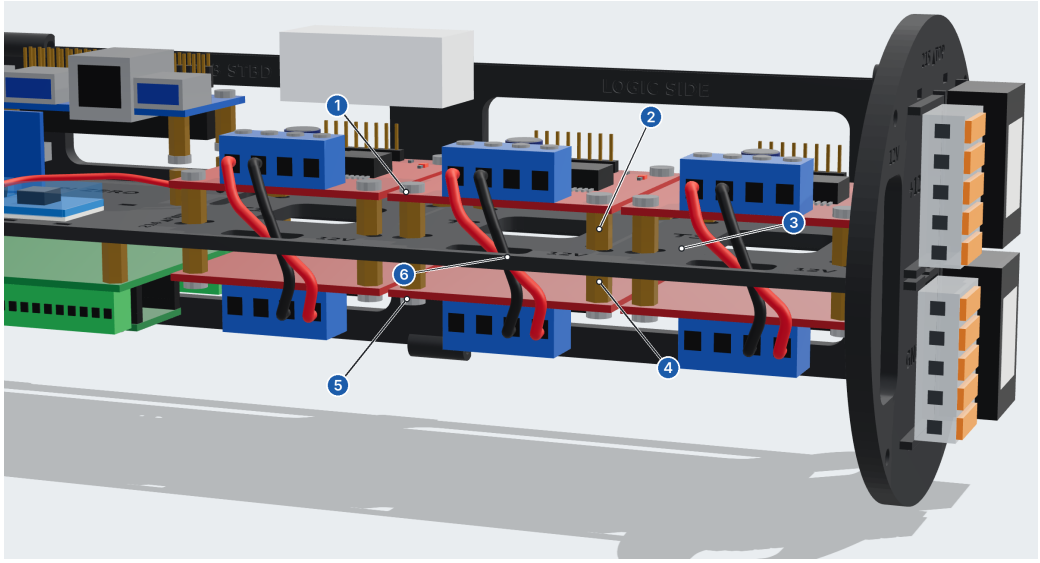
Bottom face, seen from below, front to the left. T4-T6 are the rear starboard, vertical port and vertical starboard thrusters.

1 Pico on its screw-terminal breakout, 10 mm standoffs, centred between the rails: USB to the front, pins 1-20 on the starboard terminals

2 Terminals face the rails: every wire comes in through the rail windows (front ends opened up)

3 Pico plugged into the breakout from below

4 Terminals face port on this face too



One stacked pair from the port side: the standoffs pass through the deck slots, and the 16 AWG jumpers carry 12 V to the lower board.

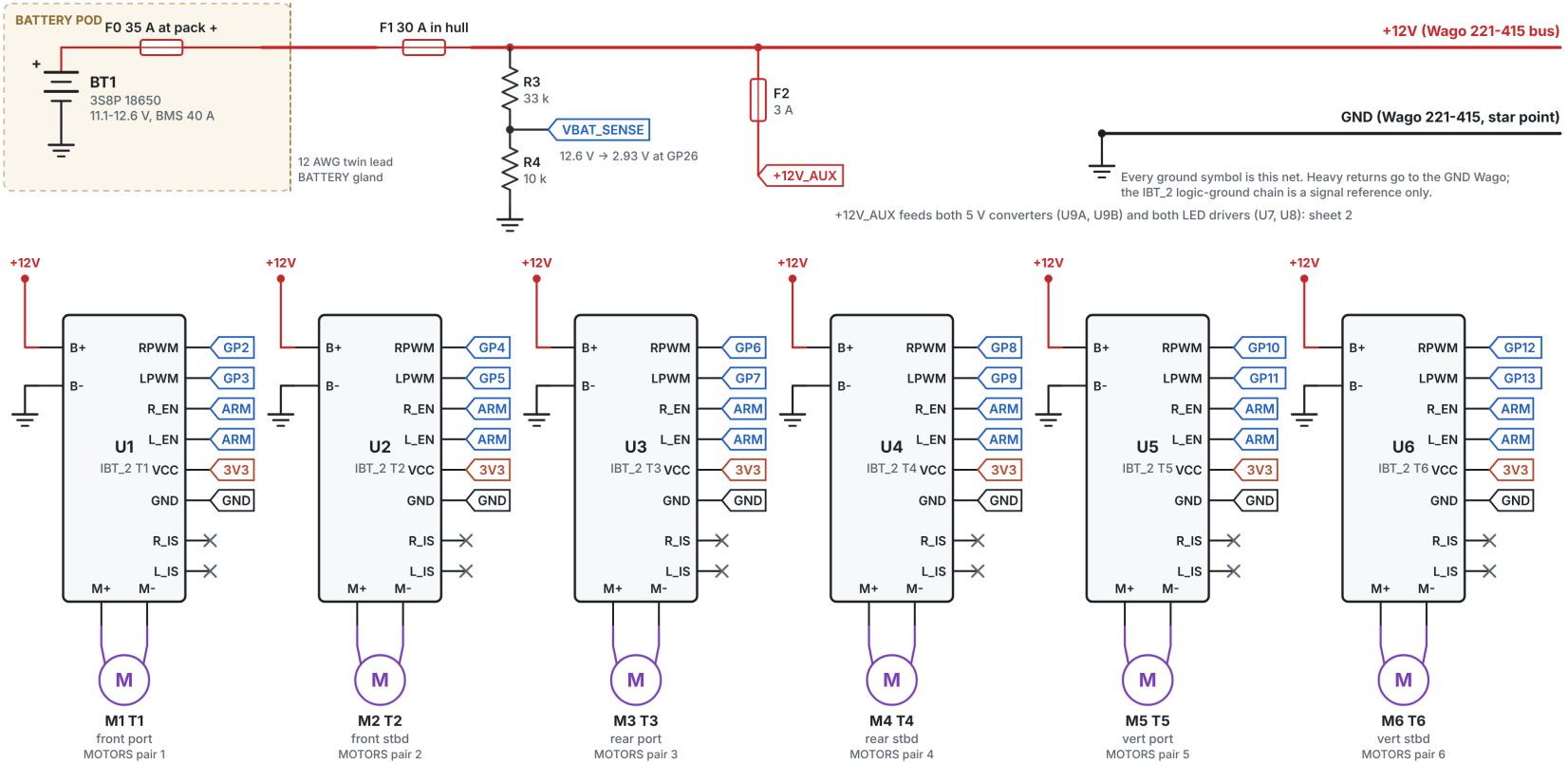
1 Top board (T2) on M3 x 6 screws
2 M3 x 8 male-female standoff through the deck slot
3 Deck 214, 3 mm
4 M3 x 8 female-female standoff below
5 Bottom board (T5): same standoffs, terminals to port
6 16 AWG B+ / B- jumpers through the deck's port slot

9.3 Schematic

Two sheets. Sheet 1 is the power path and the six motor drivers; sheet 2 is the 5 V supply, the Pico, the Pi, the camera, the servos and the LED drivers. Nets with the same name are connected (net labels instead of long wires). The pages are turned sideways: turn the guide clockwise to read them. Full-size copies are in Images/Electronics in the package.

Schematic 1 of 2: power and motor drivers

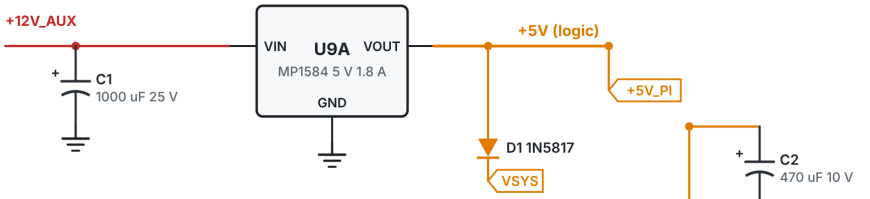
Battery, fuses, 12 V bus, battery sense, six IBT_2 drivers (U1-U6) and the six 1200 GPH motors (M1-M6)



IBT_2 boards: VCC chained from the Pico's 3V3 OUT (never 5 V or 12 V); R_EN + L_EN bridged on each board and chained to ARM (GP15, R1 10 k to GND, sheet 2); R_IS, L_IS not connected.
Each stacked pair (T1/T4, T2/T5, T3/T6) takes one 14 AWG feed from the bus to the top board, jumped to the bottom board with 16 AWG. F1 is the only path into +12V.
M+ / M-: the motor pair in the 12-core MOTORS cable. Spin direction is set in firmware (THRUSTER_DIR), not by swapping wires.

Schematic 2 of 2: 5 V supply, control, camera, servos and lights

U9A / U9B 5 V converters, U10 Pico, U11 Pi Zero 2W, U12 ETH/USB HUB HAT, J1 camera, SV1-SV2 servos, U7-U8 LED drivers and the two lights

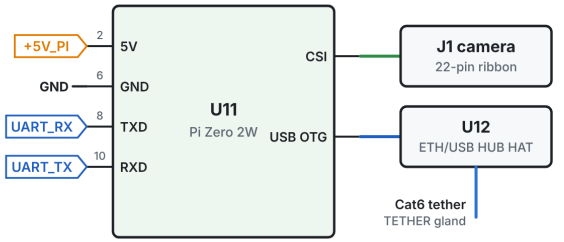


D1 lets the Pico run from USB on the bench without back-feeding the converter.

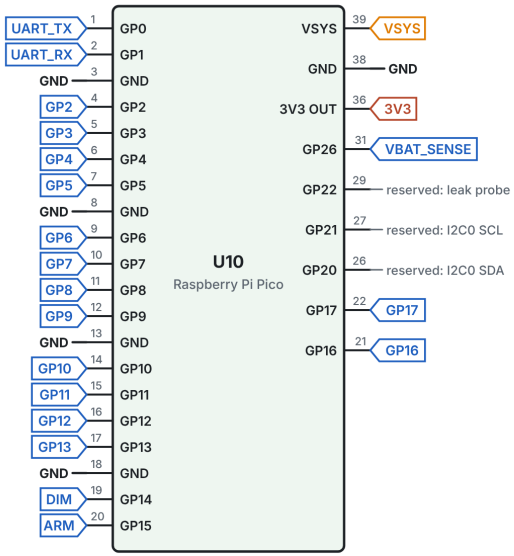


U9A feeds the Pi, HAT, camera and Pico; U9B feeds only the servos, so a stalled servo can't pull the Pi's 5 V down. Never join the two 5 V outputs; the grounds are common.

Signal 3.3 V straight from the Pico (no level shifter); power from +5V_SERVO (U9B) only, C2 at the servo plugs.

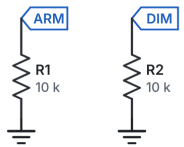


Pi TXD (GPIO14, pin 8) → Pico GP1 RX; Pi RXD (GPIO15, pin 10) ← Pico GP0 TX

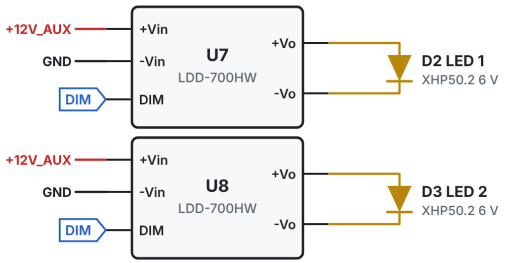


GP2-GP13: RPWM / LPWM of U1-U6 (sheet 1). GND pin 13: the IBT_2 logic-ground chain; pin 3: the Pi; pin 18: R1 and R2.

Nets with the same name are connected. 3V3 = Pico 3V3 OUT (pin 36) to VCC of all six IBT_2. No Pico pin ever touches 12 V.



R1 holds all six bridges off until firmware raises ARM.
R2 holds the lights off at power-up (an open DIM is full on).



Each driver's +Vo / -Vo is its own pair in the 4-core LIGHTS cable. Never join -Vo to GND, -Vin or the other driver's -Vo.

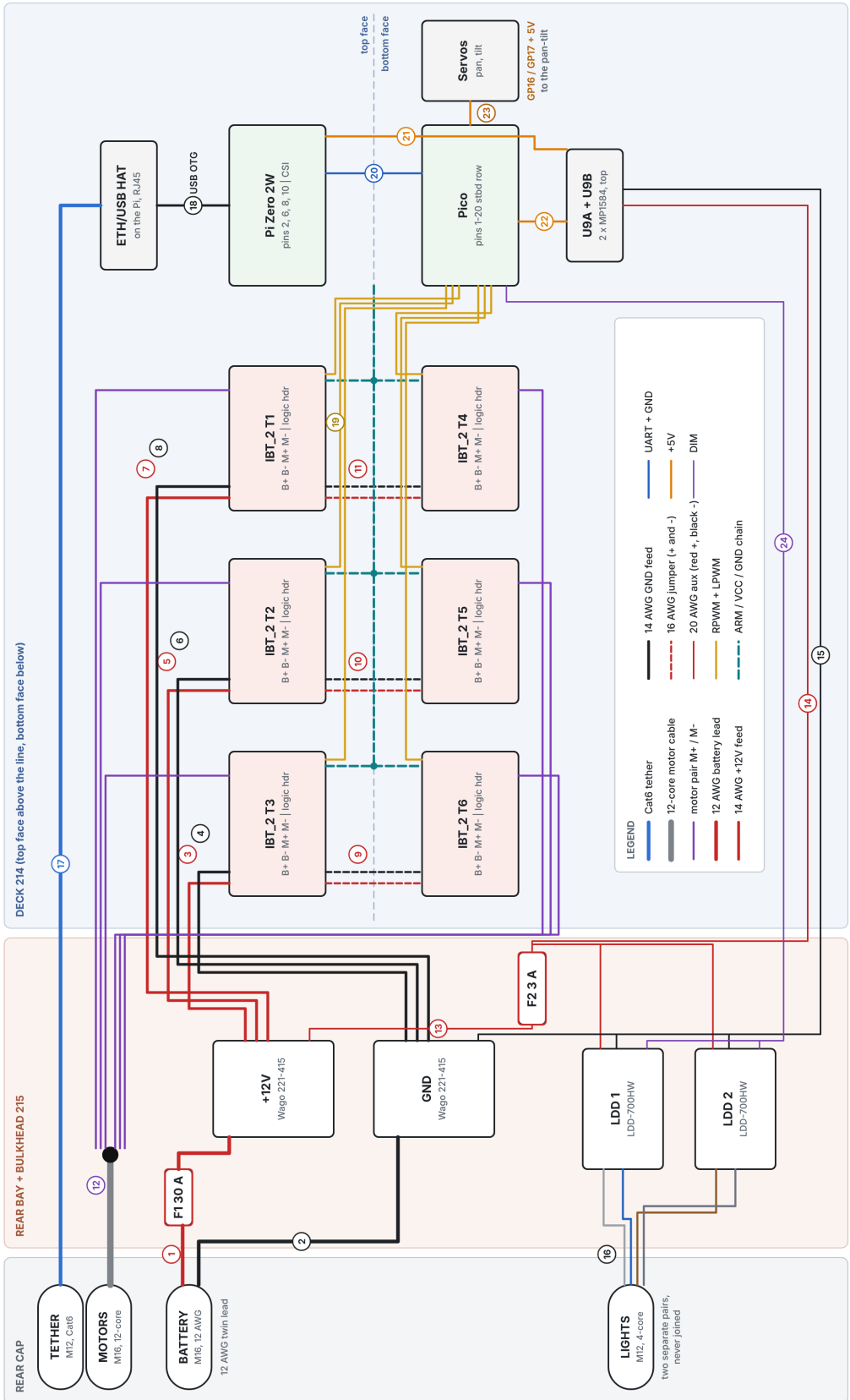
9.4 Wiring diagram and wire list

The wiring diagram shows each harness from the rear cap through the rear bulkhead to the boards, with each top board drawn above its bottom partner. The numbers on the runs match the wire list below.

#	From	To	Wire
1	BATTERY gland	F1 -> +12V Wago, slot 1	12 AWG red (twin lead)
2	BATTERY gland	GND Wago, slot 1	12 AWG black (twin lead)
3	+12V Wago	T3 B+ (top)	14 AWG red
4	GND Wago	T3 B- (top)	14 AWG black
5	+12V Wago	T2 B+ (top)	14 AWG red
6	GND Wago	T2 B- (top)	14 AWG black
7	+12V Wago	T1 B+ (top)	14 AWG red
8	GND Wago	T1 B- (top)	14 AWG black
9	T3 B+ / B-	T6 B+ / B- (below)	16 AWG red + black
10	T2 B+ / B-	T5 B+ / B- (below)	16 AWG red + black
11	T1 B+ / B-	T4 B+ / B- (below)	16 AWG red + black
12	MOTORS gland	M+ / M- of T1-T6	12-core, one pair per board
13	+12V Wago, slot 5	F2 (3 A)	20 AWG red
14	F2	LDD 1 +Vin, LDD 2 +Vin, U9A + U9B IN+	20 AWG red, spliced; C1 across the converter feed
15	GND Wago, slot 5	LDD 1 -Vin, LDD 2 -Vin, U9A + U9B IN-	20 AWG black, spliced
16	LIGHTS gland	LDD 1 +Vo/-Vo, LDD 2 +Vo/-Vo	4-core: two separate pairs
17	TETHER gland	HAT RJ45 (on the Pi)	Cat6, inline coupler + 90-degree patch lead
18	HAT hub input	Pi Zero USB OTG	micro-USB link board (comes with the HAT)
19	IBT_2 logic headers	Pico breakout terminals (see the pin table)	RPWM + LPWM: own wire each (12); ARM, VCC, GND: daisy-chained board to board, one wire each to the Pico
20	Pi pins 6, 8, 10 (HAT stacking pins)	Pico terminals 3, 2, 1	soldered at the Pi: GND, TXD->GP1, RXD<-GP0
21	U9A OUT+	Pi pin 2 (5 V, HAT stacking pin)	via the 5 V splice behind the Pi
22	5 V splice; U9A OUT-	D1 -> Pico VSYS (39); Pico GND (38)	red + 1N5817; black
23	Pico GP16 / GP17; U9B OUT+ / OUT-	Pan / tilt servo plugs	servo leads; C2 470 uF at the plugs
24	Pico GP14 (pin 19)	LDD 1 DIM, LDD 2 DIM	spliced; R2 10 k to GND at the terminal

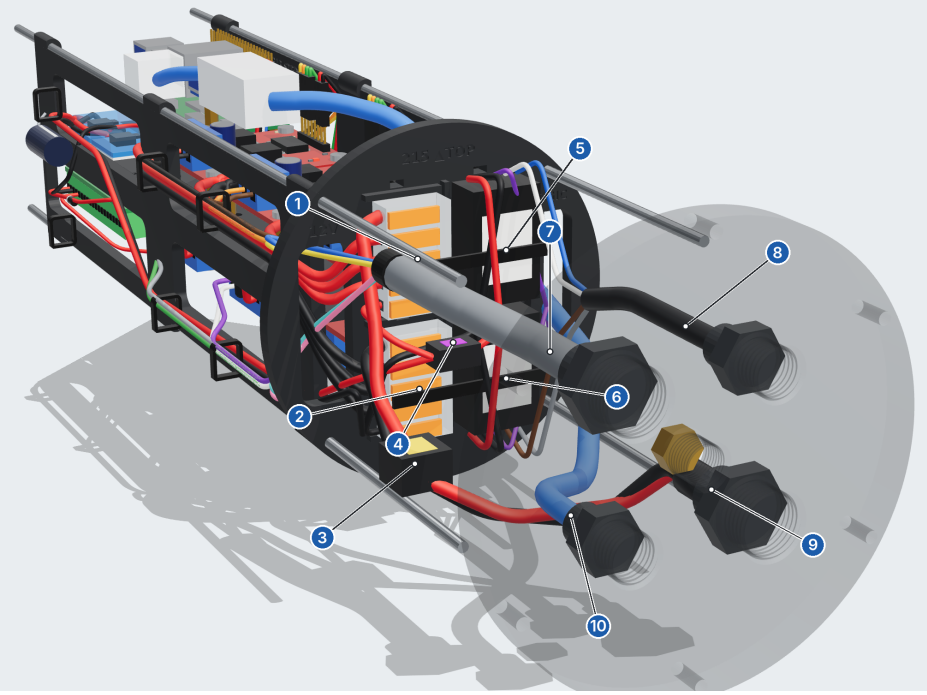
Wiring diagram: every wire, harness by harness

Glands on the left, the rear bulkhead (215) in the middle, the deck (214) on the right with each top board drawn over its bottom partner. Numbers = wire list.



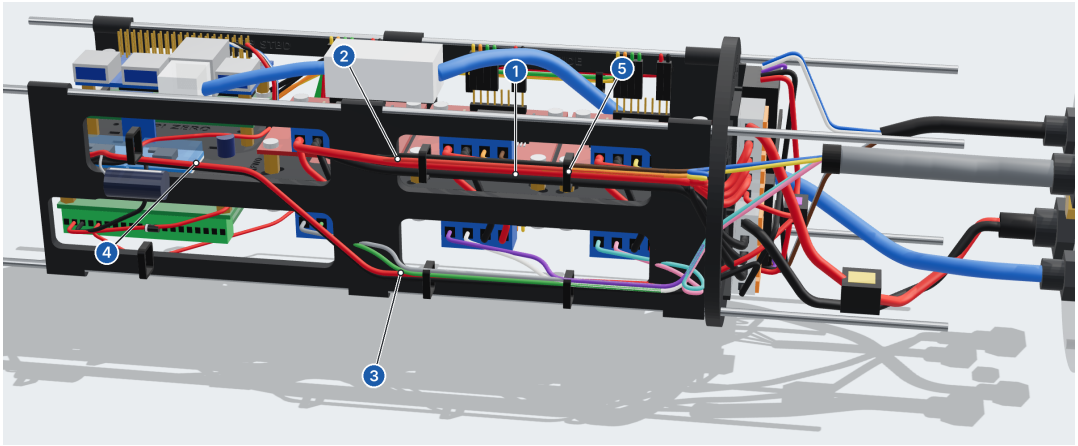
9.5 How the wires run

12 V feeds and the motor cores run in the curved gap between the port rail (212) and the hull wall and enter through the rail windows; the logic harness does the same on the starboard side (213). Tie each harness to the rail at the window bars. The cables from the rear cap pass the rear bulkhead (215) through its port window (motors, battery feeds) and starboard window (tether, DIM).



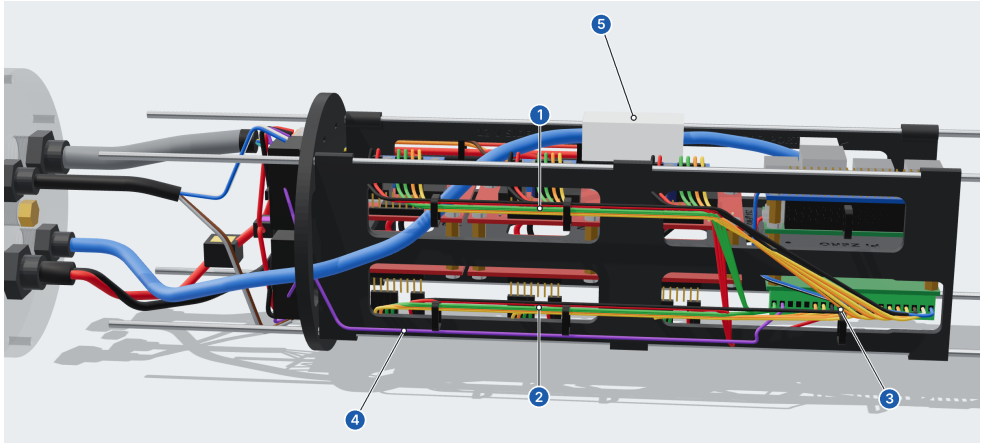
Rear bay, seen from aft and above with the rear cap see-through: the bus, both fuses and the LED drivers on 215, and the four glands.

1 +12V Wago 221-415: F1 in, three pair feeds, aux out
2 GND Wago 221-415: battery -, three pair returns, aux -
3 F1 30 A inline fuse holder
4 F2 3 A: LED drivers and the 5 V converter
5 LDD 1 (LDD-700HW)
6 LDD 2 (LDD-700HW)
7 MOTORS gland: 12-core, six pairs
8 LIGHTS gland: 4-core, two separate pairs
9 BATTERY gland: 12 AWG twin lead
10 TETHER gland: Cat6



Port side: the 12 V harness. Power feeds above, the bottom-board motor cores below, and the feed up to the two 5 V bucks.

1 14 AWG +12 V / GND feed to each top board, outside the port rail
2 Motor cores M+ / M- through the upper rail window (T1-T3)
3 Motor cores to T4-T6 through the lower window
4 12 V feed via F2 up to both mini 5 V bucks, C1 across it in the trough
5 Harness tied to the rail



Starboard side: the logic harness from every header to the Pico, the DIM line aft, and the tether over the boards.

1 Dupont at each IBT_2 header (RPWM, LPWM, R_EN + L_EN, VCC, GND), bare ends into the Pico terminals

2 Bundles from T4-T6 through the lower window

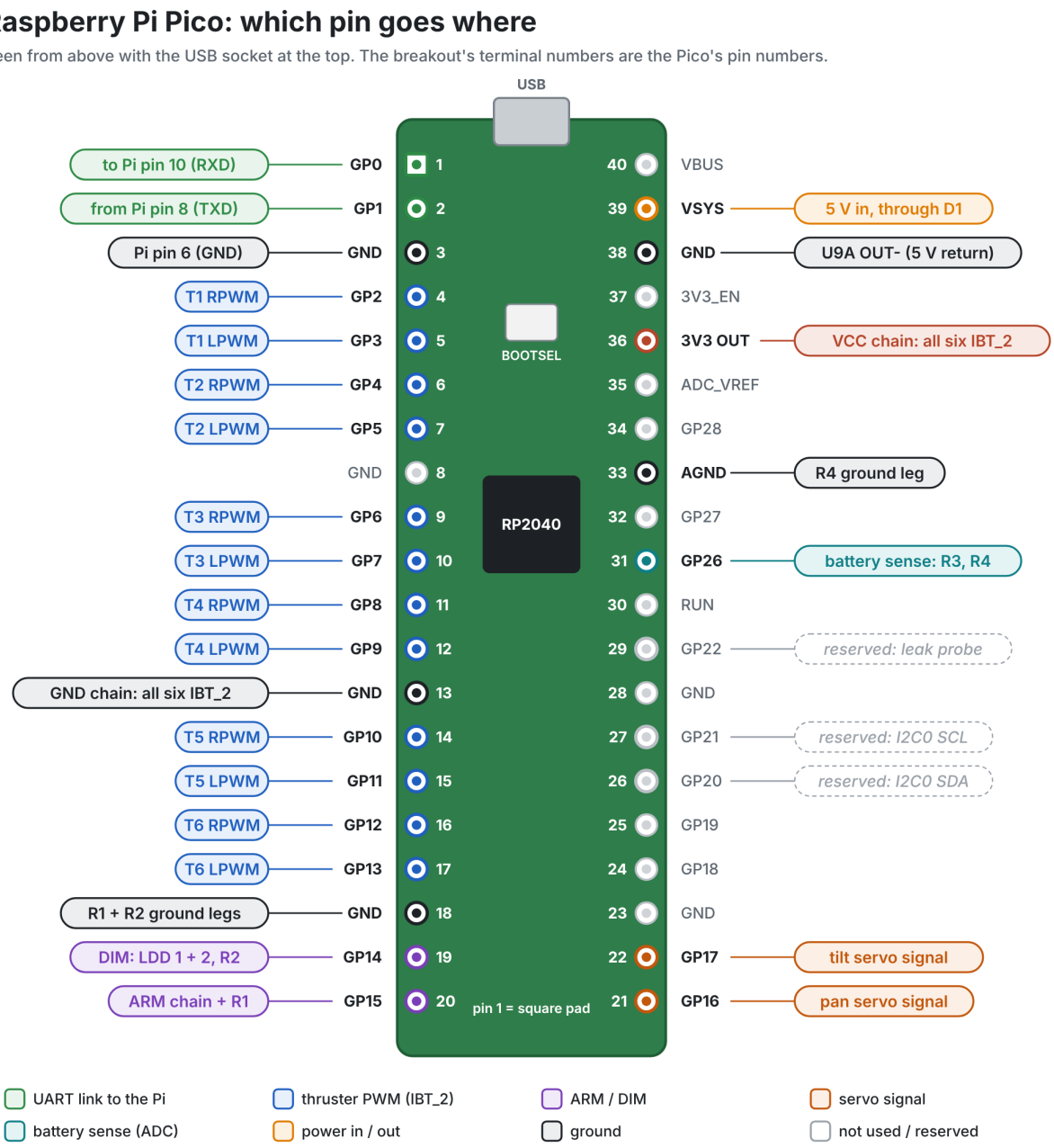
3 Into the Pico breakout terminals through the lower rail window: one PWM pair per thruster, ARM on GP15

4 DIM (GP14) aft to both LDDs

5 Cat6 over the boards, inline coupler, then a short 90-degree patch lead to the HAT

9.6 Pico and Pi Zero pins

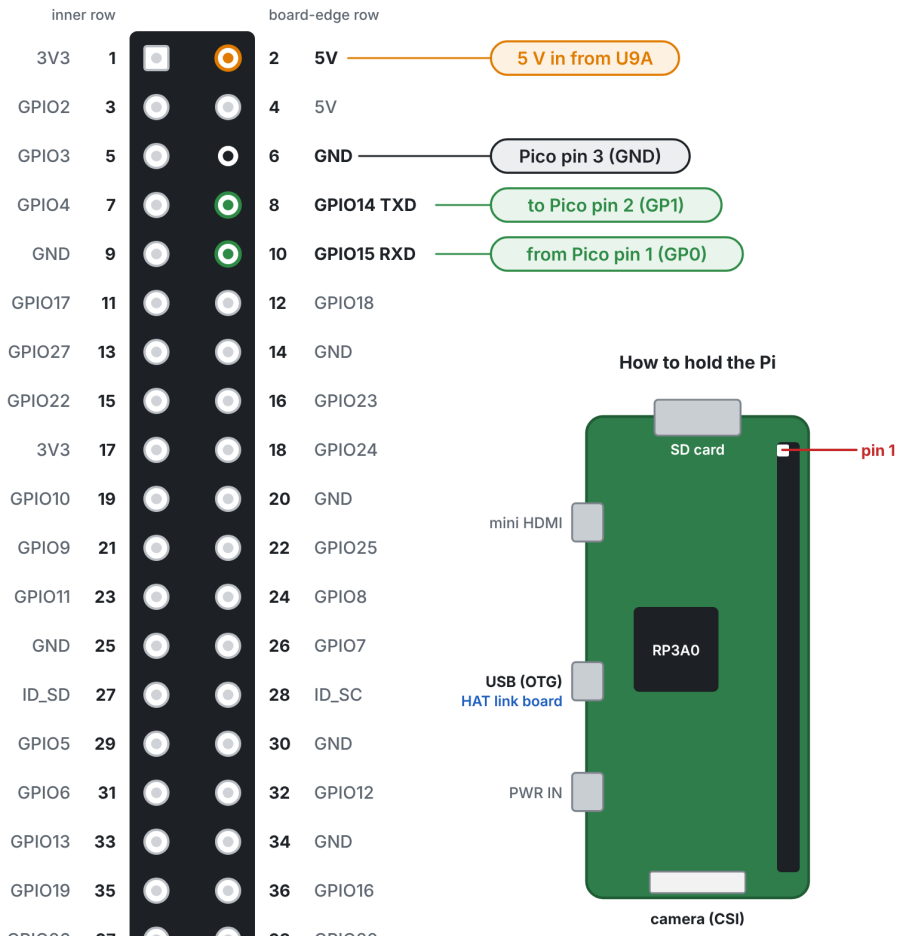
Which pin goes where, board by board, then every connection on its own line. The Pico breakout's terminal numbers are the Pico's pin numbers. On the sled the Pico's USB faces forward with pins 1-20 on the starboard side; the Pi's header runs fore and aft.



Pico pin map, seen from above with the USB socket at the top.

Pi Zero 2W: which pin goes where

Header seen from above, board turned so the header runs top to bottom (SD card at the top). The HAT's stacking pins carry the same numbers.



Only four header pins are used. Pins 2, 6, 8 and 10 are soldered at the HAT's stacking pins, 2-3 mm above the HAT.
The camera ribbon goes in the CSI socket at the far end from the SD card; the HAT's link board plugs into the middle micro-USB (OTG).

UART link to the Pico · 5 V in · ground · not used

Pi Zero 2W header, seen from above with the SD card at the top.

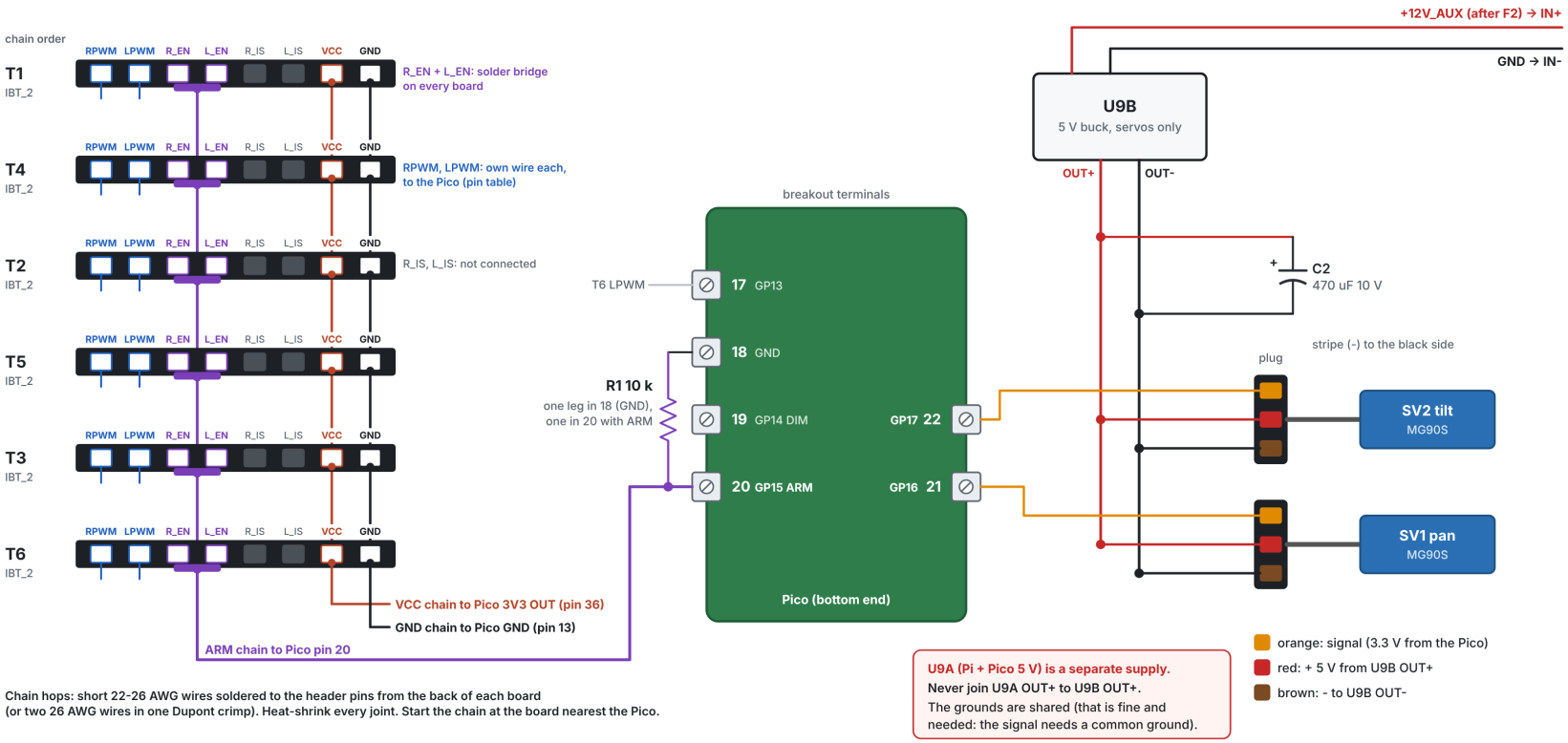
Pin connections, one line per connection. Wire # is the wire list number (9.4); a dash is a resistor soldered at the terminal.

Board	Pin	Pin name	Connects to	Wire #
Pico	1	GP0 (UART TX)	Pi Zero pin 10 (RXD)	20
Pico	2	GP1 (UART RX)	Pi Zero pin 8 (TXD)	20
Pico	3	GND	Pi Zero pin 6 (GND)	20
Pico	4	GP2 (PWM)	T1 header RPWM	19
Pico	5	GP3 (PWM)	T1 header LPWM	19
Pico	6	GP4 (PWM)	T2 header RPWM	19
Pico	7	GP5 (PWM)	T2 header LPWM	19
Pico	9	GP6 (PWM)	T3 header RPWM	19
Pico	10	GP7 (PWM)	T3 header LPWM	19
Pico	11	GP8 (PWM)	T4 header RPWM	19
Pico	12	GP9 (PWM)	T4 header LPWM	19
Pico	13	GND	GND chain: logic GND of all six IBT_2 headers, daisy-chained T1-T4-T2-T5-T3-T6	19
Pico	14	GP10 (PWM)	T5 header RPWM	19
Pico	15	GP11 (PWM)	T5 header LPWM	19
Pico	16	GP12 (PWM)	T6 header RPWM	19
Pico	17	GP13 (PWM)	T6 header LPWM	19
Pico	18	GND	R1 10 k, ground leg (other leg on pin 20)	-
Pico	18	GND	R2 10 k, ground leg (other leg on pin 19)	-
Pico	19	GP14 (DIM)	LDD 1 DIM	24
Pico	19	GP14 (DIM)	LDD 2 DIM	24
Pico	19	GP14 (DIM)	R2 10 k (other leg on pin 18)	-
Pico	20	GP15 (ARM)	ARM chain: R_EN + L_EN of all six IBT_2 (bridged on each header), daisy-chained	19
Pico	20	GP15 (ARM)	R1 10 k (other leg on pin 18)	-
Pico	21	GP16 (PWM)	Pan servo signal (orange lead)	23
Pico	22	GP17 (PWM)	Tilt servo signal (orange lead)	23
Pico	31	GP26 (ADC0)	R3 33 k, other leg to the +12V Wago (hull side of F1)	-
Pico	31	GP26 (ADC0)	R4 10 k (other leg on pin 33)	-
Pico	33	AGND	R4 10 k, ground leg	-
Pico	36	3V3 OUT	VCC chain: VCC of all six IBT_2 headers, daisy-chained	19
Pico	38	GND	U9A OUT- (5 V return)	22
Pico	39	VSYS	D1 1N5817, band end (D1's other end to the 5 V splice from U9A OUT+)	22
Pi Zero	2	5V	5 V splice from U9A OUT+ (soldered to the HAT's stacking pin)	21
Pi Zero	6	GND	Pico pin 3 (GND)	20
Pi Zero	8	GPIO14 (TXD)	Pico pin 2 (GP1, UART RX)	20
Pi Zero	10	GPIO15 (RXD)	Pico pin 1 (GP0, UART TX)	20
Pi Zero	CSI	camera connector	Camera ribbon, 22-pin (contacts toward the board)	-
Pi Zero	USB	USB OTG (middle port)	ETH/USB HAT micro-USB link board	18
ETH/USB HAT	RJ45	Ethernet	Cat6 tether via the inline coupler and 90-degree patch lead	17

Each thruster owns one PWM slice at 20 kHz (channel A = RPWM, B = LPWM). The lights dim from GP14 at 500 Hz; the servos run at 50 Hz on slice 0. GP26 reads the battery through the 33 k / 10 k divider (12.6 V reads 2.93 V). GP20, GP21 and GP22 are kept free for a depth sensor, an IMU and a leak probe.

ARM chain and servo wiring

Left: ARM, VCC and GND daisy-chained across the six IBT_2, one wire each to the Pico, R1 at terminal 20. Right: the servos run from U9B only; signals from 21 and 22.



Chain hops: short 22-26 AWG wires soldered to the header pins from the back of each board (or two 26 AWG wires in one Dupont crimp). Heat-shrink every joint. Start the chain at the board nearest the Pico.

Bench checks before plugging the servos in: U9B OUT+ to OUT- 4.9-5.2 V. Terminal 20 to GND: 0 V disarmed, about 3.3 V armed (1-2 V disarmed = an ARM joint not reaching R1).

Wire numbers: the logic chain and the RPWM / LPWM wires are wire 19; servo power and signals are wire 23.

9.7 Wire and test the sled

1. Flash the Pico before wiring: hold BOOTSEL, plug it into your computer's USB and copy Firmware/build/pico_rov.ino.uf2 onto it. Set up the Pi Zero with Firmware/pi_bridge/SETUP.md (static address 192.168.2.2 on the tether, video server, bridge service).
2. Bus (wires 1-2): fit the F1 holder on the battery lead's red core and push it into the top slot of the +12V Wago; the black core goes into the top slot of the GND Wago. Leave the 30 A fuse out for now.
3. Pair feeds (3-8): 14 AWG red from +12V slots 2-4 and black from GND slots 2-4 to B+ and B- of T3, T2 and T1. Run them through 215's port window, along the outside of the port rail and in through the upper rail windows. Tie them to the rail.
4. Jumpers (9-11): 16 AWG red and black from each top board's B+ / B- down through the deck's port slot to the same terminals on the board below. Tighten every screw terminal and tug-test each wire.
5. Aux (13-15): 20 AWG red from the +12V Wago's bottom slot through F2 (3 A); splice it to LDD 1 +Vin, LDD 2 +Vin and the IN+ of both mini bucks. 20 AWG black from the GND Wago's bottom slot, spliced the same way to both -Vin and both bucks' IN-. The buck feed runs forward in the port trough, up through the upper rail window, and splits to the two IN ends. Solder C1 across that feed in the trough, stripe (-) to GND. Heat-shrink every splice.
6. 5 V (21-22): U9A OUT+ to a splice just above the deck slot behind the Pi. From there: Pi pin 2 (soldered to the HAT's stacking pin), and down through the slot and the port lower window through D1 (band toward the Pico) to the VSYS terminal (pin 39). U9A OUT- to the Pico GND terminal (pin 38). U9B's output feeds only the servos (next steps). Never join the two 5 V outputs; the grounds are common.
7. Logic (19): first bridge R_EN to L_EN on every IBT_2 with a blob of solder across the two header pins (pins 3 and 4). Then daisy-chain three wires across the six boards in the order T1, T4, T2, T5, T3, T6 (each stacked pair next to each other): ARM (the bridged EN pins), VCC and GND, each a short 22-26 AWG hop from one board's header pin to the next, soldered to the pins from the back of the board (or two 26 AWG wires in one Dupont crimp). From the board nearest the Pico, one wire of each runs along the starboard side, outside rail 213, through the lower rail window to the Pico: ARM to pin 20, VCC to 3V3 OUT (pin 36, crossing under the breakout to the port side) and GND to pin 13. RPWM and LPWM stay separate: one wire each from every board to its own terminal in the 9.6 table, Dupont at the IBT_2 and bare ends into the terminals. R_IS and L_IS stay empty. The ARM chain drawing before 9.7 shows the chain.
8. Pull-downs and sense (ARM and R1: the ARM chain drawing before 9.7): R1 10 k from ARM (pin 20) to GND (pin 18) and R2 10 k from DIM (pin 19) to the same GND terminal (pin 18), soldered at the Pico end. Battery sense: R3 33 k from the +12V Wago (hull side of F1) to GP26 (pin 31), R4 10 k from GP26 to AGND (pin 33).
9. Pi to Pico (20): Pi pin 8 (TXD) to Pico pin 2 (GP1), Pi pin 10 (RXD) to Pico pin 1 (GP0), Pi pin 6 to Pico pin 3. With the HAT on, solder these three (and the 5 V on pin 2) to the HAT's stacking pins 2-3 mm above the HAT and heat-shrink them; a Dupont housing there would stand too tall for the hull. The wires drop through the slot just behind the Pi, out through the starboard lower rail window and forward to the GP0, GP1 and GND terminals. Check the HAT's micro-USB link board is pushed fully into the Pi's USB OTG port (18).
10. DIM (24): Pico pin 19 to both LDD DIM inputs, along the starboard side and through 215's starboard window.
11. Camera and servos (23): the camera ribbon from the Pi's CSI connector forward through 207's top window. Pan servo signal to GP16 (pin 21), tilt to GP17 (pin 22); servo + and - to U9B's OUT+ / OUT-, with C2 (470 uF) across them at the plugs (same drawing). Measure U9B's output (4.9-5.2 V) before plugging the servos in.
12. The motor cores (12), the two light pairs (16) and the tether (17) come in through the rear cap and are connected in section 16.

Bench test, before the sled goes in the hull

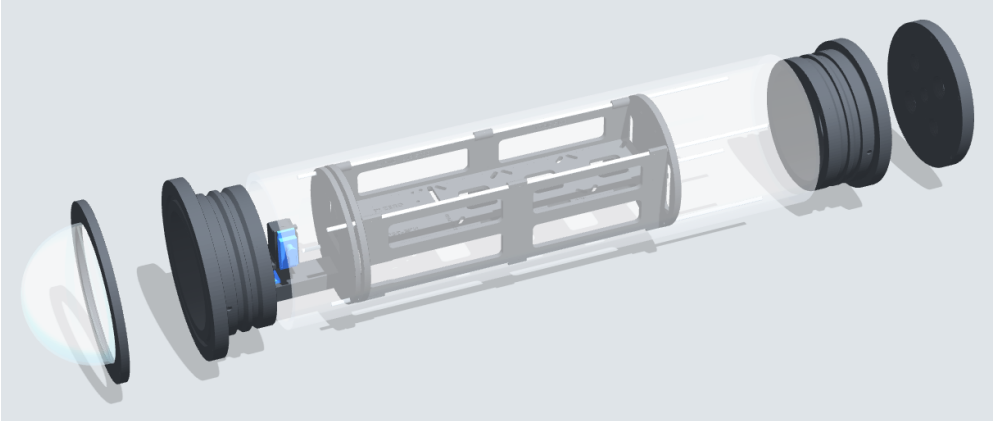
1. Meter, no power: +12V to GND on the Wagos must not read a short. Each IBT_2 VCC must read continuity to Pico pin 36 and none to +12V.
2. Pico on USB only (no battery): the servos centre, the lights stay off, ARM (pin 20) reads 0 V.
3. First 12 V: use a bench supply limited to 1-2 A on the battery lead, or the pack with F1 fitted and the motors not connected. Check +5V at both buck outputs (4.9-5.2 V), 3.3 V on every IBT_2 VCC pin, and that the Pi boots and answers on 192.168.2.2.
4. With the topside program connected: arm, then close the program and check ARM drops within half a second. The rest of the bench procedure (Firmware/README.md) follows once the motors are on in section 16.

	Before you go on
[ ]	Polarity checked with a meter; no Pico pin connected to anything above 3.3 V; every splice heat-shrunk; harnesses tied outside the rails; the bench test passed.

10 Build and test the hull

Parts for this section

201	Hull rear flange	x1	**203**	Dome flange	x1
202	Hull rear cap: 4 glands + vent	x1	**204**	Dome clamp ring	x1



Hull, exploded: dome and retaining ring, front flange, hull pipe (see-through), sled, rear flange and the 4-gland end cap.

Both flanges (201 rear, 203 front) push into the pipe ends: two radial O-rings seal against the bore, and three M3 screws through the pipe wall stop them backing out. The rear cap (202) and the dome ring (204) screw into the flange faces. Close up and test the hull EMPTY first, so a leak costs you a paper towel, not the electronics; the sled goes in afterwards.

10.1 Pipe and flanges

1. Cut the 4" pipe square at 16" (406 mm) and deburr inside and out. Chamfer the inside edges lightly so the O-rings don't nick.
2. Orientation: each flange goes in with one of its six face screw holes at the very top (12 o'clock). Its four 4 mm sled holes (inside, at the end of the skirt) then sit at 45 deg to match the rods, at both ends. Draw a pencil line along the top of the pipe at each end as your 12 o'clock.
3. Mark the radial holes: on each flange, put a pencil tick on the outer edge of the collar straight in line with each of the three radial holes in the skirt. Push the flange into its pipe end without O-rings, face hole on the top line, and copy the three ticks onto the pipe, 5 mm in from the pipe end (that is the hole centre). Pull the flange out, drill 3.4 mm through the pipe wall at the marks and deburr inside. Write F and R on the pipe so each flange goes back in its own end in the same position. These screws sit outboard of both O-rings, on the wet side, so they never need sealing.
4. Check the three radial inserts in each flange sit flush with the skirt: brass standing proud scores the pipe. Tap the six 2.5 mm holes in each flange face M3 (or let the screws cut their own thread, slowly). Leave the four sled holes plain.
5. Grease two AS568-238 O-rings into each flange's radial grooves.

10.2 Front: dome

6. Press the dome flange (203) into the front of the pipe, face hole on the top line, until the pipe end meets its collar; turn it slightly until the radial inserts show through the pipe holes and fit the 3 M3 x 10.
7. Seat an AS568-240 face O-ring in 203's face groove (it is a snug fit in the groove; that is intended). The dome is a 100 mm acrylic dome with a 115 mm flange: if its flange has small holes over the O-ring, fill them flush with epoxy or acrylic cement first and let it cure, and file any edge tab flush with the 115 mm rim.
8. Lay the dome flange side down on the O-ring, then fit the retaining ring (204) and tighten its 6 M3 x 12 evenly in a star pattern, a little at a time, until the ring just bottoms. Don't overtighten: acrylic cracks.

10.3 Rear cap and glands

Hole	Thread	Position	Use
MOTORS	M16 x 1.5	upper port	12-core from the junction box (motors)
LIGHTS	M12 x 1.5	upper starboard	4-core from the junction box (lights)
BATTERY	M16 x 1.5	lower starboard	12 AWG lead from the battery pod
TETHER	M12 x 1.5	lower port	Cat6 tether
VENT	M10 x 1.0	centre	Bolt + bonded washer; hose barb for the vacuum test

9. The threads in the rear end cap (202) are printed: no tapping. Run each gland body and the vent bolt in by hand once and back out, to clear any stringing; if one is stiff, work it in and out a few times. Each hole is labelled on the cap's outside face.
10. Thread each gland body in from the outside with its sealing washer on the outside face and a thin smear of silicone on the threads; snug, not tight. Fit the vent bolt with a bonded washer on the outside face (rubber lip toward the cap) and snug it, no more.
11. Push the real cables through their glands now, nothing connected yet, from outside in: 12-core (MOTORS, about 400 mm inside: it has to reach T1 at the front of the deck), 4-core (LIGHTS, 200 mm), the 12 AWG battery lead (BATTERY, 250 mm) and the Cat6 tether (TETHER, 350 mm). A cable is what seals a gland: an empty gland leaks. Short 150 mm stubs of the same cables do for the test if the real ones aren't ready. Tighten the gland nuts.

10.4 Test the empty hull

12. Rear flange (201): O-rings greased, press it into the rear of the pipe (face hole at 12 o'clock) and fit its 3 radial screws. Lay a folded dry paper towel inside.
13. Seat the AS568-237 face O-ring on 201 and bolt the cap on with 6 M3 x 16, labels the right way up (MOTORS upper port), in a star pattern.
14. Vacuum test (section 18.1): -10 inHg must hold for 15 minutes.
15. Dunk: strap about 4 kg to the hull (it floats hard) and sink it in a pool or tub for 30 minutes, deeper is better. Dry it off, open the cap and check the paper towel. Any damp spot shows you which end leaked.

10.5 Then open it again

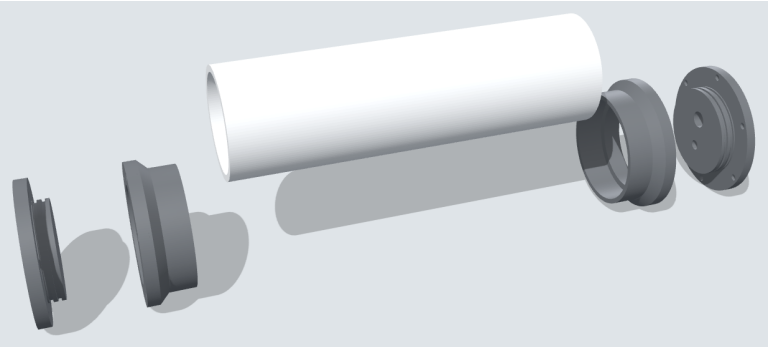
16. Unbolt the cap, take out 201's three radial screws and pull the flange out again. Keep the open end covered until the sled goes in (section 16): it is wired and tested on the bench first, where the Pico's USB is still in reach.

11 Build the battery pod

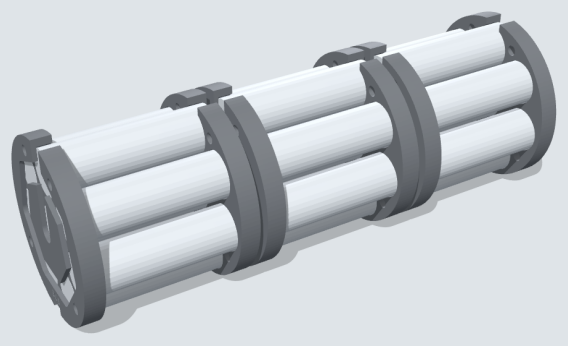
Parts for this section

301	3" pod flange, one at each end	x2	**305**	18650 holder plate	x6
302	3" pod rear cap: M16 gland + vent	x1	**306**	Rod spacer between groups	x8
303	3" pod front cap, blank (no holes)	x1			

The battery lives in its own 3" tube under the hull, low and close to the middle, which keeps the ROV level and stable. Both ends get the same printed flange; the front is closed by a blank printed cap and the rear by a cap with the battery gland and vent. The 24 cells are packed as 3S8P: three groups of 8 in parallel, the groups in series.



Pod, exploded: blank front cap, front flange, 3" pipe, rear flange, rear cap with gland + vent. Both flanges are the same part.



3S8P pack: six holder plates (305), two per group, on four M4 rods with 3 mm spacers (306) between the groups.

1. Cut the 3" pipe to 300 mm (11-13/16"), square. Deburr and chamfer the inside of both ends.
2. Epoxy a flange onto each end with marine epoxy, pushed on until the pipe hits the flange's stop ledge. Line up the bolt holes of the two flanges. Let it cure fully.
3. Front: grease two 72 x 2.5 O-rings onto the blank cap's plug (303, marked FRONT) and bolt it on with 6 M5 x 16.
4. Build the pack by docs/Battery_Pack_Build.md (cell matching, welding, BMS wiring, first charge, voltage limits): test and match the cells, fish-paper ring on every + end, six plates (305) grooved faces out, all 24 cells + toward the rear, 8 mm pure nickel in the grooves with the centre-to-edge run doubled, series links folded across the 3 mm gaps, four M4 rods with a 3 mm spacer (306) on each rod in both gaps. Every plate goes on with its tab notch at the top. The pack is about 213 mm long and reads about 11.1 V.
5. The rear cap's (302) threads are printed: run the M16 gland body (BATTERY) and the M10 x 1.0 vent bolt (VENT) in once by hand to clear them. Fit the gland body (sealing washer outside, silicone on the threads) and the vent bolt with a bonded washer.
6. Test the EMPTY pod before the pack goes in: push a 150 mm stub of the 12 AWG lead through the gland and tighten it, put a dry paper towel inside, grease the rear cap's two 72 x 2.5 O-rings and bolt it on. Vacuum test it (section 18.1), then strap on about 2 kg, sink it for 30 minutes and check the towel. Only then open it again.

7 Mount the BMS on group C's rear plate, balance leads along the notch in the bottom edge. Fit the 35 A inline fuse holder at pack + (F1, 30 A, is the second fuse, in the hull). Pass the free end of the 12 AWG battery lead (its other end is already through the hull's BATTERY gland, section 10.3) in through the pod gland from outside and solder it to the BMS output (P+ / P-). Check polarity with a meter now and mark the red core.

8 Put a block of foam against the front cap, slide the pack in from the rear, grease the rear cap's two 72 x 2.5 O-rings and bolt it on with 6 M5 x 14 (the rear cap is 2 mm thinner than the front: an M5 x 16 bottoms out), then tighten the gland on the lead. The pack can also come out the front if you ever need it.

Charging: open the rear cap, or fit an inline waterproof connector on the battery lead and charge through it. Always charge with the BMS in circuit, and never charge a wet pack.

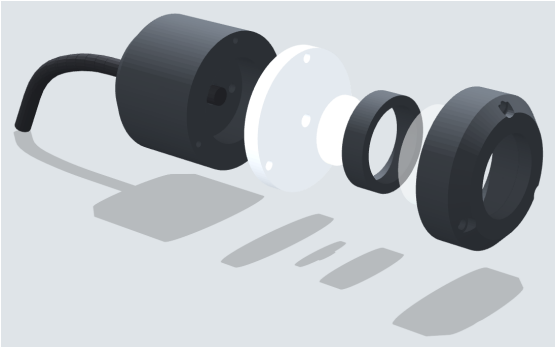
	Before you go on
[ ]	Pack reads 11-12.6 V at the lead, BMS balance leads correct, both caps on, vacuum holds (section 18.1).

12 Build the lights

Parts for this section

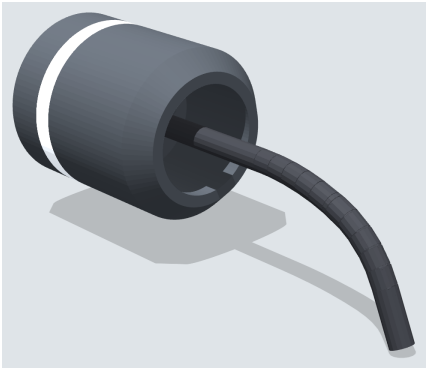
401	Light head rear body	x2	**403**	Window spacer	x2
402	Light head front ring	x2			

Each light is a 42 mm head: an XHP50.2 LED on a 20 mm star, bonded to an aluminium disc whose rim is in the water and carries the heat. The cable leaves straight out of the back so it can follow the head through its sleeve in the cowl nose (section 17).



Light head, exploded along the beam: rear body, aluminium disc, LED, spacer, window, front ring.

1. Aluminium disc (42 x 5 mm, 6061): drill a 4 mm centre hole and 3 x 2.9 mm holes on a 37 mm circle. Deburr.
2. Look into the rear body (401): the 6.4 mm centre hole is for the cable, the two 3 mm holes either side are the potting fill and vent, and the three 2.1 mm holes take the screws (no inserts).
3. Cut about 350 mm of 2-core 18 AWG cable (OD 5-6 mm) for each light. Solder it to the LED star (red to +), pass the wires back through the disc's centre hole and bond the star to the disc's front face with thermal adhesive, covering the hole.
4. Thread the cable through the rear body's centre hole from the front, so it comes straight out of the back (the round socket). Sit the disc on the front face, its holes over the three screw holes.
5. Seat the 30 x 1.5 O-ring in the front ring's (402) back-face groove and the 24 x 1.5 O-ring under its lip. Drop in the glass window, then the spacer (403), and screw the ring and disc to the rear body with 3 M2.5 x 25: they pass 16.5 mm of ring and disc, then cut their own thread in the PETG, so drive them slowly, snug, and stop.
6. Pot the head: stand it nose-down, pull the cable taut and dab silicone round it where it enters the centre hole. Syringe potting compound into one 3 mm hole until it comes out of the other. When that has set, fill the round socket at the back to the brim around the cable: that is the cable's strain relief and second seal.
7. Test briefly in air at low PWM only: the disc needs water to shed the heat.



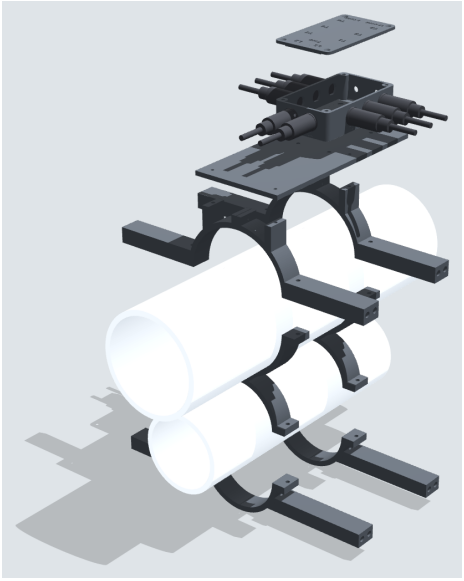
Finished light head from behind: the cable leaves straight out of the back, potted into the round socket.

13 Station frames: join hull and pod

Parts for this section

601	Station top, port half: hull top + upper arm	x2	**604**	Station bottom, port half: pod bottom + lower arm	x2
602	Station top, starboard half	x2	**605**	Station bottom, starboard half	x2
603	Station middle: hull bottom + pod top	x2			

Two stations, 60 mm either side of the hull centre, clamp the hull and pod together. Each is 5 printed parts: a top in two halves (port and starboard) with the upper arms, a middle piece between the hull and the pod, and a bottom in two halves with the lower arms. The side plates bolt to the arm ends.



Both stations exploded around the hull and pod; the junction deck spans both, the box sits over the rear station.

1 Push M4 nuts into the hex pockets under the middle piece's hull ears and under the bottom halves' pod ears.

2 Mark 60 mm ahead of and behind the hull's centre on the hull and the same on the pod. Hull upside down on a towel: lay a middle piece (603) at each mark, then the pod in the middle pieces' lower cradles, gland end toward the rear.

3 Fit the two bottom halves (604 PORT, 605 STBD) under the pod at each station and bolt the pod joints: M4 x 20 through each ear into the nut.

4 Turn it over. Fit the two top halves (601 PORT, 602 STBD) over the hull; bolt the halves to each other through the centre rib (M3 x 20 + nyloc, no washers: the hull clamp ring is right under it) and the hull joints with M4 x 30 down through each upper arm and ear into the middle piece's nuts.

5 Snug evenly so the hull and pod stay parallel. The arm ends of both stations must line up straight along each side: check with a straight edge.

Before you go on

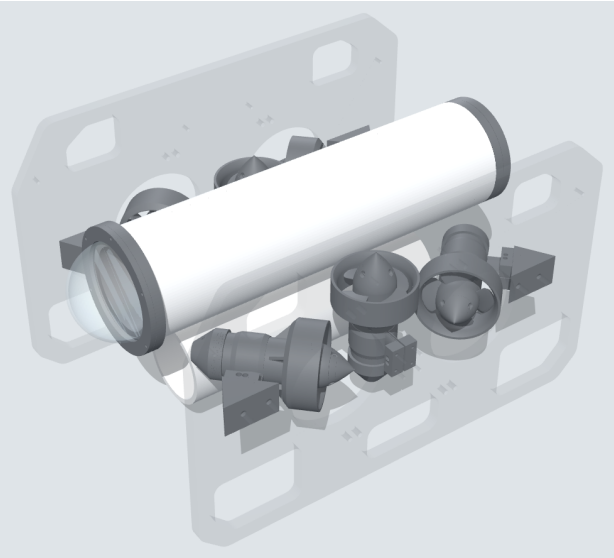
[ ] Hull and pod parallel, all four arm ends of each side in line, no bolts torqued so hard they crack the PETG.

14 Side plates and thrusters

Parts for this section

607	Front-port + rear-starboard thruster bracket	x2	**902**	Side plate, starboard (HDPE)	x1
608	Front-starboard + rear-port thruster bracket	x2	**T1-T6**	Finished thrusters (section 7)	x6
901	Side plate, port (HDPE)	x1			

Four horizontal thrusters sit in the corners turned 45 deg in a diamond: together they drive forward/back, sideways and turn on the spot. The two verticals sit on the plates at the estimated centre of gravity, so going up or down doesn't pitch the nose.



Corner thrusters on their 45 deg brackets, verticals straight on the plates (plates shown see-through).

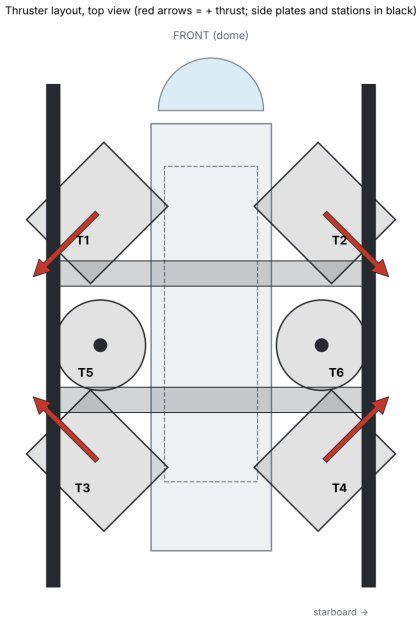
1. Bolt each corner pylon (106) to its bracket with 4 M3 x 16 (heads in the hex pockets) (607 bracket A, marked A FP/RS = front-port and rear-starboard; 608 bracket B, marked B FS/RP = front-starboard and rear-port). Prop end faces out, tail cone toward the hull.
2. Bolt the four brackets to the plates (901 port, 902 starboard) at the holes you marked 607 A / 608 B: 2 M5 x 20 from outside each plate.
3. Vertical thrusters (T5 port, T6 starboard): pylon straight on the plate's inside face between the two big windows, 4 M3 x 30 through the plate with a 1 mm shim washer on each and a nyloc outside. Prop up.
4. Offer each plate up to the station arms and bolt it with 8 M5 x 20 (2 per arm end). Do one side, then the other.
5. Slide each thruster in its pylon until the duct clears everything, then tighten the pinch bolts. Spin every prop by hand: nothing may touch. The corner jets leave through the round windows in the plates.

Mixing (topside software)

Coefficients for a + command (thrust toward the nose cone). Surge + = forward, sway + = starboard, heave + = up, yaw + = nose to port. Per thruster: command = surge x Fx + sway x Fy + yaw x Mz (+ heave for T5/T6), then scale so no output exceeds 1. Flip the sign for any motor wired or propped the other way.

Thruster	x, z (mm)	Surge	Sway	Heave	Yaw
T1 front port	108, 118	-0.71	-0.71	-0.00	+1.00
T2 front starboard	-108, 118	-0.71	+0.71	-0.00	-1.00
T3 rear port	108, -118	+0.71	-0.71	+0.00	-1.00
T4 rear starboard	-108, -118	+0.71	+0.71	+0.00	+1.00

Thruster	x, z (mm)	Surge	Sway	Heave	Yaw
T5 vertical port	105, -8	+0.00	-0.00	+1.00	-0.00
T6 vertical starboard	-105, -8	+0.00	-0.00	+1.00	+0.00

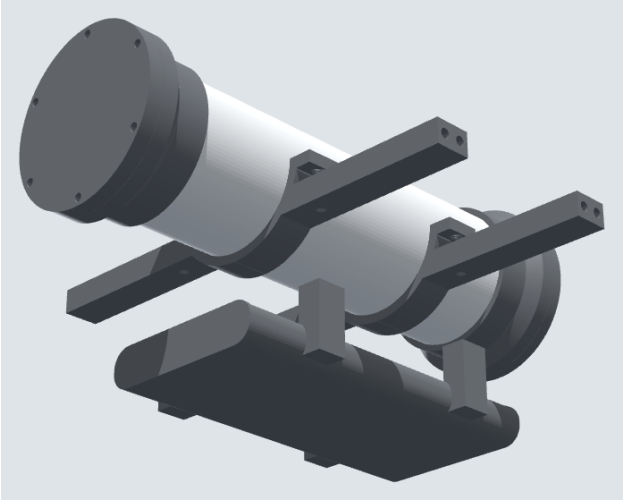


Thruster layout, top view.

15 Belly pan

Parts for this section

701	Lead ballast tray under the pod	x1



Belly pan hanging from the four lower station arms under the pod.

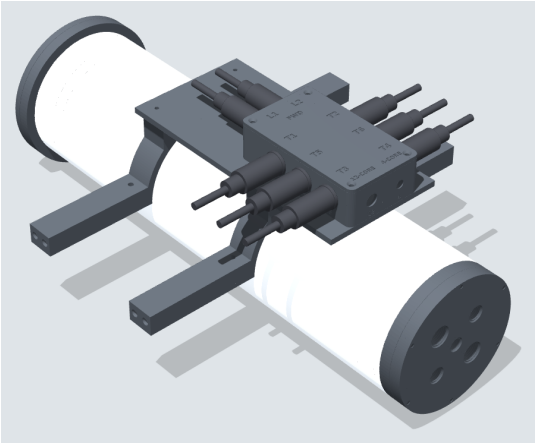
1. Lay lead shot in sealed bags (or bar) along the pan floor: start with 1.6 kg spread evenly front to back.
2. Hold the pan (701) up under the four lower arms and fit an M4 x 50 up through each of its four posts into the arm inserts.
3. The pan is open at the top and floods; the pod above keeps the lead in place.

16 Junction deck and wiring

Parts for this section

606	Carries the junction box (501)	x1	503	Junction box lid with the socket map	x1
501	Junction box: 6 thruster + 2 light SP13 sockets, 12-core + 4-core out	x1			

The thrusters and lights plug into one junction box (501) on the frame, so only two cables enter the hull: the 12-core for the motors and the 4-core for the lights. Each thruster lead ends in an SP13 2-pin plug, each light lead in an SP13 3-pin plug, so a thruster or light can be unplugged and swapped without opening anything. The lid (503) covers the wiring.



Junction box over the rear station: thruster sockets on both sides, light sockets at the front, cable exits at the back.

Face	Sockets (front to back)	Type
Port	T1 front, T5 vertical, T3 rear	SP13 2-pin
Starboard	T2 front, T6 vertical, T4 rear	SP13 2-pin
Front	L1 port, L2 starboard	SP13 3-pin (pins 1 +, 2 -)
Back	12-CORE (motors), 4-CORE (lights)	cable exits, potted

1. Mount the sockets: push each SP13 panel socket through its hole from outside, flat at the bottom so it only goes in one way, and tighten the rear nut inside. The name under each hole (T1-T6, L1, L2) is the one to use.
2. Pass the 12-core and 4-core in through the back exits. Solder the 12-core pairs to the motor socket cups, one pair per socket, and the 4-core to the light sockets as two separate pairs, L1 +/- and L2 +/-. Never join the two LED - wires: each LDD-700HW output is its own current loop. Sleeve every cup with heat-shrink and note which core colours go to which socket.
3. Screw the deck (606) to the ribs of both stations with 4 M3 x 10 (2 into each station top), then the box to the deck where it says JUNCTION BOX, 4 M3 x 8 down through the floor.
4. Test every socket for continuity to the hull end of its cable before potting. Then pour potting compound in thin layers, tapping out bubbles, until it covers the socket cups and the floor screws but stays a few mm below the lid bosses. The potting is what keeps the box waterproof: the sockets are only splash-rated. Let it cure fully.
5. Screw the lid (503) on with 4 M3 x 8 countersunk screws. The socket map on the lid matches the labels on the sides.
6. Thruster leads: from each tail gland, up the inside of the plate and over to its socket. Zip-tie to the station arms and leave a drip loop at each thruster. Cut to length, solder on the 2-pin plug (pin 1 to the motor's red), and slide the plug's gland nut and seal on first.
7. Light leads are plugged in when the lights go into the cowl (section 17, Mount the lights).

8. Put a smear of silicone dielectric grease on each plug's pins and O-ring, push it home and tighten the coupling ring hand-tight. Wipe off the excess.
9. Multicores: the 12-core and 4-core already pass through the MOTORS (M16) and LIGHTS (M12) glands (section 10.3). Run them forward from the box exits along the rear station, zip-tied, to the hull's rear cap, which hangs behind the open hull on its cables.
10. Battery: the 12 AWG lead already runs from the pod's rear gland to the hull's BATTERY gland. Route it straight up between the two and tie it to the rear station.
11. Tether: through the TETHER gland, with a strain relief tied to a station arm so the gland never takes the pull. Inside, crimp an RJ45 on it.
12. Thread all four cable ends through the rear flange's (201) bore, O-rings already on it. Set the sled on the bench right behind the open hull, camera end toward the hull, and connect it: each motor pair from the 12-core to its IBT_2's M+ / M- (wire 12, one pair per board, following the colour note you made at the box); each 4-core pair to its LDD's +Vo / -Vo (wire 16); the battery lead red through the F1 holder into the +12V Wago and black into the GND Wago (wires 1-2), with the 30 A fuse still OUT; the tether into the inline coupler over T1-T2, and the 0.3 m 90-degree patch lead from the coupler to the HAT's RJ45, plug turned aft. Dress every cable through 215's port or starboard window and tie it.

First power-up, with the sled still on the bench

Do this before the sled goes in: the Pico's USB faces forward at the front of the deck and can't be reached once the sled is in the hull, and every setting that needs a new firmware build is found here.

1. Stand the ROV so all six props are clear to spin in air. Topside laptop on the tether at 192.168.2.1, gain at 25 %.
2. Fit F1 (30 A). Telemetry should show the battery voltage and 'armed 0'.
3. Work through the bench procedure in Firmware/README.md: VBAT_CAL against a meter at the Wago, link loss stops the thrusters within 0.5 s, thruster directions (THRUSTER_DIR, one per motor: the left-hand props come out reversed) with short bursts only (the shaft seals and oil like water, not air), camera directions (PAN_DIR, TILT_DIR).
4. Each change is a line in Firmware/pico_rov/config.h: edit it, rebuild in the Arduino IDE (Raspberry Pi Pico / arduino-pico core), pull F1, and copy the new UF2 to the Pico (hold BOOTSEL, plug in USB). Set directions in software only: never swap motor wires afterwards, or the mixing table and the code disagree. Repeat until every direction is right.
5. I_FULL_A needs a thruster in water: put the ROV in a tub or pool with the sled still outside the hull on its cables (keep it dry and the hull end up), or come back to it after the first pool session (that means pulling the sled out again).
6. Lights: both dim smoothly from off to full on the topside slider; run them only a few seconds in air.

Sled in and close up

7. Pull F1. Slide the sled into the hull from the rear, camera first and Pi on top, rods at 45 deg, feeding the cables in behind it. Push it forward until the front rod tips drop into 203's four sled holes: it moves another 6 mm, stops, and can no longer turn. Watch through the dome: the camera must not touch anything at full pan and tilt.
8. Press 201 in behind it, guiding the four rear rod tips into its sled holes, and refit its 3 radial screws. The sled should have only 1-2 mm of end play. Tuck the slack cables into the rear bay so none crosses the 12 V / logic divide or presses on the camera ribbon.
9. Fit F1 (the ROV is now live: see below). Wipe and re-grease the AS568-237 O-ring, seat it on 201 and bolt the cap on (MOTORS upper port, 6 M3 x 16 in a star). Tighten every gland nut last, after the cables are dressed, so no cable is twisted in its gland.

Power on and off

There is no main switch yet. The ROV is live whenever both fuses are in: the 35 A fuse at the pack (in the pod) and F1 (30 A, in the hull at the BATTERY gland). Fit F1 last, just before the rear cap goes on, and pull it whenever the hull is open. Idle, the Pi, the Pico and the drivers draw roughly 0.3 A, so a charged pack lasts a few days switched on, but the ROV is never truly off while sealed. A waterproof 2-pin connector rated 30 A or more in the battery lead between the pod and the hull would give an outside on/off and a charging point without opening anything: worth adding once the ROV is working.

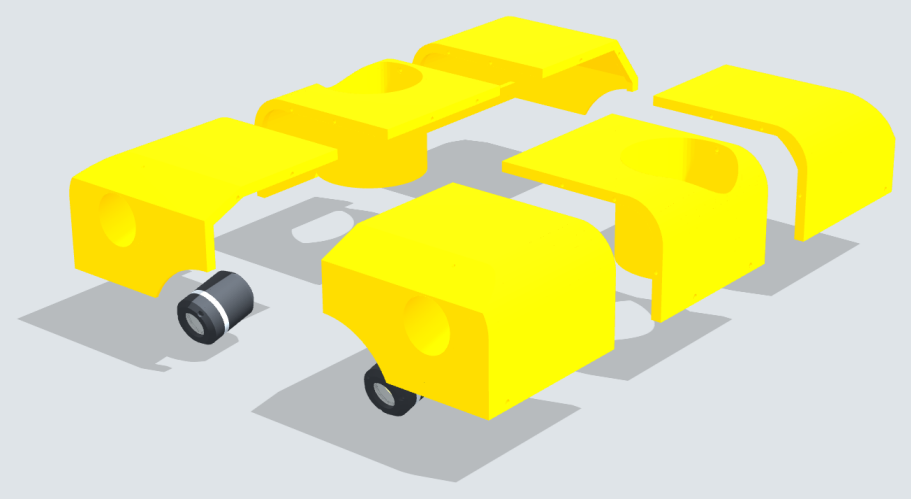
	Before you go on
[ ]	Every thruster spins the right way from the topside software (props off the water only briefly), both lights dim with PWM.

17 Top cowl

Parts for this section

801	Top cowl, rear, port	x1	**804**	Top cowl, middle, starboard	x1
802	Top cowl, rear, starboard	x1	**805**	Top cowl, front, port	x1
803	Top cowl, middle, port	x1	**806**	Top cowl, front, starboard	x1

The cowl is the only yellow part: six sections (front, middle, rear, each in a port and starboard half). It floods, and holds the buoyancy foam and the two light heads. It is a fairing, not a lifting point: see Lifting and carrying at the end of this section.



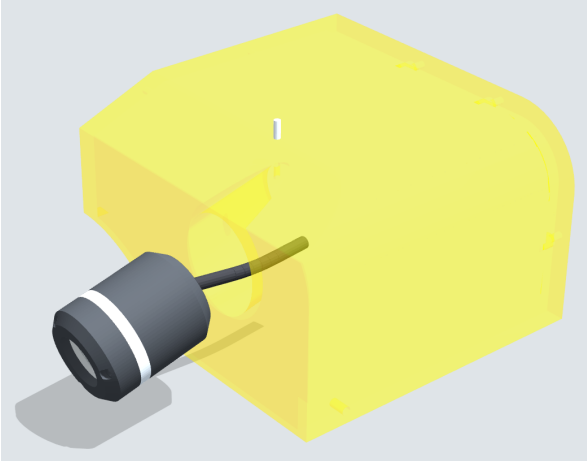
Cowl sections pulled apart, with the light heads at the nose.

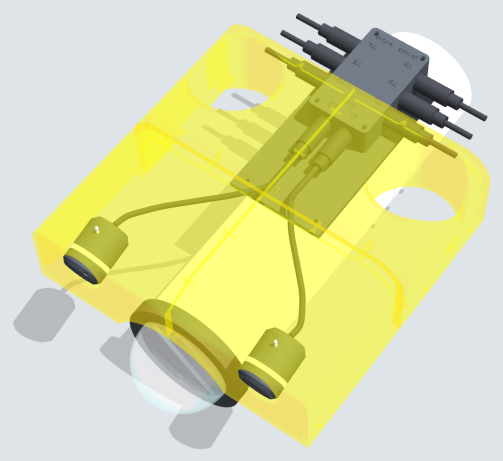
1. Lay the six sections out by their labels: 801 REAR PORT, 802 REAR STBD, 803 MID PORT, 804 MID STBD, 805 FRONT PORT, 806 FRONT STBD. Bolt the port and starboard halves of each section together through the centre flange (M3 x 20 + nyloc), then the three sections to each other through the seam flanges. 18 bolts in all.
2. Foam: cut depth-rated closed-cell foam to fit inside the cowl roof, either side of the centre flange and clear of the vertical thruster sleeves. Start with about 1.0 L. Glue it in with epoxy so it can't shift.
3. Fit the lights into the cowl first (below). Then lower the cowl between the plates, plugging L1 and L2 into the junction box as it goes down, and fix it with 4 M3 x 20 through each plate.

Mount the lights

Each light head slides into the sleeve in its corner of the cowl nose: 805 FRONT PORT takes L1, 806 FRONT STBD takes L2. The sleeve is a 43 mm bore, 44 mm deep and open at the back. The head is 42 mm, so it slides in by hand with its last 7 mm standing out behind the sleeve, inside the cowl, and the cable carries on aft from there. One M3 set screw, driven down through the cowl roof and the sleeve wall, holds it.

Both sleeves are aimed 15 deg nose-down. A bare XHP50.2 throws a beam about 80 deg wide underwater (40 deg either side), so aimed straight ahead it ended right where the camera looks at full tilt (45 deg down). Aimed down 15 deg it covers from about 25 deg above level to 55 deg below, the camera's whole tilt range, with the brightest part slightly below level where you look most. For anything off to the side, turn the ROV rather than panning: the lights turn with it.

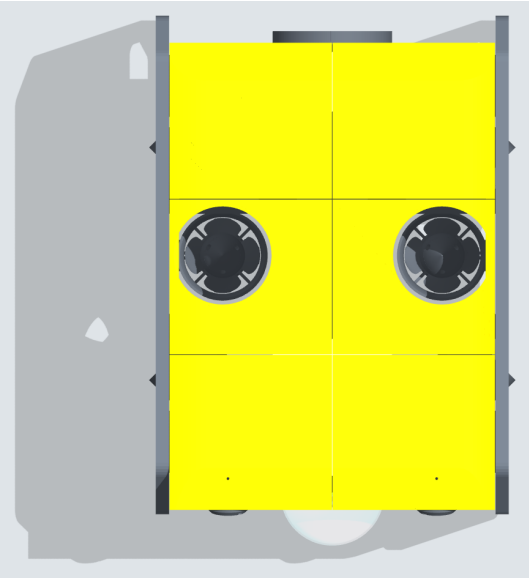




Left: port head pulled forward out of its sleeve, cable first; the set screw goes in from the top of the cowl. Right: cowl shown clear, both leads running back to L1 and L2 on the junction box.

1. Tap the set-screw hole M3 all the way through: it starts in the cowl roof about 30 mm behind the nose, where the roof and the top of the sleeve are one piece about 6 mm thick. One pass of an M3 tap does it.
2. Feed the cable into the front of the sleeve and out of the back, inside the cowl, then push the head in after it until the bottom edge of its front face sits about 4 mm proud of the nose (the head is tilted 15 deg, so the top edge sits about flush).
3. Drive an M3 x 8 set screw down through the roof hole until it bites on the head, then a quarter turn more. Don't crank it: it presses on printed plastic. A dab of threadlocker keeps it there.
4. Run each lead aft inside the cowl, under the foam and inboard of the vertical-thruster ducts: port head to L1, starboard head to L2. Hold it in place inside the cowl with a few adhesive zip-tie mounts. Leave about 60 mm of slack behind each head so it can be pulled out of the front for service.
5. Sit the cowl in place on the ROV to measure, then cut each lead to reach its socket with a little slack. Slide on the SP13 3-pin plug's nut and seal, and solder: pin 1 to the LED + (red), pin 2 to LED - (black); pin 3 is not used. Grease the plug and screw it onto its socket hand-tight.
6. To take a light out later: unscrew the cowl and lift it enough to unplug the light at the box, back the set screw off two turns and pull the head out of the front of the sleeve.

Lifting and carrying: never by the cowl, a thruster or the dome. Stand at one end and hook your fingers through the lower corner windows of both side plates (the 98 x 67 mm windows next to the skids), lifting on the bottom strip of each plate: there is about 100 mm of clear space behind those windows. The upper rear windows have the cowl right behind them; don't use them. The plates carry the whole ROV this way exactly as they do when it sits on the ground. Sand the edges inside those four windows smooth first.



Finished ROV from above.

18 Leak test, trim and first dive

Test in stages, so a leak costs as little as possible. The vacuum test is safe with everything inside, so do it before every dive: a vacuum can't hurt the electronics or the cells, it only proves the seals.

18.1 Vacuum test

1. Swap the VENT bolt for the M10 x 1.0 brass barb (with a bonded washer), push the hand pump's 6 mm hose on and pump to about -10 inHg.
2. Watch the gauge for 15 minutes. It must not move. A slow drop is a leak: re-seat and re-grease the O-rings, check each gland is tight on its cable, and repeat. To find it, re-seat one seal at a time and test again.
3. Let the air back in, swap the barb for the bolt and bonded washer, snug it, no more.
4. Never test by blowing air INTO the hull or pod: pressure from inside pushes the dome and caps off their seals and can crack them. -10 inHg is about the same pressure difference as 3.4 m (11 ft) of water, in the same direction as diving, so it proves the seals but not the depth.

18.2 Order of tests

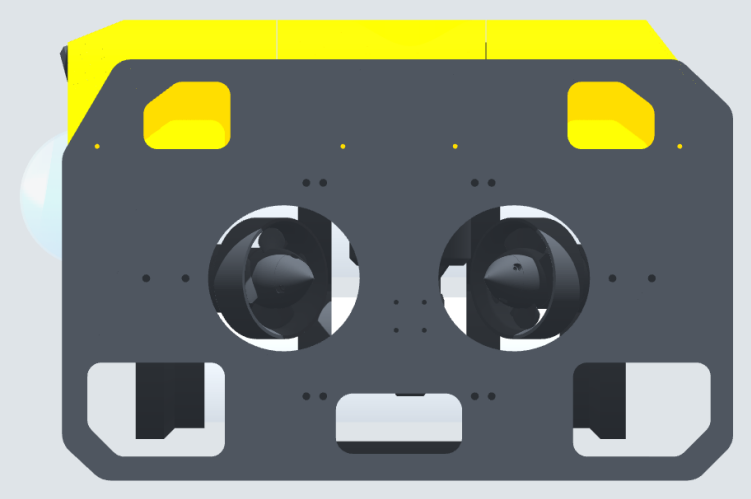
1. Empty hull and empty pod: vacuum and a 30-minute dunk with a paper towel inside (sections 10.4 and 11). Done before anything valuable goes in.
2. Whole ROV, everything installed, F1 in: vacuum test the hull and the pod.
3. Thrusters in water: the ROV in a tub or pool at the edge, each thruster at low power for a minute, then all together. Check for oil weeping at the tail cones and that each one pushes the way the mixing table says.
4. First wet run: pool, tethered, no deeper than 1-2 m, 10 minutes. Lift it out, dry it, pull F1 and open the hull and the pod: look for any water at the O-rings and glands before going on.
5. Trim (below), then 10 ft, 30 ft and 60 ft dives on separate days, a vacuum test before each and a look inside after each.
6. Measure I_FULL_A in the pool now if you didn't on the bench (section 16).

18.3 Trim

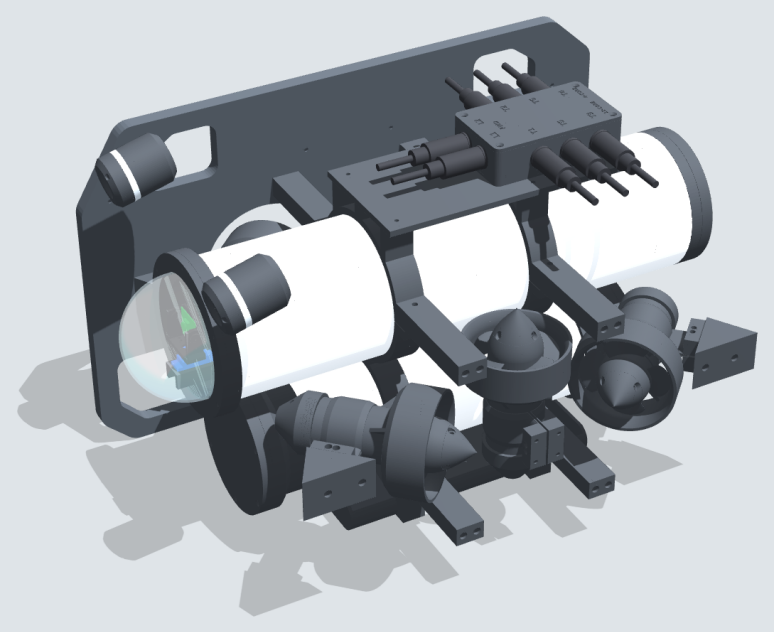
1. In the pool it should float level and just barely positive (0.2-0.4 kg): a gentle push down and it comes back up slowly.
2. Too buoyant: add lead in the belly pan. Rolls or wobbles: add lead low AND foam high (both make it steadier). Nose up or down: move lead along the pan.

Starting points from the 3D model are in section 1 (Key numbers).

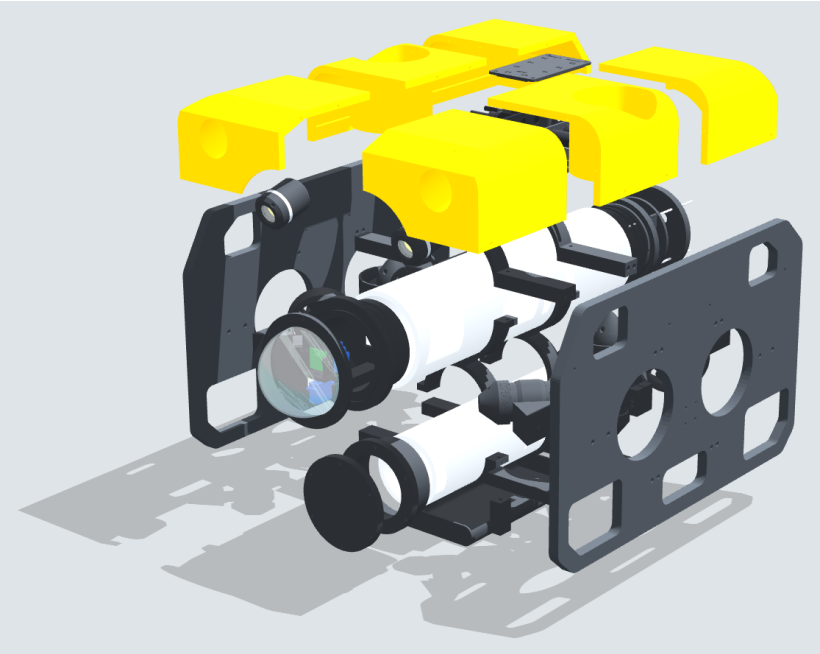
19 Reference views and layout



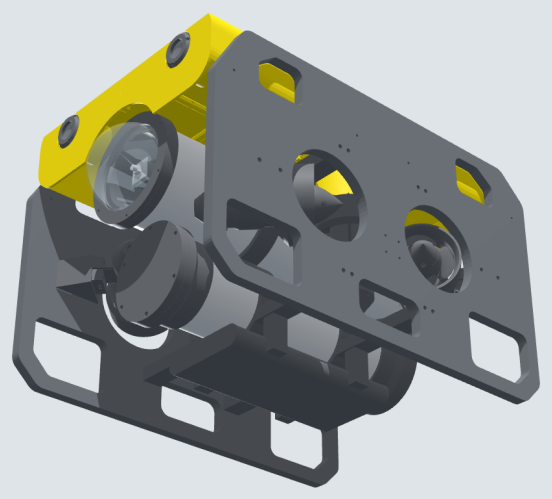
Side view.



Cowl and port plate removed.



Exploded.



From below.

Item	z (mm)	Notes
Side plates (HDPE)	-238 to +240	x +/-156.5 to 169.2 (12.7 thick), y -202 (skid edge) to +98 (300 mm tall)
Rear station	-72 to -48	Junction deck on top, under the box
Front station	+48 to +72	Front end of the junction deck
Vertical thrusters	-8	T5 / T6 on the plates
Corner thrusters	about +/-118	T1-T4 on 45 deg brackets
Top cowl	-212 to +236	Splits at -62 and +87, and on the centreline
Belly pan	-112 to +112	Under the pod
Hull / dome	-223 to +270	Electronics only
Battery pod	-162 to +188	Axis 116 mm below the hull; printed flange + cap at both ends

Appendix A Side plate cutting templates (1:1)

The next 9 pages (pages 60-68) are the full-size cutting templates for side plates 901 (port) and 902 (starboard): an instruction page with the sheet layout, then four tiles per plate (A1, A2, B1, B2), every hole marked with its drill size and part tag. Print only these pages, landscape on Letter, and cut the plates as section 6 describes.

TinkerSub KB-1 side plates: cutting templates for one 24 x 24 in sheet of 1/2 in HDPE

1 Print every page at 100 % / Actual size, NOT 'Scale to fit'. Measure the 100 mm bar on a page: it must be exactly 100 mm (about 3 15/16 in).

2 Each plate is several tiles (rows A, B, C from the top, columns 1, 2, 3 from the left). Trim a tile on its grey dashed line, overlap the next tile, line up the red circles and tape the joints on the front.

3 Lay both templates on the sheet as shown on the right, 5 mm apart (the sheet has only ~2 mm to spare at each edge: measure it first). Spray adhesive (3M Super 77) or glue stick holds them; peel off afterwards.

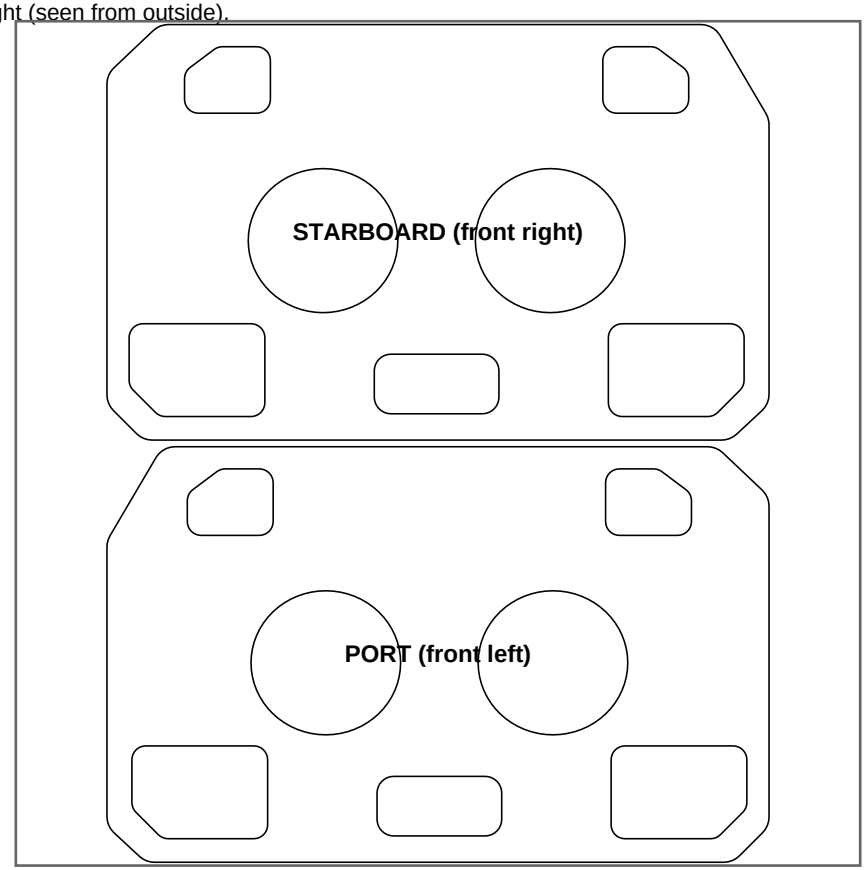
4 Centre-punch every crosshair, then drill: 3.5 mm (M3), 4.5 mm (M4), 5.5 mm (M5) as printed beside each hole. Drill straight (drill press if you have one), slow speed, back the sheet with scrap wood.

5 Drill the 10 mm blue starter hole in each window, then cut the windows with a jigsaw: fine-tooth blade, slow speed, so the HDPE does not melt and re-weld behind the blade. Cut just outside the line.

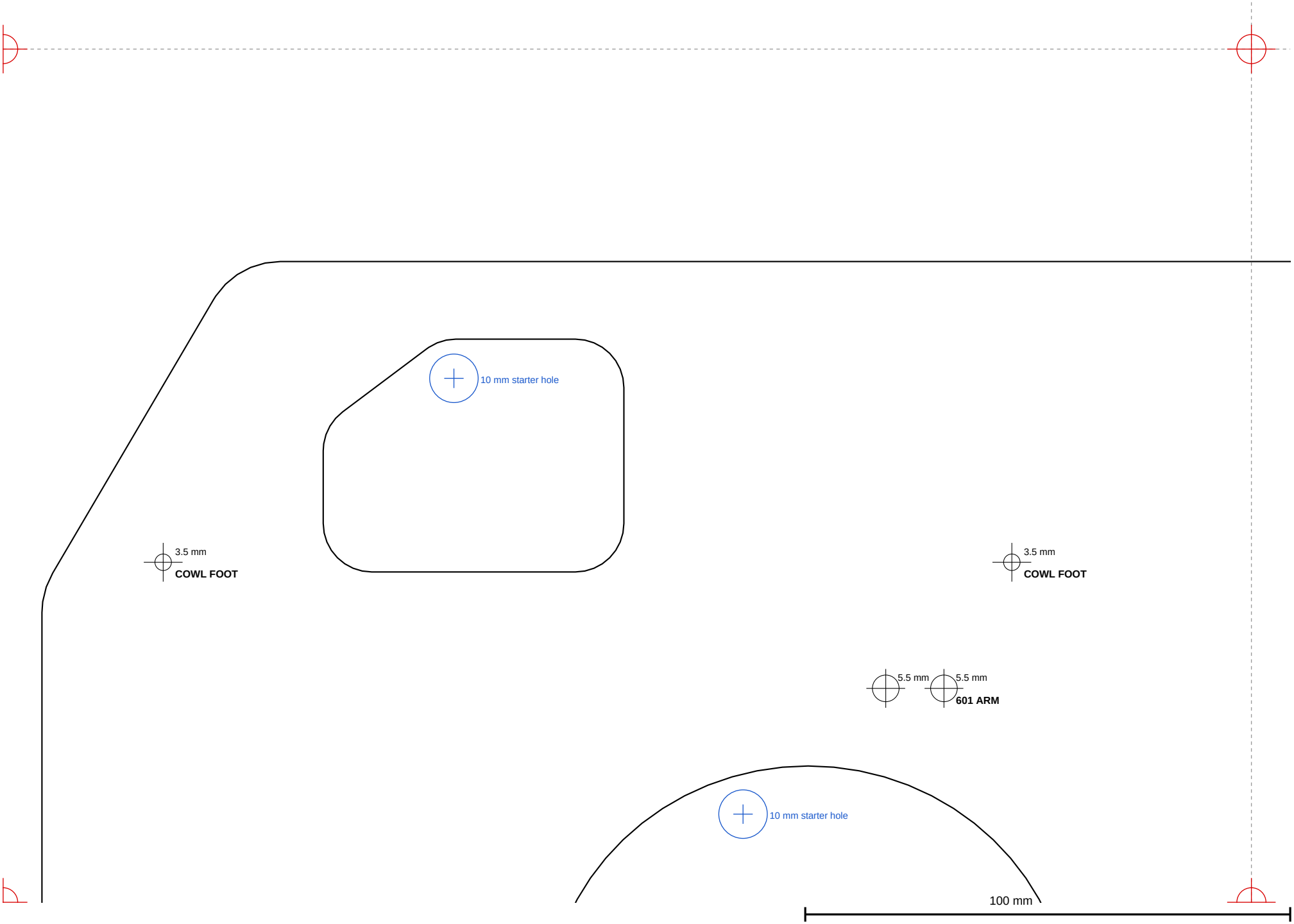
6 Cut the two plate outlines, then file / sand every edge smooth and break the sharp corners.

7 The two plates are mirror images: PORT has the front of the ROV on the left, STARBOARD on the right (seen from outside). Mark each plate's outside face before you peel the template off.

Tools: jigsaw + fine blades, drill + 3.5 / 4.5 / 5.5 / 10 mm bits, centre punch, clamps, file or sandpaper.

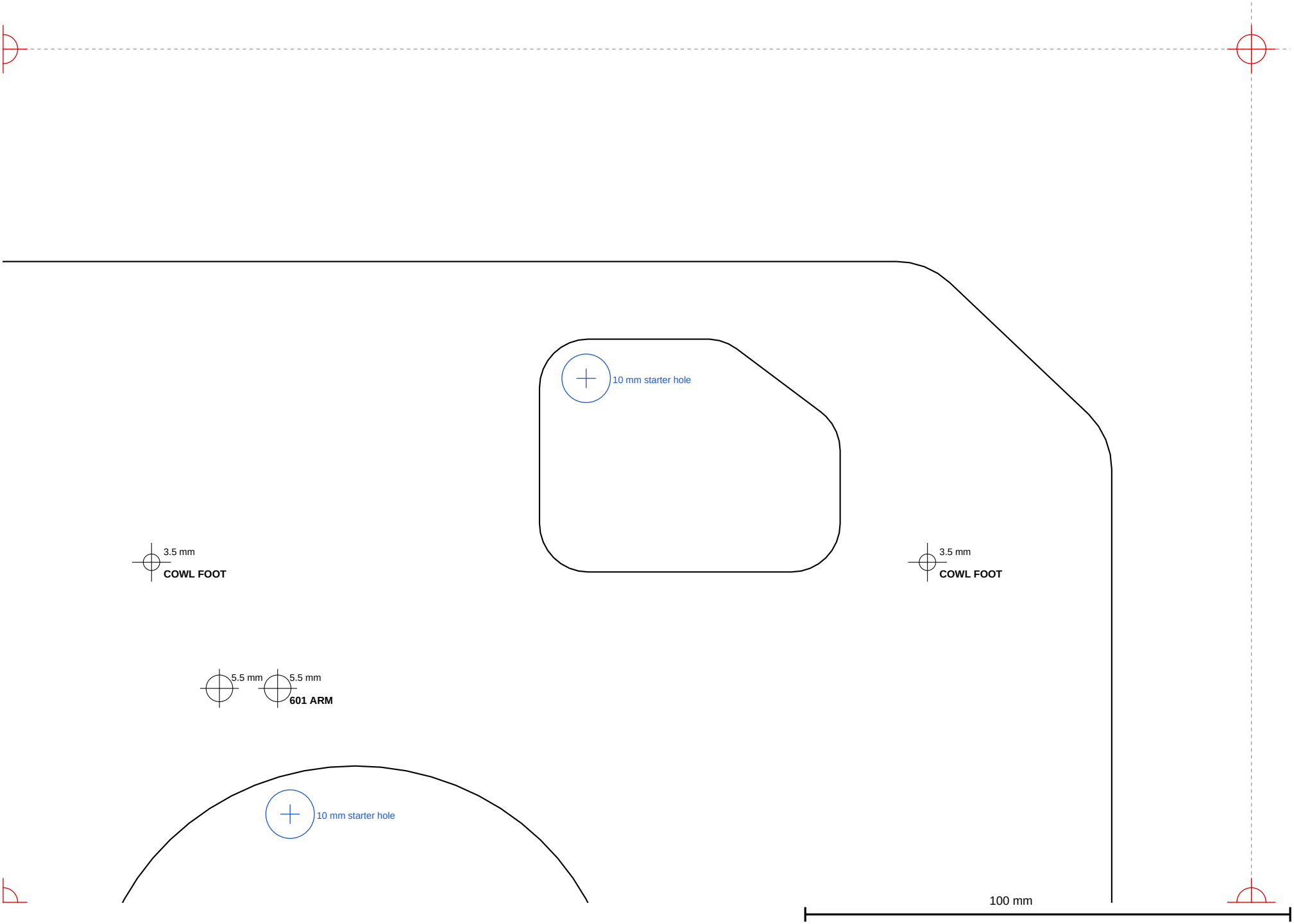


24 x 24 in sheet (609.6 mm), both plates outside face up



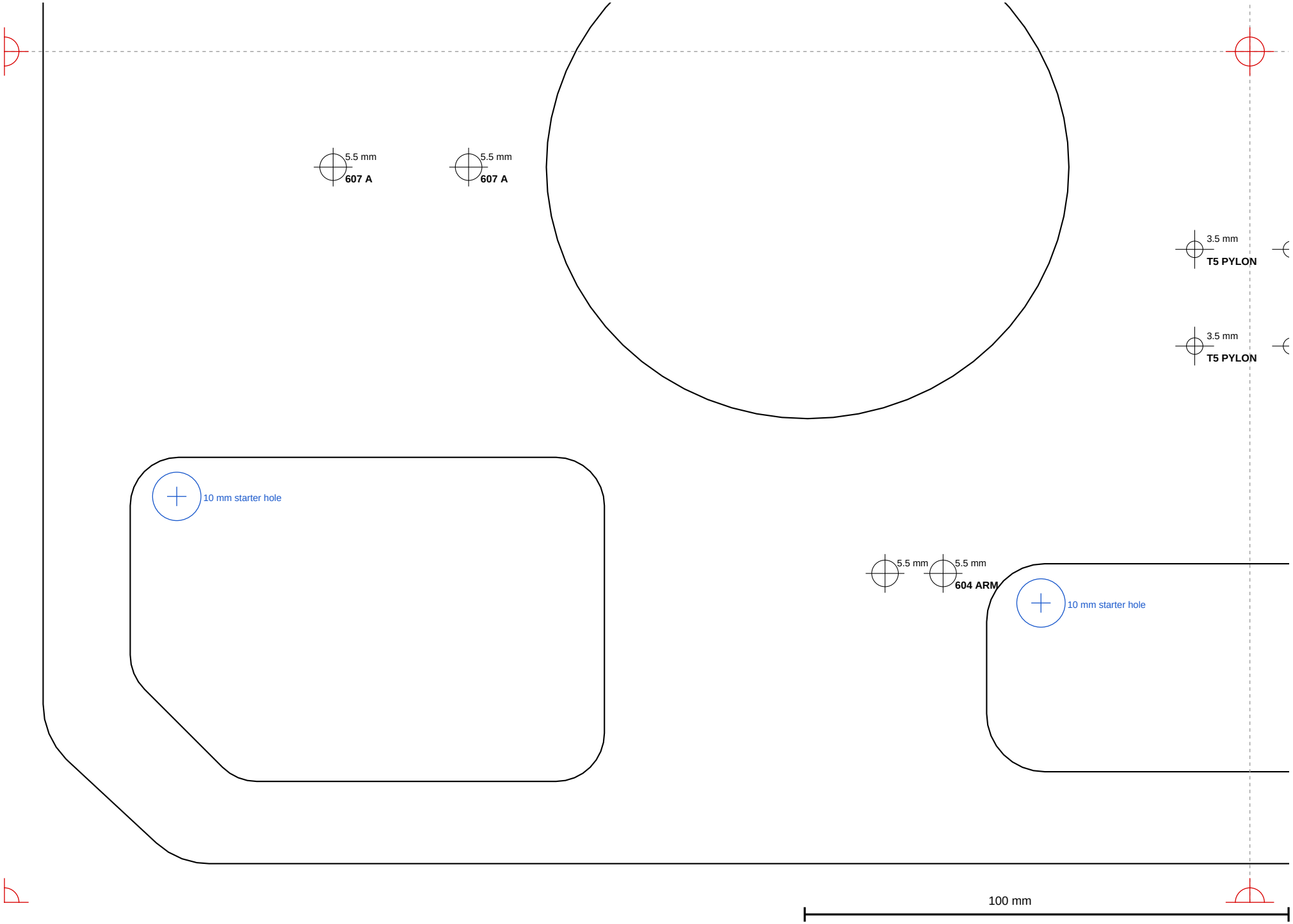
TinkerSub KB-1 side plate 901 PORT - tile A1 of 4 - seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



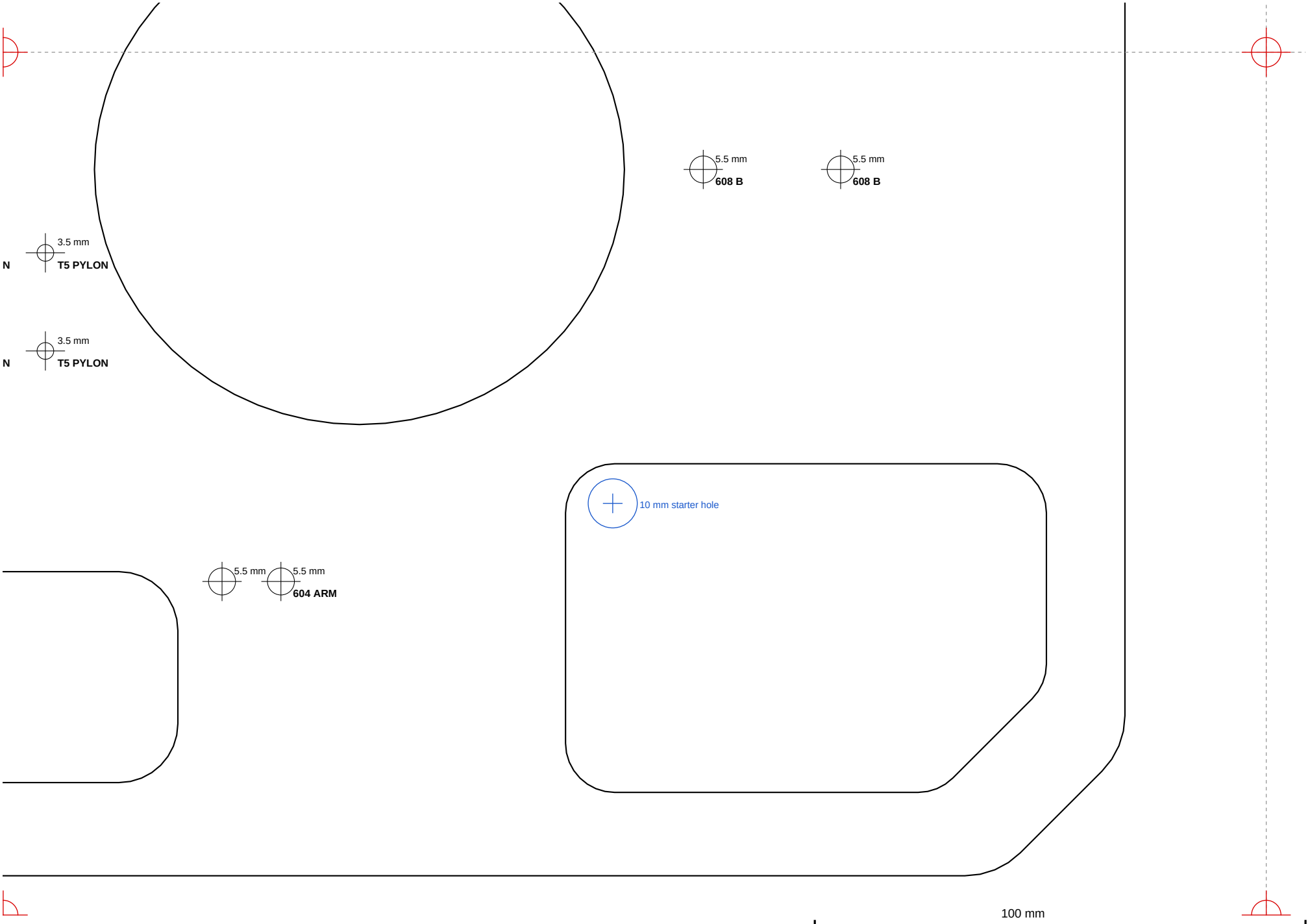
TinkerSub KB-1 side plate 901 PORT - tile A2 of 4 - seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



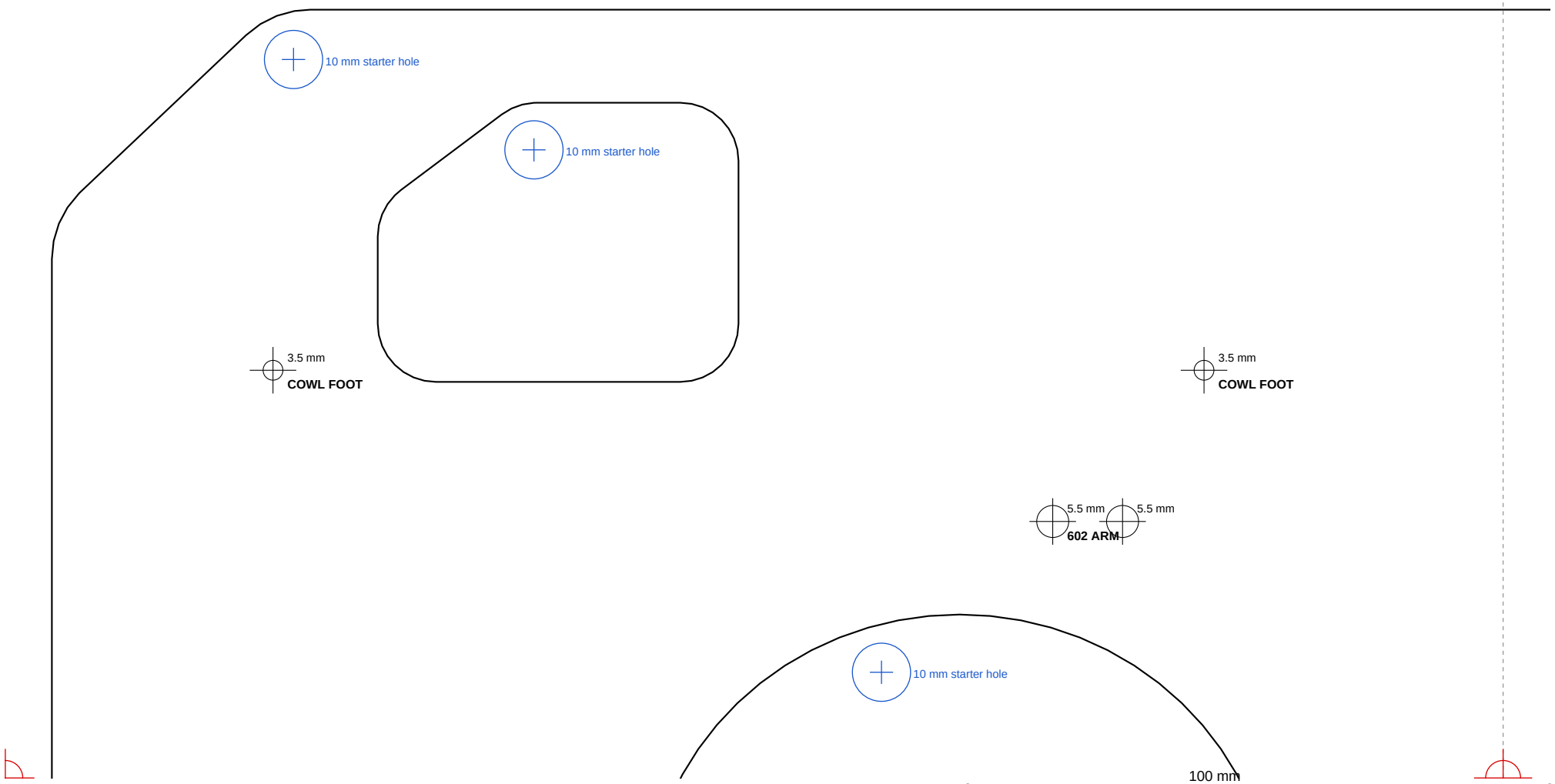
TinkerSub KB-1 side plate 901 PORT - tile B1 of 4 - seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



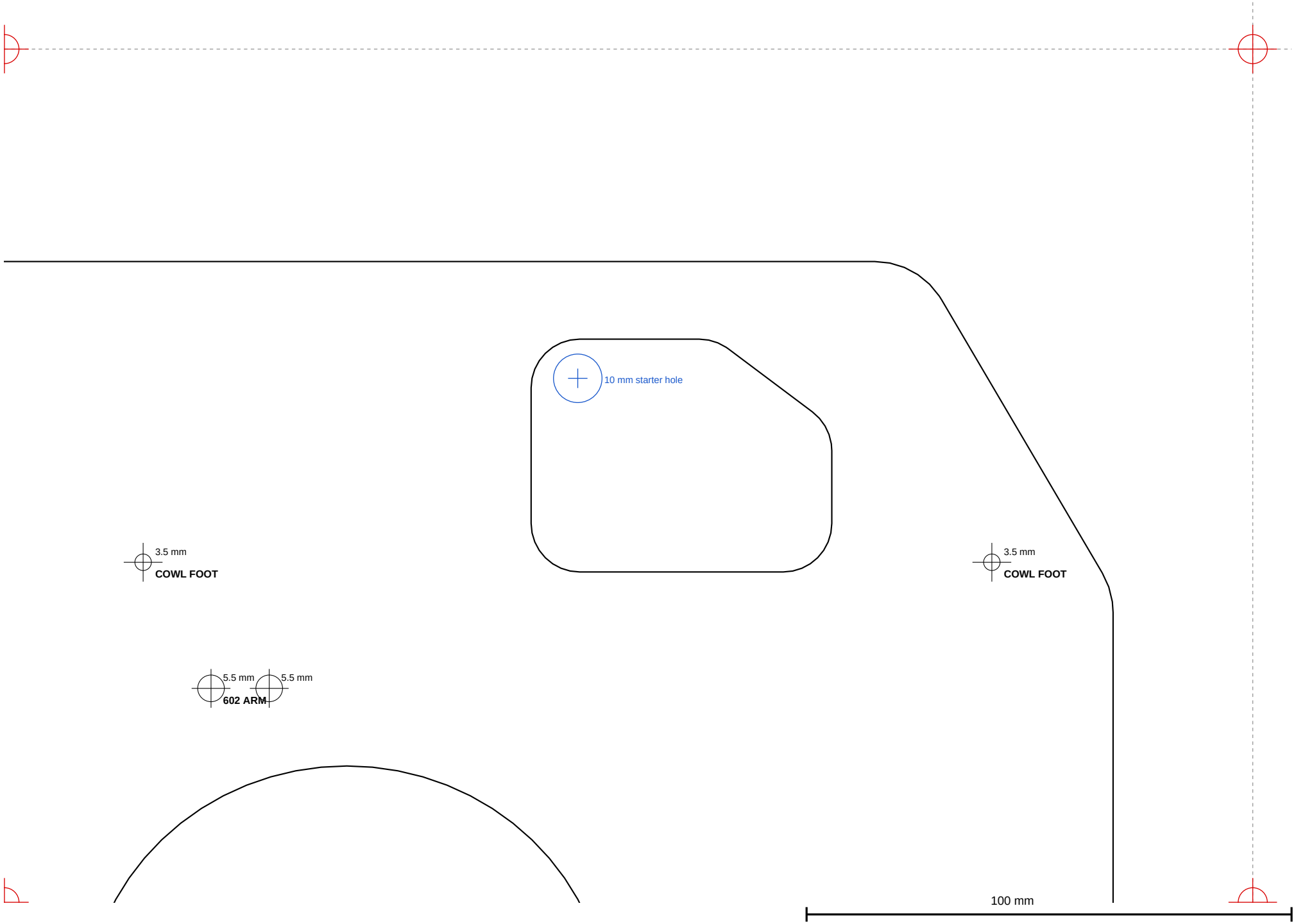
TinkerSub KB-1 side plate 901 PORT - tile B2 of 4 - seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



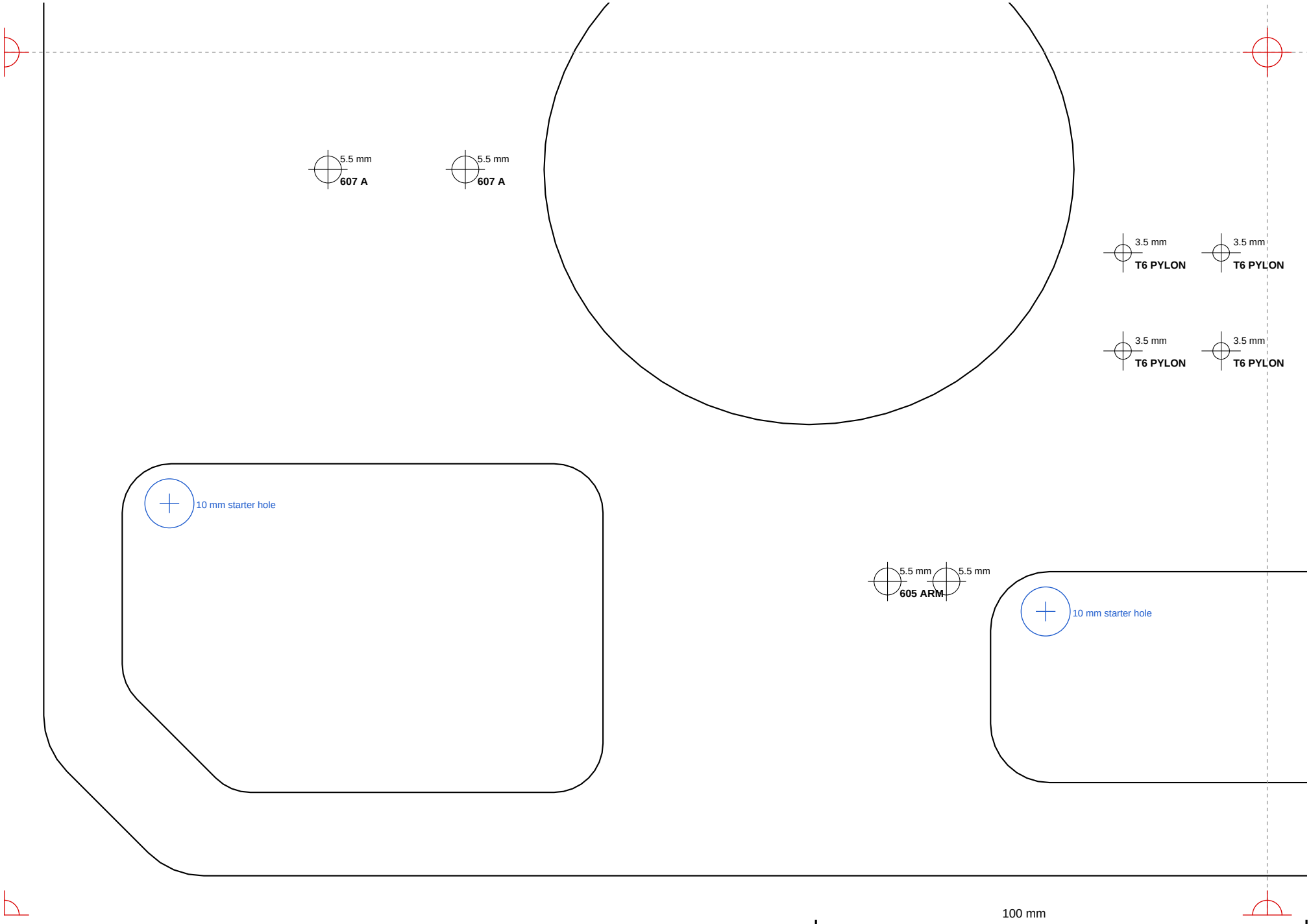
TinkerSub KB-1 side plate 902 STARBOARD - tile A1 of 4 - seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



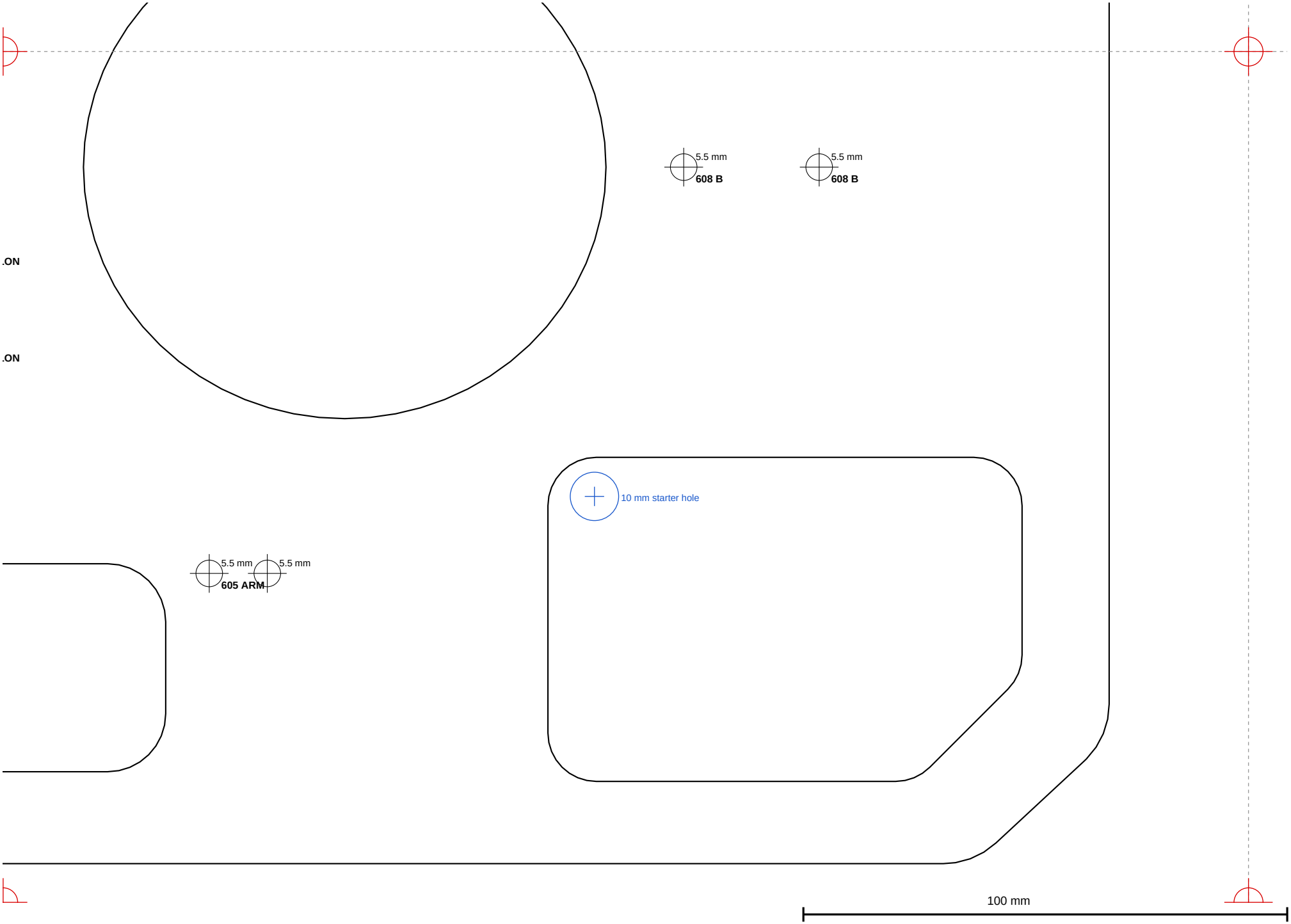
TinkerSub KB-1 side plate 902 STARBOARD - tile A2 of 4 - seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



TinkerSub KB-1 side plate 902 STARBOARD - tile B1 of 4 - seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



TinkerSub KB-1 side plate 902 STARBOARD - tile B2 of 4 - seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.

Appendix B Heat-set insert chart and picture-by-picture assembly

Every printed part, by part number. Arrows point into each hole that takes a heat-set insert, coloured by size; parts in grey boxes take none. Press inserts with the iron at 230-245 C and a light push, keep the insert square, and stop flush (proud brass stops parts sitting flat or scores the pipe). Count per part, not per set: x2 parts need it twice.

M3 insert hole 4.0 mm M4 insert hole 5.6 mm M5 insert hole 6.4 mm Totals for one ROV: **42 x M3, 16 x M4, 36 x M5**

101



Thruster body + Kort duct **x6**

No inserts

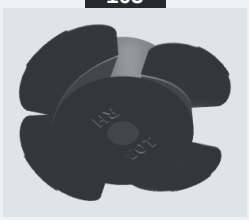
102



Tail cone with oil compensator **x6**

No inserts

103



Right-hand prop **x3**

No inserts

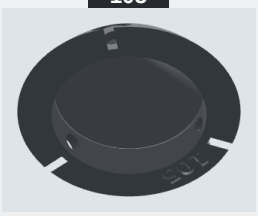
104



Left-hand prop **x3**

No inserts

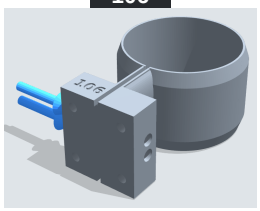
105



Prop spinner **x6**

No inserts

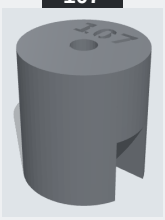
106



Clamps each thruster body **x6**

2 x M4
pinch ears, pressed in from the side the arrows show

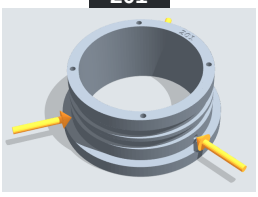
107



Sets the coupler height **x1**

No inserts

201



Hull rear flange **x1**

3 x M3
radial holes in the skirt, from outside, flush. The 4 sled holes stay plain

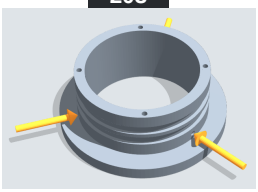
202



Hull rear cap: 4 glands + vent **x1**

No inserts

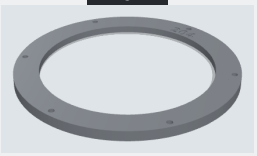
203



Dome flange **x1**

3 x M3
radial holes in the skirt, from outside, flush. The 4 sled holes stay plain

204



Dome clamp ring **x1**

No inserts

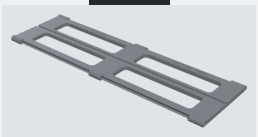
207



Sled front bulkhead **x1**

No inserts

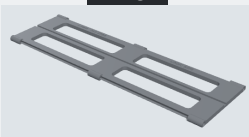
212



Sled side rail, port (12 V side) **x1**

No inserts

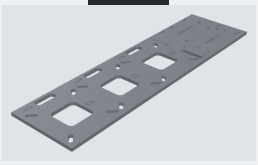
213



Sled side rail, starboard (logic side) **x1**

No inserts

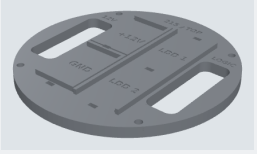
214



Sled deck: boards on both faces **x1**

No inserts

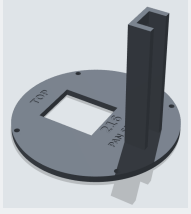
215



Sled rear bulkhead: 12 V bus + LED drivers **x1**

No inserts

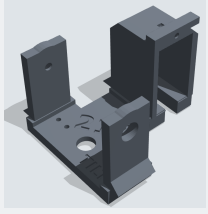
216



Pan-tilt base v2: on the sled rods, carries the pan servo **x1**

No inserts

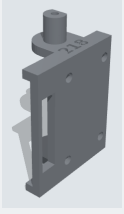
217



Pan-tilt yoke v2: on the pan horn, tilt servo in a closed pocket **x1**

No inserts

218



Pan-tilt camera cradle **x1**

No inserts

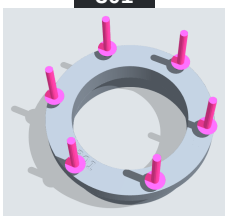
219



Tilt lever on the cradle axle **x1**

No inserts

301



3" pod flange, one at each end **x2**

6 x M5
flange face, around the bore

302



3" pod rear cap: M16 gland + vent **x1**

No inserts

303



3" pod front cap, blank (no holes) **x1**

No inserts

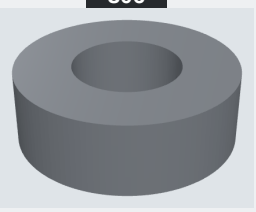
305



18650 holder plate **x6**

No inserts

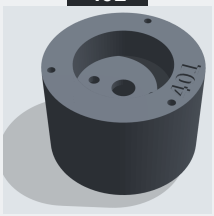
306



Rod spacer between groups **x8**

No inserts

401



Light head rear body **x2**

No inserts

402



Light head front ring **x2**

No inserts

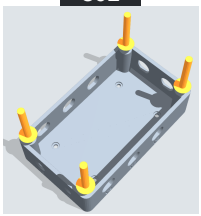
403



Window spacer **x2**

No inserts

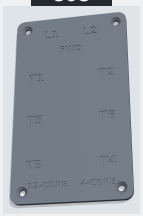
501



Junction box: 6 thruster + 2 light SP13 sockets, 12-core + 4-core out **x1**

4 x M3
lid bosses in the 4 top corners

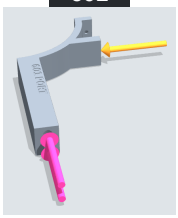
503



Junction box lid with the socket map **x1**

No inserts

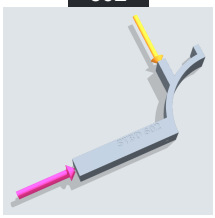
601



Station top, port half: hull top + upper arm **x2**

1 x M3 + **2 x M5**
M5: arm end, from the outer face.
M3: top of the deck rib

602



Station top, starboard half **x2**

1 x M3 + **2 x M5**
M5: arm end, from the outer face.
M3: top of the deck rib

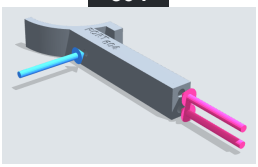
603



Station middle: hull bottom + pod top **x2**

No inserts

604



Station bottom, port half: pod bottom + lower arm **x2**

1 x M4 + **2 x M5**
M5: arm end, from the outer face.
M4: under the lower arm (belly pan hanger)

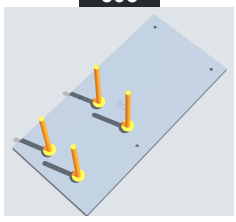
605



Station bottom, starboard half **x2**

1 x M4 + **2 x M5**
M5: arm end, from the outer face.
M4: under the lower arm (belly pan hanger)

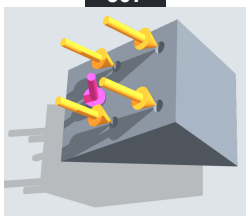
606



Carries the junction box (501) **x1**

4 x M3
from the top, for the junction box floor screws

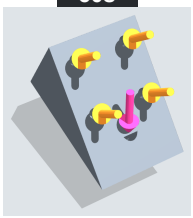
607



Front-port + rear-starboard thruster bracket **x2**

4 x M3 + **2 x M5**
M3: the 45 deg face (pylon). M5: the plate face

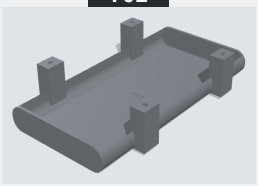
608



Front-starboard + rear-port thruster bracket **x2**

4 x M3 + **2 x M5**
M3: the 45 deg face (pylon). M5: the plate face

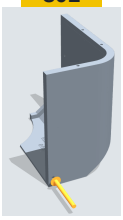
701



Lead ballast tray under the pod **x1**

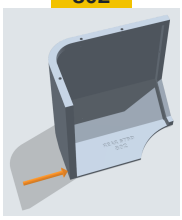
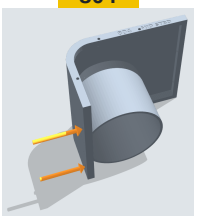
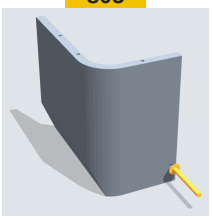
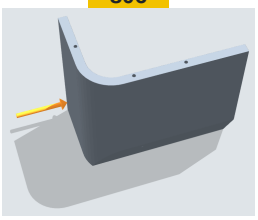
No inserts

801



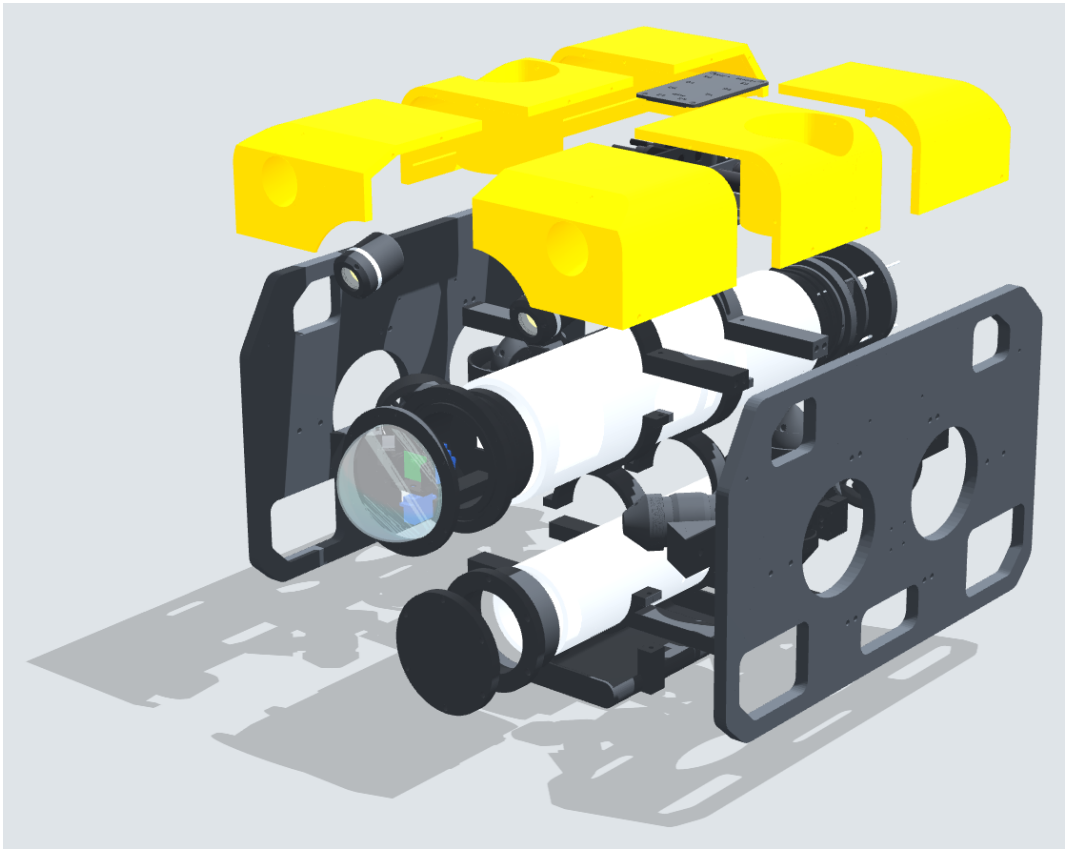
Top cowl, rear, port **x1**

1 x M3
cowl foot, from the outside face

802	803	804	805	806
				
Top cowl, rear, starboard **x1**	Top cowl, middle, port **x1**	Top cowl, middle, starboard **x1**	Top cowl, front, port **x1**	Top cowl, front, starboard **x1**
1 x M3 cowl foot, from the outside face	**2 x M3** cowl feet, from the outside face	**2 x M3** cowl feet, from the outside face	**1 x M3** cowl foot, from the outside face	**1 x M3** cowl foot, from the outside face

Picture-by-picture assembly

A quick visual reference, one panel per stage in build order: the new parts in colour, pulled out along dashed lines to where they go, with what is already built in light grey, and what you need for that stage. The steps are in the section named on each panel.

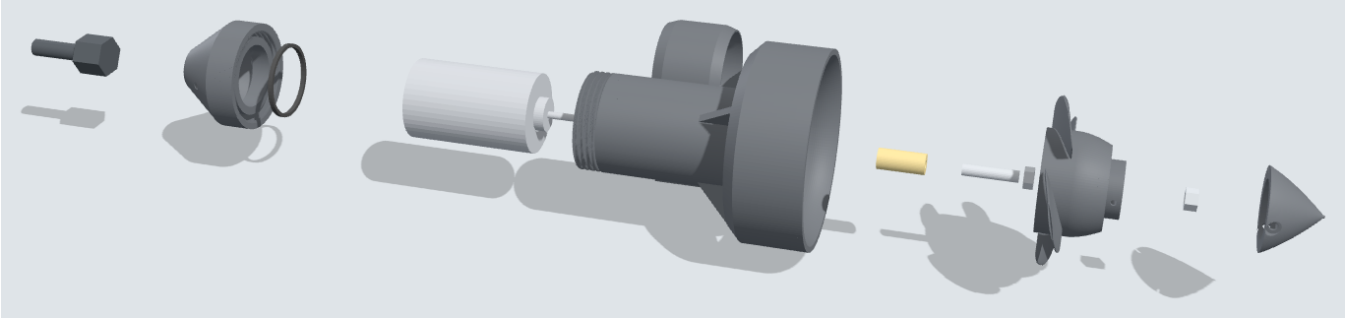


The whole ROV pulled apart. Build order: thrusters, sled, pan-tilt, hull, pod, lights, stations, plates, belly pan, junction box, wiring and sled, cowl.

1 Thrusters

x6

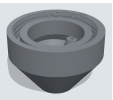
Full instructions: section 7



YOU NEED



101 x1



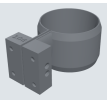
102 x1



103/104 x1



105 x1



106 x1

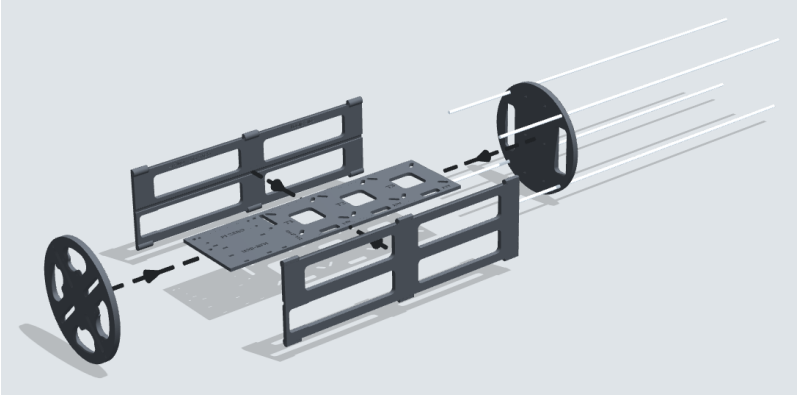


107 gauge

1200 GPH motor	Brass coupler + M5 x 20 set screw	M5 jam nut + M5 nyloc	3 x M3 x 8 button (nose)
M12 x 1.5 gland	32 x 2 O-ring	2 x M4 set screw (oil ports)	2 x M4 x 30 (pylon)
Nitrile finger cot + 5 x 1.5 O-ring	Mineral oil		

2 Electronics sled

Full instructions: section 8.1 and 9



YOU NEED



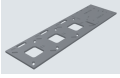
207 x1



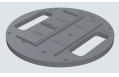
212 x1



213 x1



214 x1

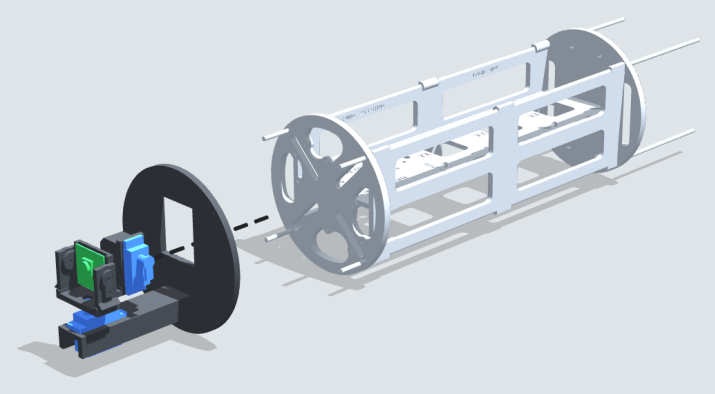


215 x1

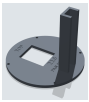
4 x M3 threaded rod, 340 mm	16 x M3 nut	IBT_2 boards, Pi, Pico (section 9)	Wago 221-415 x2, LDD-700HW x2

3 Camera pan-tilt

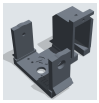
Full instructions: section 8.2



YOU NEED



216 x1



217 x1



218 x1

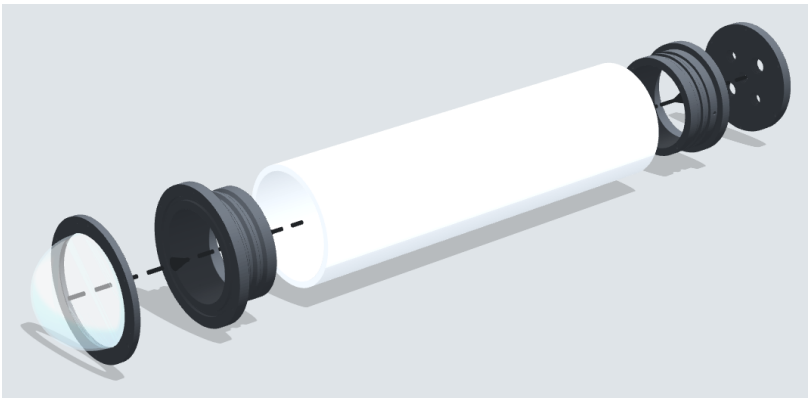


219 x1

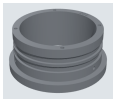
2 x MG90S servo	Arducam + 4 x M2 x 6, 2 mm standoffs	M2 screws (servo, axle)	1 mm steel wire pushrod
8 x M3 nut			

4 Hull: build it and test it empty

Full instructions: section 10



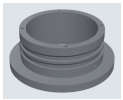
YOU NEED



201 x1



202 x1



203 x1

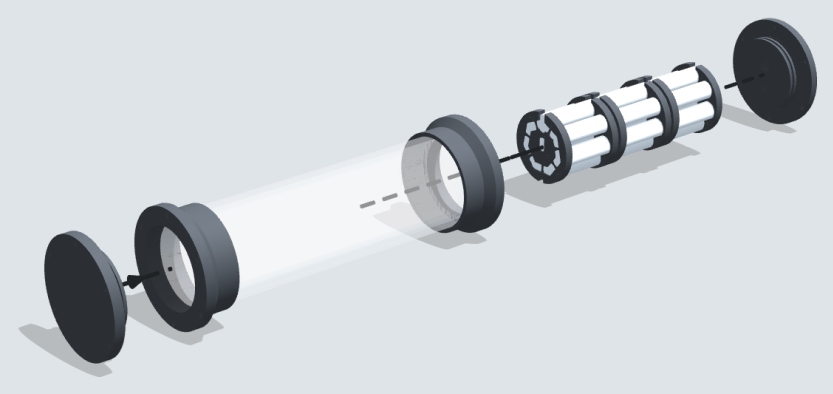


204 x1

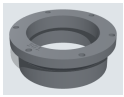
4" pipe, 406 mm	4 x AS568-238 O-ring	AS568-240 + -237 O-rings	6 x M3 x 10 (radial)
6 x M3 x 16 (cap)	6 x M3 x 12 (ring)	100 mm acrylic dome	2 x M16 + 2 x M12 gland, M10 vent bolt

5 Battery pod

Full instructions: section 11



YOU NEED



301 x2



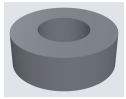
302 x1



303 x1



305 x6



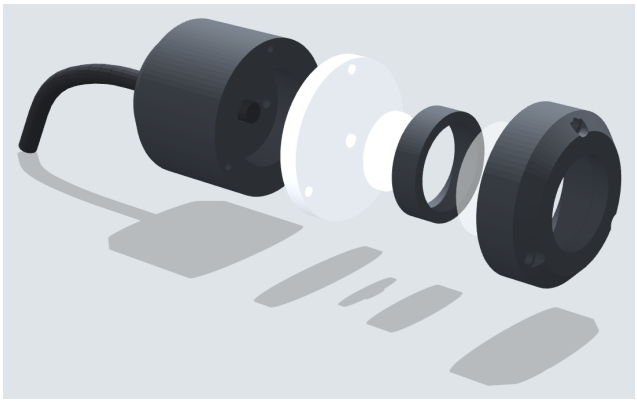
306 x8

3" pipe, 300 mm	4 x 72 x 2.5 O-ring	6 x M5 x 16 (front)	6 x M5 x 14 (rear)
4 x M4 rod, about 225 mm	24 x 18650 cells (3S8P)	BMS + 35 A fuse	M16 gland, M10 vent bolt
Marine epoxy			

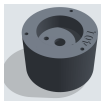
6 Light heads

x2

Full instructions: section 12



YOU NEED



401 x1



402 x1

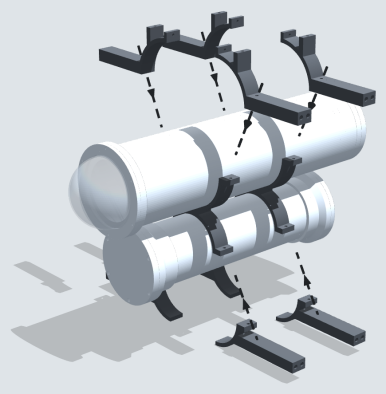


403 x1

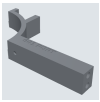
Aluminium disc 42 x 5	XHP50.2 LED on 20 mm star	Glass window	30 x 1.5 + 24 x 1.5 O-ring
3 x M2.5 x 25	2-core 18 AWG, 350 mm	Thermal adhesive, potting	

7 Station frames: join hull and pod

Full instructions: section 13



YOU NEED



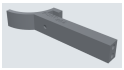
601 x2



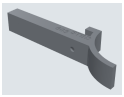
602 x2



603 x2



604 x2

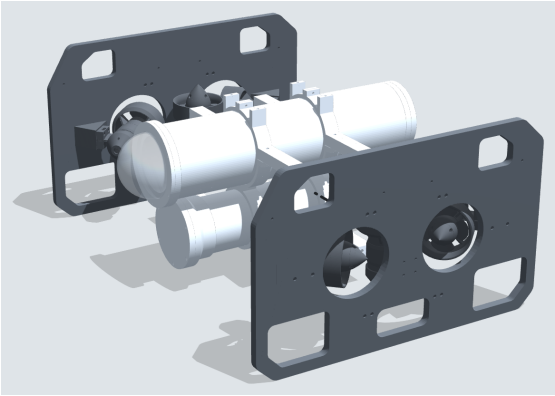


605 x2

M4 x 30 (hull joints)	M4 x 20 (pod joints)	M3 x 20 + nyloc (centre rib)	M4 nuts in the hex pockets

8 Side plates and thrusters

Full instructions: section 14



YOU NEED



607 x2



608 x2

901 x1

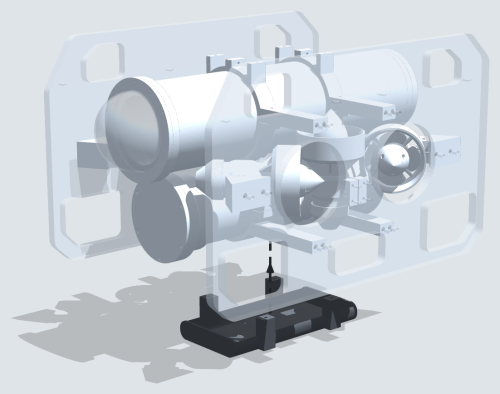
902 x1

T1-T6 x6

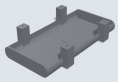
16 x M3 x 16 (pylons to brackets)	8 x M5 x 20 (brackets)	16 x M5 x 20 (plates to arms)	8 x M3 x 30 + nyloc + 1 mm shim (verticals)

9 Belly pan

Full instructions: section 15



YOU NEED

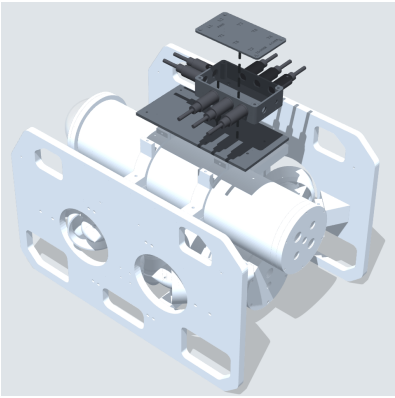


701 x1

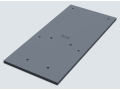
4 x M4 x 50	1.6 kg lead shot in sealed bags

10 Junction box

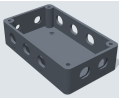
Full instructions: section 16



YOU NEED



606 x1



501 x1

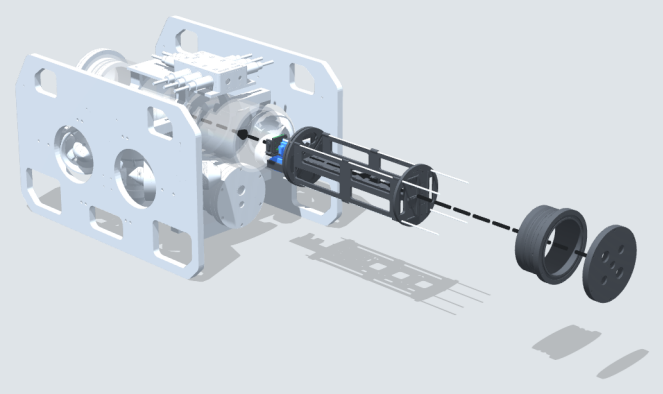


503 x1

6 x SP13 2-pin socket	2 x SP13 3-pin socket	2 x M3 x 10 (deck)	4 x M3 x 8 (box)
4 x M3 x 8 countersunk (lid)	12-core + 4-core cable	Potting compound	

11 Wire it, test it on the bench, slide the sled in

Full instructions: section 16



YOU NEED



201 x1

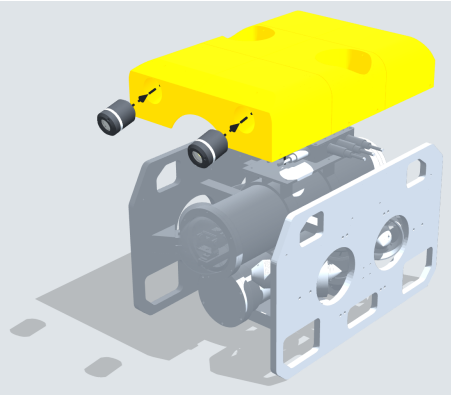


202 x1

F1 fuse, 30 A	AS568-237 O-ring (re-greased)	6 x M3 x 16	Patch lead + RJ45 coupler

12 Lights and top cowl

Full instructions: section 17



YOU NEED



801 x1



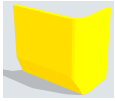
802 x1



803 x1



804 x1



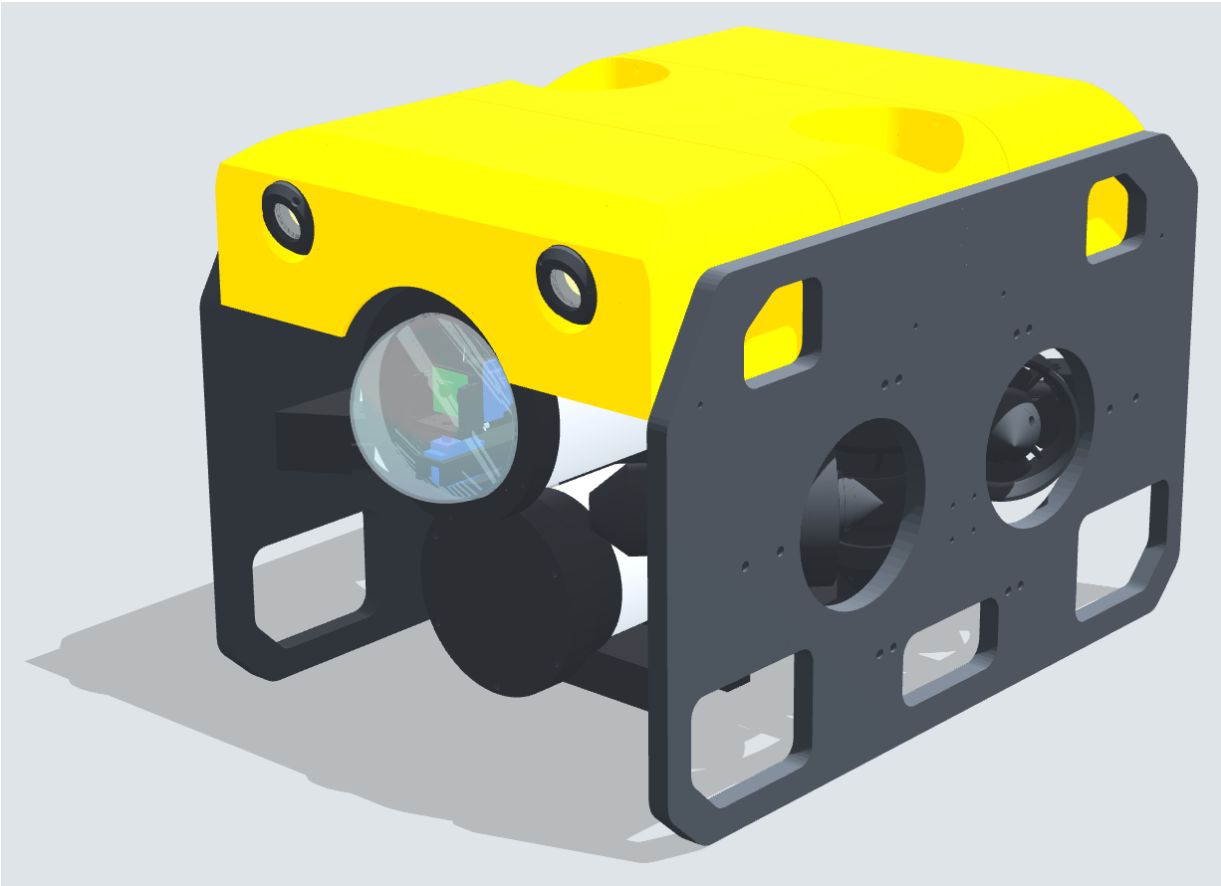
805 x1



806 x1

18 x M3 x 20 + nyloc (cowl joints)	8 x M3 x 20 (cowl to plates)	2 x M3 x 8 set screw	About 1.0 L closed-cell foam
2 x SP13 3-pin plug	Adhesive tie mounts		

Done



Before the first dive: leak tests, trim and the staged dives in section 18.

Appendix C Updating parts from an earlier build

The build steps describe only the current parts. If you printed, cut, bought or assembled anything from an earlier release, find it here: what changed and how to bring it up to date. CHANGELOG.md has the full history.

Part	If yours is	What was wrong	What to do
Rear caps 202 (hull) and 302 (pod)	Printed before 2026-10-30	Plain holes that needed M16 x 1.5, M12 x 1.5 and M10 x 1.0 taps.	Tap them as before, or reprint them: the threads are now printed in.
Hull flanges 201, 203, rear cap 202, pod flanges 301, caps 302 / 303, light front rings 402	Printed before 2026-10-20	Printed at 25 % infill: water can seep through the skin into the infill and past the O-rings.	Reprint them (TinkerSub_KB-1_Reprint_2026-10-20.3mf prints them solid), or give every outside surface that sees water two thin coats of epoxy, keeping it out of the O-ring grooves, the sealing faces and the threads.
Side plates 901 / 902	Cut from templates printed before 2026-10-20	The rearmost COWL FOOT hole is 15 mm too far back, behind the end of the cowl.	Drill one new 3.5 mm hole in each plate 38 mm from the rear end and 62 mm below the top edge. Leave the old hole empty.
Top cowl 801-806	Printed before 2026-10-23	Before 2026-10-20 the side walls stopped above the foot (a loose strip with a slot beside it); before 2026-10-23 the section-seam holes were closed by the flange ramps and some seam bolts had no room for a head.	Reprint all six sections from the current files.
Pan-tilt base 216	Printed before 2026-10-20	The pan servo shelf hung on two thin rails, the weakness that broke the first yoke.	Reprint it (it is in the Reprint project).
Pan-tilt yoke 217	Printed before 2026-10-16	The tilt servo sat in a thin frame that snaps off at its base.	Reprint it (in the Reprint project).
Sled rails 212 / 213	Printed before 2026-10-16	The lower windows stop short of the front, so the wires can't reach the Pico breakout's front terminals.	Cut a 20 x 20 mm opening with a rotary tool from the front of the lower window to 3.5 mm from the rail's front end.
Hull flanges 201 / 203, sled holes	Built from a guide before 2026-10-19	Heat-set inserts fitted in the four sled holes at the end of each skirt.	Drill each insert's thread out with a 3.2 mm bit so the rod tips slide in.
Thrusters: coupler and stud	Assembled from a guide before 2026-10-23	The M5 stud went in before the coupler gauge, so the stud hit the gauge's ceiling and the coupler sits too low.	Loosen the coupler's grub screw, back the stud out, set the height with gauge 107 (section 7.2), then refit the stud.
Station tops 601 / 602	Printed before 2026-10-23	The arm-end inserts are 0.75 mm lower than the plate holes.	Open the four upper holes in each plate to 7 mm (9/32") and put a washer under each screw head. The centre rib takes the same M3 screw as the new part.
Light rear body 401	Printed before 2026-10-23	The three screw holes are 8 mm deep; the screws bottom out.	Drill each to 12 mm with a 2.1 mm (5/64") bit.
Belly pan 701	Printed before 2026-10-23	The hanger holes stop 16 mm short of the bottom of each post.	Drill each through from the top with a 4.5 mm (11/64") bit; the hole guides the drill.
Tail cones 102	Printed before 2026-10-25	The compensator nipple is tucked under the spigot lip, where a finger cot is very hard to fit.	Reprint them, about 25 g each.
Junction deck 606	Printed before 2026-10-26	Shorter: it reaches only the rear station and flexes.	Reprint it (Frame plate 5); it screws to both stations.
Screws	Bought from a list before 2026-10-23	Some lengths were wrong.	Use the shopping list in section 4. Changed: plates to arms and brackets 20 mm (was 25), pylons to brackets 16 mm (was 12), vertical pylons 30 mm (was 25), dome ring 12 mm (was 16), pod rear cap 14 mm, nose cones 8 mm (was 6), belly pan 50 mm (was 30), centre rib now M3 + nyloc.
O-rings	Bought from a list before 2026-10-25	Wrong sizes for their grooves.	Hull radial seals AS568-238 (was -240), rear cap face AS568-237 (was -239), pod caps 72 x 2.5 (was 74), light front ring 30 x 1.5 (was 31). The dome face AS568-240 and the others were right.